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650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
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Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The present document is part of a multi-part TR as described below:

- Part 1: LI for IMS based services. See TR 33.929-1 [10].
- Part 2: LI for IMS based STIR/SHAKEN. See TR 33.929-2 [11].
- Part 3: LI for messaging services. See TR 33.929-3 [12].
- Part 4: LI for data in 5G core. See TR 33.929-4 [13].
- Part 5: LI for ID association caching. See TR 33.929-5 [14].
- Part 6: LI for IMS based RCS (present document).
- Part 7: LI location acquisition capabilities. See TR 33.929-7 [15].
- Part 8: LI for MMS. See TR 33.929-8 [16].
- Part 9: LI for PTC. See TR 33.929-9 [17].

In Part 1, the illustrations of LI for IMS-based services focus on the LI aspects of IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (local break-out and home-routed roaming).

In Part 2, the illustrations of LI for IMS based STIR/SHAKEN focus on various STIR/SHAKEN related LI reporting scenarios.

In Part 3, the illustrations of LI for messaging services focus on SMS over NAS and SMS over IP.

In Part 4, the illustrations of LI of data in 5G core focus on the LI aspects of data interception focussing on the PDU session related events including the handover scenarios.

In Part 5, the illustrations of LI for ID association caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

In Part 6 (present document), the illustrations of LI for IMS based RCS focus on the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider.

In Part 7, the illustrations of LI location acquisition capabilities focus on the conceptual overview and the flow diagrams for location reporting, LALS and location acquisition.

In Part 8, the illustrations of LI for MMS focus on the conceptual overview and the flow diagrams for MMS.

In Part 9, the illustrations of LI for PTC focus on the conceptual overview and the flow diagrams for MCPTT and PoC, the two commonly referred to as Push to Talk over Cellular (PTC).

Introduction

Unlike the previous generation of LI technical specifications, the latest LI technical specifications (TS 33.126 [2], TS 33.127 [3], TS 33.128 [4]) contain only the normative part of the requirements. However, the implementers may need additional information such as architectural diagrams, conceptual scenarios, flow-diagrams and additional background information to better understand the requirements in those specifications. The present document collects the relevant informative annexes from the previous generation of LI technical specifications and then expands them to include the similar level of details as for latest technologies such as 5G.

1 Scope

The present document provides architectural diagrams, conceptual scenarios, flow-diagrams, examples, and other background information which can be useful to implement the LI functions defined in TS 33.126 [2], TS 33.127 [3] and TS 33.128 [4].

The present document covers the LI for IMS based RCS that include the illustrations covering the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider. The LI for IMS based services are illustrated in TR 33.929-1 [10].

The LI for IMS-based RCS is agnostic to the packet core network.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.126: "Lawful Interception requirements".

[3] 3GPP TS 33.127: "Lawful Interception (LI) architecture and functions".

[4] 3GPP TS 33.128: "Protocol and procedures for Lawful Interception (LI)".

[5] 3GPP TS 24.147: "Conferencing using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3".

[6] 3GPP TS 23.501: "System architecture for the 5G System".

[7] 3GPP TS 24.174: "Support of multi-device and multi-identity in the IP Multimedia Subsystem (IMS); Stage 3".

[8] 3GPP TS 23.228: "IP Multimedia Subsystem (IMS), Stage 2".

[9] 3GPP TR 33.928: "ADMF logic for provisioning Lawful Interception (LI)".

[10] 3GPP TR 33.929-1: "Lawful Interception (LI) implementation guidance; LI for IMS-based services".

[11] 3GPP TR 33.929-2: "Lawful Interception (LI) implementation guidance; LI for IMS based STIR/SHAKEN".

[12] 3GPP TR 33.929-3: "Lawful Interception (LI) implementation guidance; LI for messaging services".

[13] 3GPP TR 33.929-4: "Lawful Interception (LI) implementation guidance; LI for data in 5G core".

[14] 3GPP TR 33.929-5: "Lawful Interception (LI) implementation guidance; LI for ID association caching".

[15] 3GPP TR 33.929-7: "Lawful Interception (LI) implementation guidance; LI location acquisition capabilities".

[16] 3GPP TR 33.929-8: "Lawful Interception (LI) implementation guidance; LI for MMS".

[17] 3GPP TR 33.929-9: "Lawful Interception (LI) implementation guidance; LI for PTC".

[18] GSMA RCC.07: "Rich Communication Suite – Advanced Communications Services and Client Specification".

[19] IETF RFC 4975: "The Message Session Relay Protocol (MSRP)".

[20] 3GPP TS 23.040: "Technical realization of the Short Message Service (SMS)".

[21] 3GPP TS 29.540: "5G System; SMS Services; Stage 3".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

None.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

3GPP	3rd Generation Partnership Program
ADMF	Administration Function
AMF	Access and Mobility Management Function
APN	Access Pointe Name
AS	Application Server
BBIFF	Bearer Binding Intercept and Forward Function
BBIFF-C	Bearer Binding Intercept and Forward Function Control plane
BBIFF-U	Bearer Binding Intercept and Forward Function User plane
BGCF	Border Gateway Control Function
CC	Content of Communication
CPM	Converged IP Messaging
CSCF	Call Session Control Function
CSP	Communication Service Provider
CUPS	Control and User Plane Separation
DNN	Data Network Name
eCNAM	Enhanced Calling NAMe
E-CSCF	Emergency Call Session Control Function
EPC	Evolved Packet Core
H-SMF	Home SMF
H-UPF	Home UPF
HPLMN	Home PLMN
HSS	Home Subscriber Server
I-CSCF	Interrogating CSCF
IBCF	Interrogating Border Control Function
ID	Identity

IMPI	IP Multimedia Private Identity
IMPU	IP Multimedia Public Identity
IMS	IP Multimedia Subsystem
IMS-AGW	IMS Media Gateway
IRI	Intercept Related Information
LALS	Lawful Access Location Services
LBO	Local Break Out
LEA	Law Enforcement Agency
LEMF	Law Enforcement Monitoring Facility
LI	Lawful Interception
LICF	Lawful Interception Control Function
LI_HI1	LI_Handover Interface 1
LI_HI2	LI_Handover Interface 2
LI_HI3	LI_Handover Interface 3
LIPF	LI Provisioning Function
LI_T1	Lawful Interception Triggering Interface 1
LI_T3	Lawful Interception Triggering Interface 3
LI_X1	Lawful Interception Internal Interface 1
LI_X2	Lawful Interception Internal Interface 2
LI_X3	Lawful Interception Internal Interface 3
LMISF	LI Mirror IMS State Function
LMISF-CC	LMISF for the handling of CC
LMISF-IRI	LMISF for the handling of IRI
LTE	Long Term Evolution
MCPTT	Mission Critical Push To Talk
MDF	Mediation and Delivery Function
MDF2	Mediation and Delivery Function 2
MDF3	Mediation and Delivery Function 3
MGCF	Media Gateway Control Function
MMS	Multimedia Message Service
MRFC	Media Resource Function Controller
MRFP	Multimedia Resource Function Processor
N9HR	N9 Home Routed
NAS	Non-Access Stratum
NR	New Radio
PASSporT	Personal ASSertion Token
P-CSCF	Proxy Call Session Control Function
PDN	Packet Data Network
PEI	Permanent Equipment Identifier
PGW	PDN Gateway
PGW-C	PDN Gateway Control Plane
PGW-U	PDN Gateway User Plane
PLMN	Public Land Mobile Network
PoC	Push to talk over Cellular
POI	Point Of Interception
PTC	Push to Talk over Cellular
RCD	Rich Call Data
RCS	Rich Communication Suite
S8HR	S8 Home Routed
S-CSCF	Serving Call Session Control Function
SDP	Session Description Protocol
SHAKEN	Signature-based Handling of Asserted information using toKENs
SMF	Session Management Function
SGW	Serving Gateway
SGW-C	Serving Gateway Control Plane
SGW-U	Serving Gateway User Plane
SIP	Session Initiation Protocol
SMS	Short Messaging Service
STIR	Secure Telephony Identity Revisited
SUPI	Subscriber Permanent Identifier
TF	Triggering Function
TrGW	Transit Gateway

UDM	Unified Data Management
UE	User Equipment
UPF	User Plane Function
VPLMN	Visited PLMN
V-SMF	Visted SMF
V-UPF	Visited UPF
xCC	LI_X3 Communications Content
xIRI	LI_X2 Intercept Related Information

4 Illustration of LI for IMS-based RCS

4.1 Overview

4.1.1 Background

As defined in GSMA RCC.07 [18], Rich Communication Suite (RCS) provides a framework for advanced communication services that, in part, include the handling of capability discovery, instant messaging, chat sessions and file-sharing. Users can receive read receipts, see the indication of other party typing, etc. RCS is inter-operable with the other messaging types such as SMS and MMS.

Within the network, RCS Server, as an IMS-based Application Server, is used to establish the RCS sessions between the RCS users. To share a file with the other user present on an RCS session, the RCS user would upload the file to an HTTP Content Server and then inform the other RCS user about the uploaded file (with the URL) using the SIP: Message on the established RCS Session. The other RCS user then could download the file (using the received URL) from the HTTP Content Server. At recipient RCS user's end an optional File Localization Function may be present, specially, when the recipient RCS user is in a different domain.

Figure 4.1.1-1 depicts an overview of RCS related functions at a high-level. In the diagram, RCS User (A) and RCS User (B) are involved in an RCS communication.

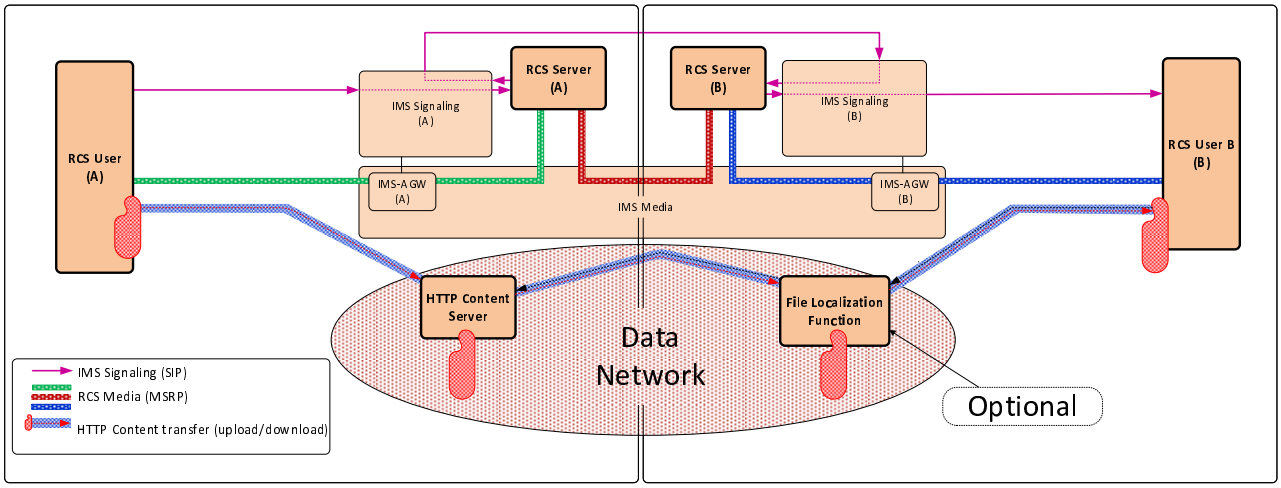


Figure 4.1.1-1: A high-level view of RCS related functions

As with the IMS, RCS sessions are also established using the SIP signalling with MSRP-based media. At the terminating end, it is possible that the RCS session may get redirected to a different RCS user.

The media may within the SIP message (e.g. SIP MESSAGE) for non-session related RCS (e.g. pager mode communications).

The file-upload and download are done using the HTTP messages.

The RCS Server (s) may be within the CSP domain or in a third party domain. In the latter case, it is assumed that the IMS signalling and IMS media related functions are also in the third party domain and in this case, depending on how the RCS user connectivity to the third party domain, several cases arise. For example, the CSP could just provide the access to the RCS users with the packet data connection that carry IMS signalling and media to the third party domain or the RCS user may not be a subscriber of a CSP.

It is also possible in some deployments, the third party domain may not have the IMS signalling functions and IMS media related functions and the SIP messages may be routed to the RCS routed via SIP proxies and media may be routed to the RCS Server. The present document assumes that third party domain has an IMS domain.

4.1.2 RCS view with LI

The LI functions for RCS are provided by the RCS Server, the HTTP Content Server and the File Localization Function (when present). Extending the diagram shown in figure 4.1.1-1, with a high-level view of LI aspects for RCS, are depicted in figure 4.1.2-1 below.

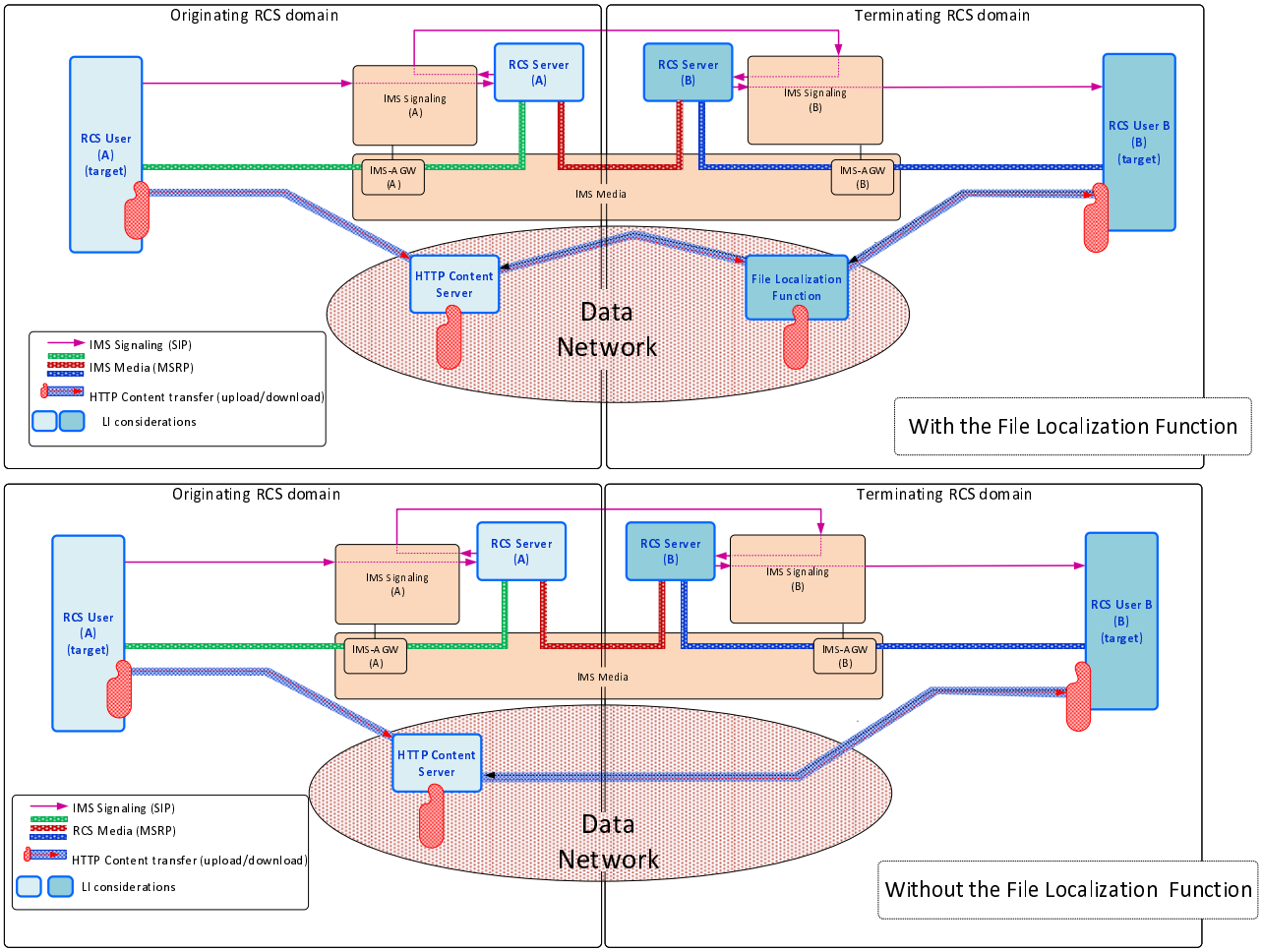


Figure 4.1.2-1: A high-level view of RCS LI functions

In the diagram shown in figure 4.1.2-1, RCS User (A) and RCS User (B) are LI targets in the respective RCS provider's network (shown as Originating RCS domain and Terminating RCS domain). The RCS Server (A) and the HTTP Content Server provide the LI functions for RCS User (A) in the originating RCS domain. The RCS Server (B) and the File Localization Function provide the LI functions for RCS User (B) in the terminating RCS domain.

When File Localization Function is not deployed, with the understanding that HTTP Content Server is not managed by the Terminating RCS domain, the LI functions for the RCS User B is limited to the LI functions provided by the RCS Server (B).

When both RCS users are in the same RCS domain, the File Localization Function is not needed as illustrated in figure 4.1.2-2 below:

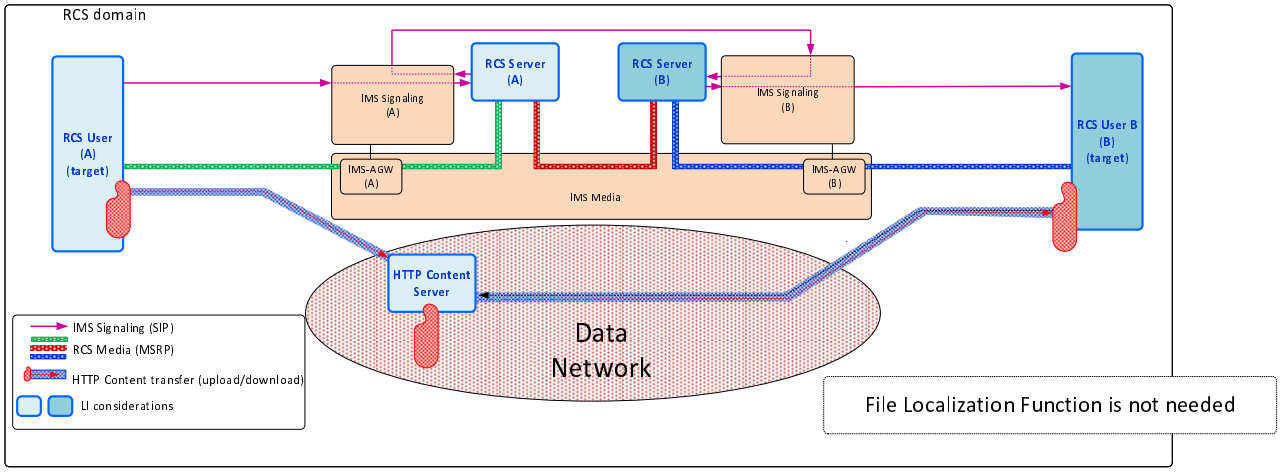


Figure 4.1.2-2: A high-level view of RCS LI functions (single RCS domain)

In the diagram shown in figure 4.1.2-2, RCS User (A) and RCS User (B) are LI targets in the same RCS provider's network (shown as RCS domain). The RCS Server (A) and the HTTP Content Server provide the LI functions for RCS User (A). The RCS Server (B) and HTTP Content Server provide the LI functions for RCS User (B).

When the RCS Servers are in third party domain, the LI functions are provided by those RCS Servers present in the third party domain. In this case, the HTTP Content Server and File Localization Function are also part of that third party domain. In this case, LI for RCS does not apply within the CSP domain that serves the RCS User.

4.1.3 RCS LI architecture

As per the LI architecture defined in TS 33.127 [3], TS 33.128 [4], the RCS Server provides the IRI-POI, CC-POI, IR-TF and CC-TF functions with the following target identifiers:

- IMPU.
- IMPI.
- IMEI.
- PEIIMEI.

Among the identities listed above, only IMPU can be a target non-local ID.

The IRI-POI triggered by the IRI-TF and the CC-POI triggered by the CC-TF are in the HTTP Content Server or the File Localization Function. The triggers allow such POIs to report the file-sharing related communications (e.g. file upload, file download) based on the URL associated with the file as the target identifier. The LI architecture for these aspects is depicted in figure 4.1.3-1.

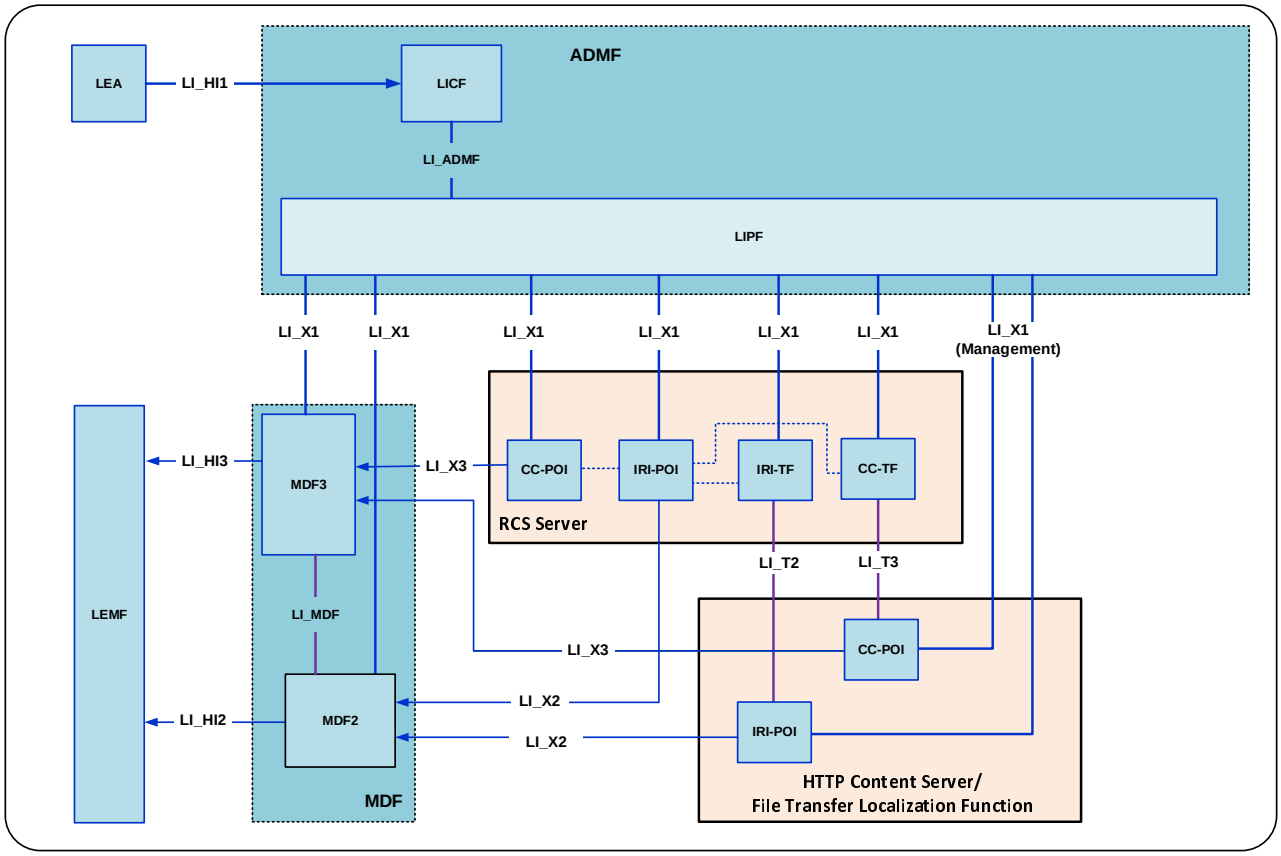


Figure 4.1.3-1: LI architecture with POIs, TFs in RCS Server and triggered POIs in HTTP Content Server/File Localization Function

The POIs in the RCS Server report the RCS communications that may or may not have an RCS session (e.g. chat, instant messaging).

As per the LI architecture defined in TS 33.127 [3], TS 33.128 [4], the HTTP Content Server and File Localization Function may also have provisioned IRI-POI and CC-POI with the following target identifiers:

- SIPURI.
- TELURI.
- GPSIMSISDN.
- MSISDN.
- GPSINAI.
- IMSI.
- SUPIIMSI.
- SUPINAI.
- Email Address.

The LI architecture for these aspects is depicted in figure 4.1.3-2.

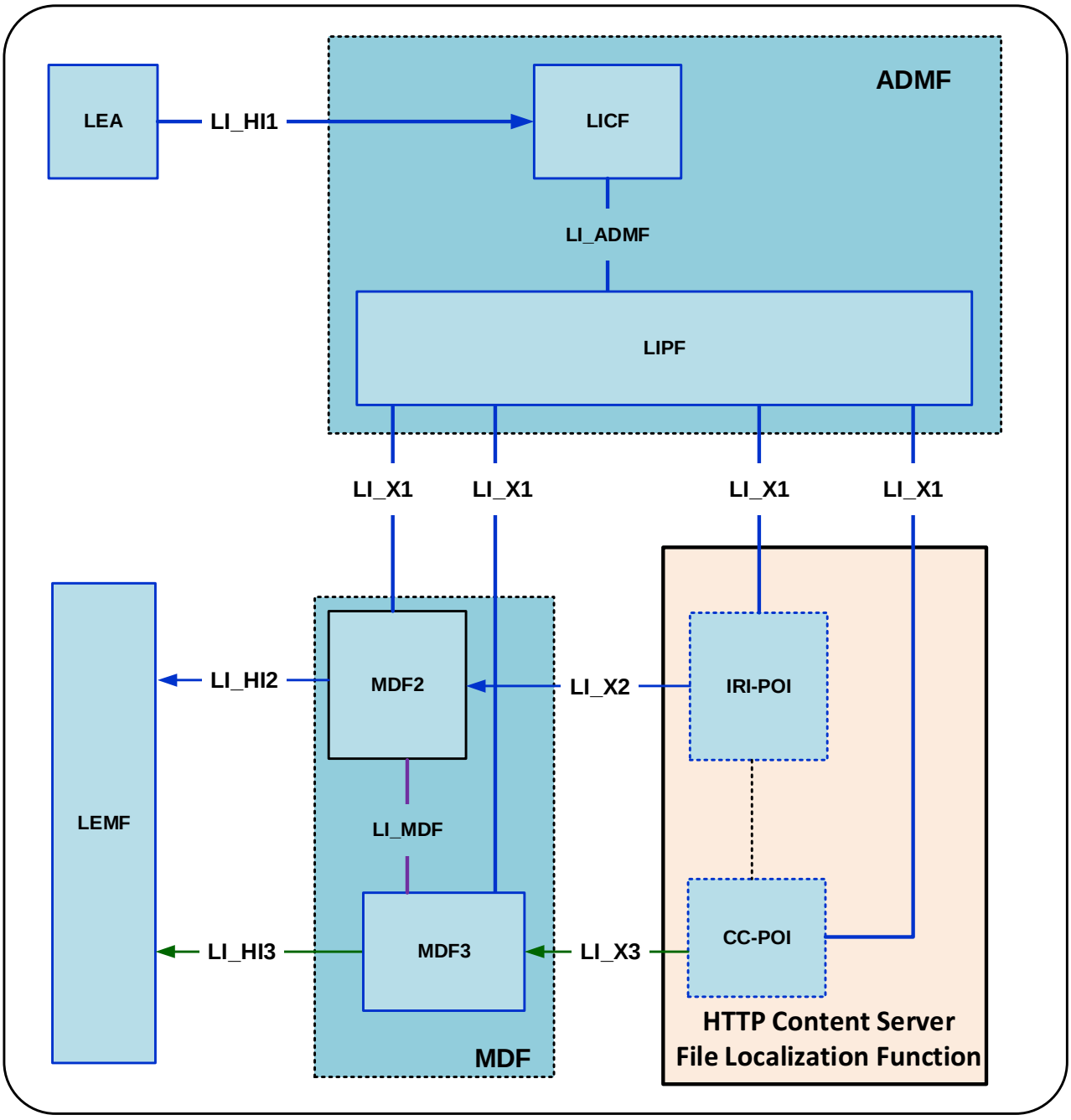


Figure 4.1.3-2: LI architecture with provisioned POIs in HTTP Content Server/File Localization Function

The provisioned POIs report file-sharing related communications (e.g. file-upload, file-download), if and only if the provisioned target identifiers are used by the RCS users during the file-upload or download.

When third party registration is used for RCS, the IRI-POI present in the S-CSCF of that CSP domain would report the RCS registration related events. To enable such reporting, the IRI-POI present in the S-CSCF is provisioned as indicated in figure 4.1.3-3 below:

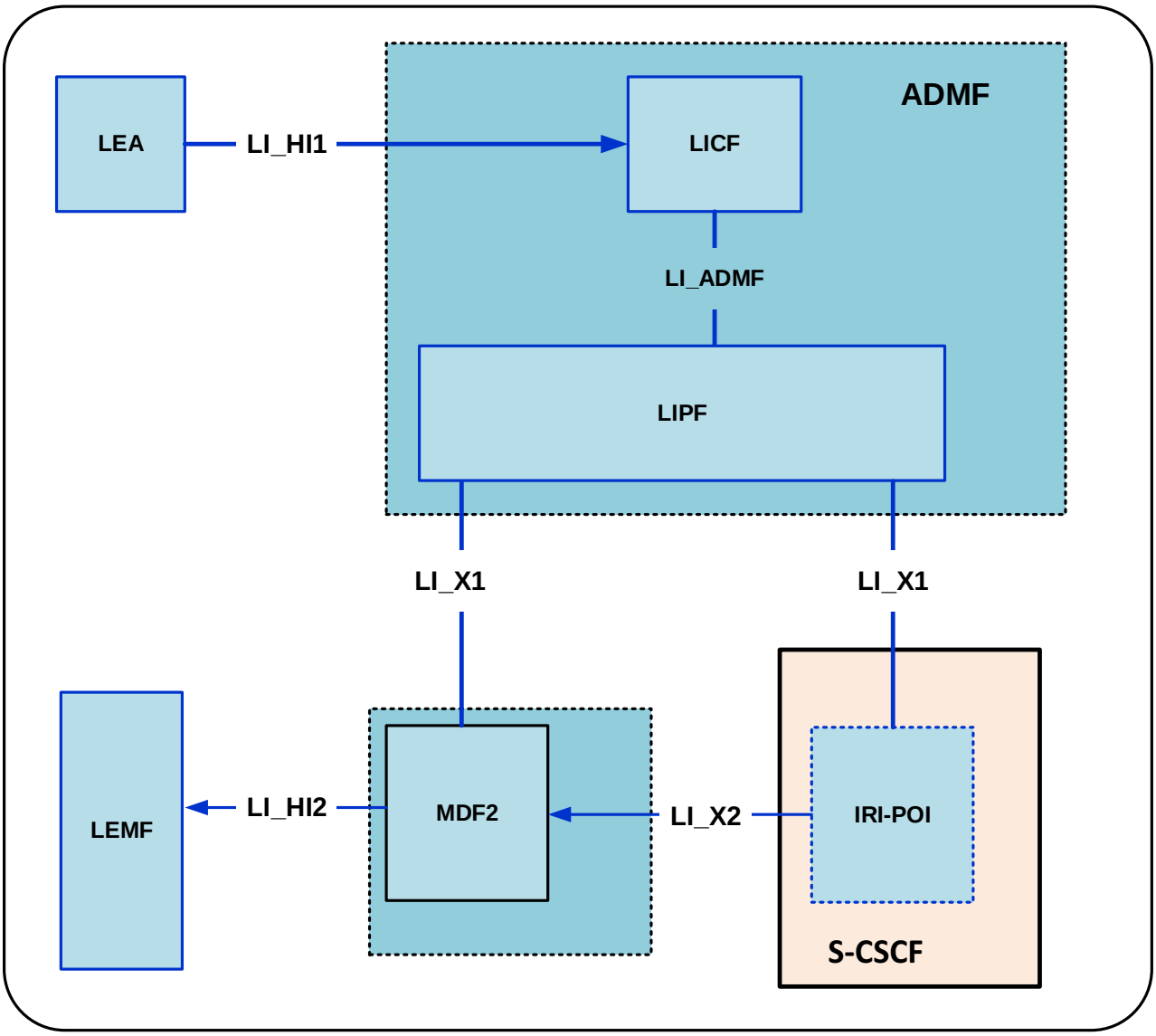


Figure 4.1.3-3: LI architecture with provisioned IR-POI in S-CSCF

In the case, when IRI-POI present in the S-CSCF is provisioned to support the interception of other IMS-based services (e.g. voice), a separate task to provision the IRI-POI in that S-CSCF may still be required with the service type of RCS. When multiple intercepts are active on the same target identifier, the IRI-POI in the S-CSCF is provisioned only once. This is true even for the case of POIs in the RCS Server and the HTTP Content Server/File Localization Server.

4.2 RCS topology scenarios

4.2.1 General

The subsequent clauses illustrate various topology scenarios for RCS which are built on the high-level views presented in clause 4.1.1 and 4.1.2.

Three cases are illustrated here:

- RCS Servers are within the CSP domain.
- RCS Servers are within a third-party RCS service provider domain.
- Hybrid of the above two.

The focus is on the illustration of LI for RCS even though a few illustrations of IMS LI for other services are shown for comparison purpose. In the illustrations that depict the LI for other IMS-based services only the default option defined in TR 33.929-1 [10] is considered. TR 33.929-1 [10] illustrates the other possible options.

The illustrations also show that the RCS LI does not apply to VPLMN.

RCS interworking with SMS and MMS are also illustrated. File transfer related topologies are also illustrated.

When third party registration is used for RCS, the S-CSCF present in the CSP domain may report the RCS registration related events. The illustrations do not show LI function at S-CSCF for RCS purpose.

4.2.2 Few conventions used

In all the illustrations, when the RCS user is a subscriber of a CSP domain, 5G radio and packet core is assumed. Within the drawings, (A) represents the ingress side and (B) represents the egress side. When ingress and egress have to be shown in multiple network domains, (A1), (A11), (B1), B (11), etc. are used.

Figure 4.2.2-1 shows some of the conventions used within the topology drawings (in a general way):

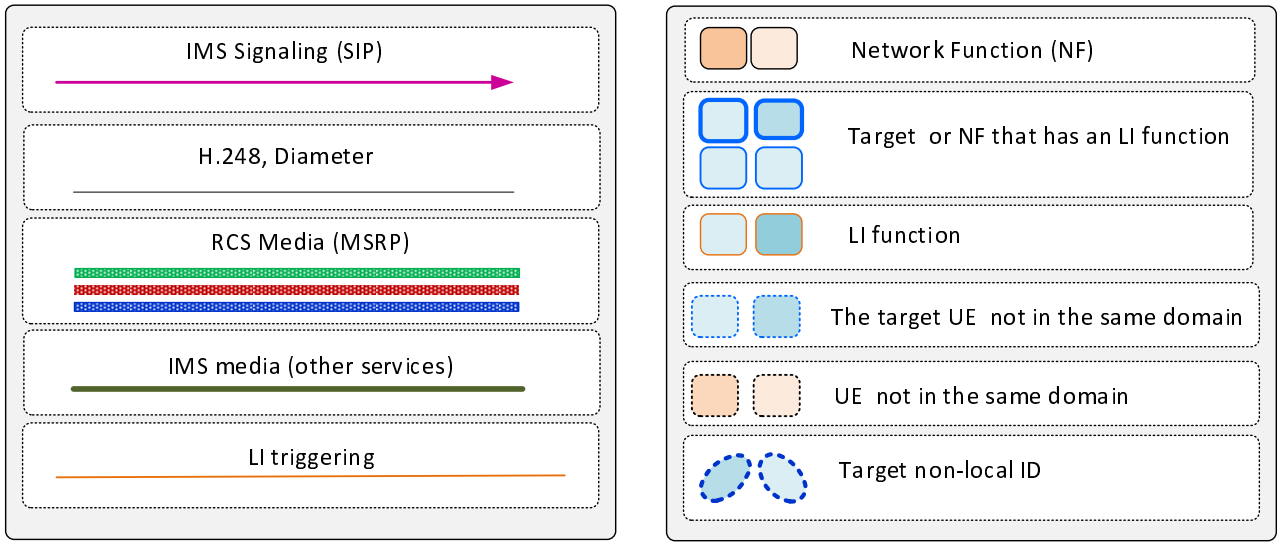


Figure 4.2.2-1: Few conventions used within the drawings

The term target non-local ID is defined and used in TS 33.127 [3], TS 33.128 [4] and other 33.929 series TRs (e.g. TR 33.929-1 [10]). As in the case of other IMS based services, for RCS as well, when a target is non-local ID, the remote user ID is checked for a target match (e.g. at the originating end, Request URI is checked whereas at the terminating end, P-Asserted ID is checked).

As defined and used in TS 33.127 [3] and TS 33.128 [4], the term "non-local" as used in "target non-local ID" does not represent any non-local nature of a target. For example, a target can be outbound roaming user and hence, can be located in another CSP domain (VPLMN). Here, the outbound roaming target is not considered or treated as a target non-local ID. Likewise, a target in a third party RCS service provider domain can be a subscriber of a CSP. Here, too the target is not considered or treated as a target non-local ID in that third party RCS service provider network.

4.2.3 RCS servers in CSP domain

4.2.3.1 General

In this case, the RCS Server is in the same CSP domain where the RCS user is a subscriber of. Two cases are considered here:

- RCS users (and hence, the RCS Servers) are in the same CSP domain.

- RCS users (and hence, the RCS Servers) are in different CSP domains.

The concept of target non-local ID applies only for the second of the two cases listed above. Limited illustrations of roaming cases (Local Breakout (LBO) and Home-routed (HR)) are shown.

4.2.3.2 RCS users in the same CSP domain

4.2.3.2.1 General

In this case, the RCS users are in the same CSP domain. In the roaming case, the two RCS users belong to the same home CSP domain.

The LI functions for other IMS-based services are shown for a comparison purpose in grasping the scope of RCS LI with respect to the LI for other IMS-based services. Note that not all the NFs that provide the LI functions for other IMS-based services are depicted in such illustrations. The subsequent clauses may focus just on the RCS LI unless illustrating LI for other IMS-based services may provide additional clarity.

Note that it is possible to have just the IMS LI, or just RCS LI, or both (figure 4.2.3.2.2-2) for a given target.

4.2.3.2.2 Non-roaming case

In the illustrations shown here, RCS User (A) and RCS User (B) are un the same CSP domain.

RCS LI

Figure 4.2.3.2.2-1 illustrates a case of non-roaming where both User (A) and User (B) are targets in the CSP domain for RCS. In this illustration, LI is applicable only for RCS.

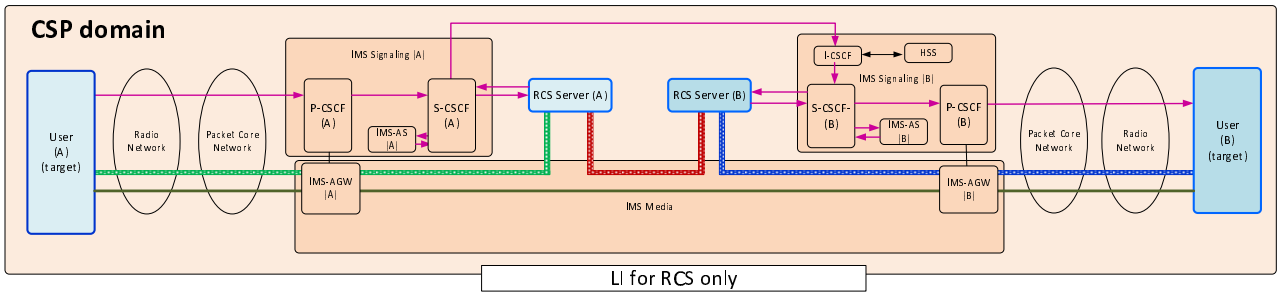


Figure 4.2.3.2.2-1: Non-roaming case - LI for RCS only

The RCS Server (A) provides the LI functions for RCS when User (A) is a target for RCS and RCS Server (B) provides the LI functions for RCS when the User (B) is a target for RCS. The RCS Servers provide both IRI-POI and CC-POI functions.

RCS LI and IMS LI

Figure 4.2.3.2.2-2 illustrates a case of non-roaming where both User (A) and User (B) are targets in the CSP domain for RCS and other IMS-based services (e.g. voice). In this illustration, LI is applicable for RCS and other IMS-based services (e.g. voice).

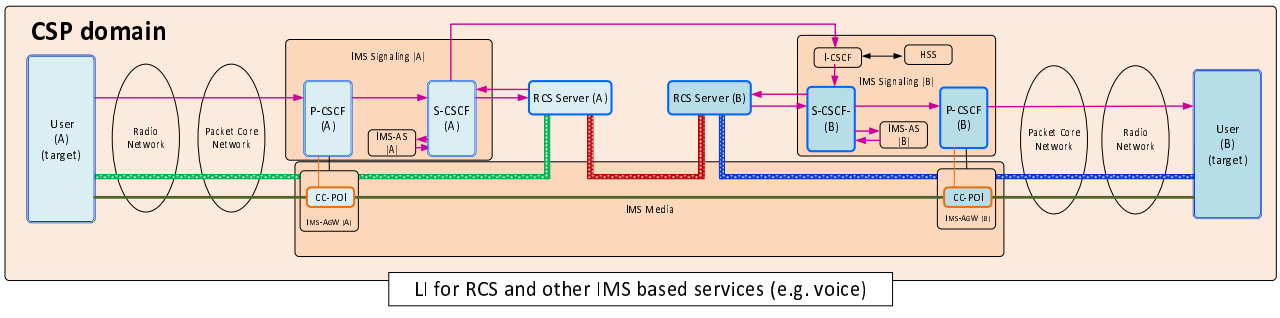


Figure 4.2.3.2.2-2: Non-roaming case - LI for RCS and other IMS-based services

For RCS LI, the RCS Server (A) provides the LI functions when User (A) is a target and RCS Server (B) provides the LI functions for when the User (B) is a target for RCS. The RCS Servers provide both IRI-POI and CC-POI functions for RCS.

For LI of other IMS-based services, S-CSCF (A), P-CSCF (A) and IMS-AGW (A) provide the LI functions when User (A) is a target for the other IM-based services and S-CSCF (B), P-CSCF (B) and IMS-AGW (B) provide the LI functions when the RCS User (B) is a target for other IMS-based services. The S-CSCFs provide the IRI-POI functions. The P-CSCFs provide the CC-TF functions and the IMS-AGWs provide the CC-POI functions.

4.2.3.2.3 Roaming case (LBO)

In these illustrations, the two outbound roaming (LBO) users belong to the same home CSP domain (HPLMN). User (A) is roaming in CSP domain A1 (VPLMN) and User (B) is roaming in CSP domain B1 (VPLMN).

RCS LI

Figure 4.2.3.2.3-1 illustrates a case of roaming (LBO) where both User (A) and User (B) are targets in the HPLMN for RCS. In this illustration, LI is applicable only for RCS in the HPLMN.

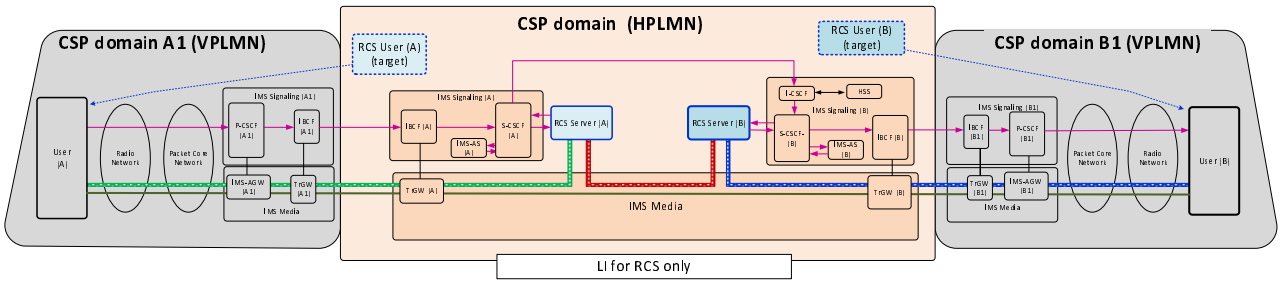


Figure 4.2.3.2.3-1: Roaming case (LBO) – LI for RCS only

The RCS Server (A) provides the LI functions for RCS when User (A) is a target as an RCS User and RCS Server (B) provides the LI functions for RCS when the User (B) is a target as an RCS User. The RCS Servers provide both IRI-POI and CC-POI functions. RCS servers are in the HPLMN. There is no RCS related LI in the VPLMNs.

RCS LI and IMS LI

Figure 4.2.3.2.3-2 illustrates a case of roaming (LBO) where both User (A) and User (B) are targets in the HPLMN for RCS and other IMS-based services (e.g. voice). In addition, User (A) is a target for other IMS-based services (e.g. voice) in the CSP domain A1 (VPLMN) and User (B) is a target for other IMS-based services (e.g. voice) in the CSP domain B1 (VPLMN). In this illustration, LI is applicable for RCS and other IMS-based services (e.g. voice).

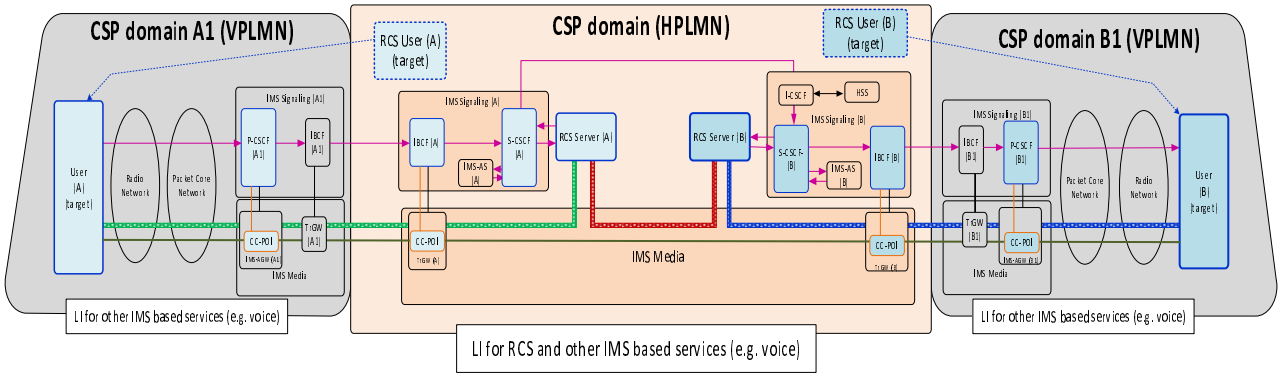


Figure 4.2.3.2.3-2: Roaming case (LBO) - LI for RCS and other IMS-based services

For RCS LI, the RCS Server (A) provides the LI functions when User (A) is a target as an RCS user in the HPLMN and RCS Server (B) provides the LI functions when the User (B) is a target as an RCS user in the HPLMN. The RCS Servers provide both IRI-POI and CC-POI functions for RCS.

For LI of other IMS-based services (in the HPLMN), S-CSCF (A), IBCF (A) and TrGW (A) provide the LI functions when the outbound roaming User (A) is a target for other IMS-based services and S-CSCF (B), IBCF (B) and TrGW

(B) provide the LI functions when the outbound User (B) is a target for other IMS-based services. The S-CSCFs provide the IRI-POI functions. The IBCFs provide the CC-TF functions and the TrGWs provide the CC-POI functions.

For LI of other IMS-based services (in the VPLMNs), the P-CSCF (A1) and IMS-AGW (A1) provides the LI functions when the inbound roaming User (A) is target for other IMS-based services in CSP domain A1 (VPLMN) for and P-CSCF (B1), IMS-AGW (B1) provide the LI functions when the inbound roaming target User (B) is target for other IMS-based services in CSP domain B1 (VPLMN). The P-CSCFs provide the IRI-POI and CC-TF functions, IMS-AGWs provide the CC-POI functions.

4.2.3.2.4 Roaming case (HR)

In these illustrations, the two outbound roaming (HR) users belong to the same home CSP domain (HPLMN). User (A) is roaming in CSP domain A1 (VPLMN) and User (B) is roaming in CSP domain B1 (VPLMN).

RCS LI

Figure 4.2.3.2.4-1 illustrates a case of roaming (HR) where both User (A) and User (B) are targets in the HPLMN for RCS. In this illustration, LI is applicable only for RCS in the HPLMN.

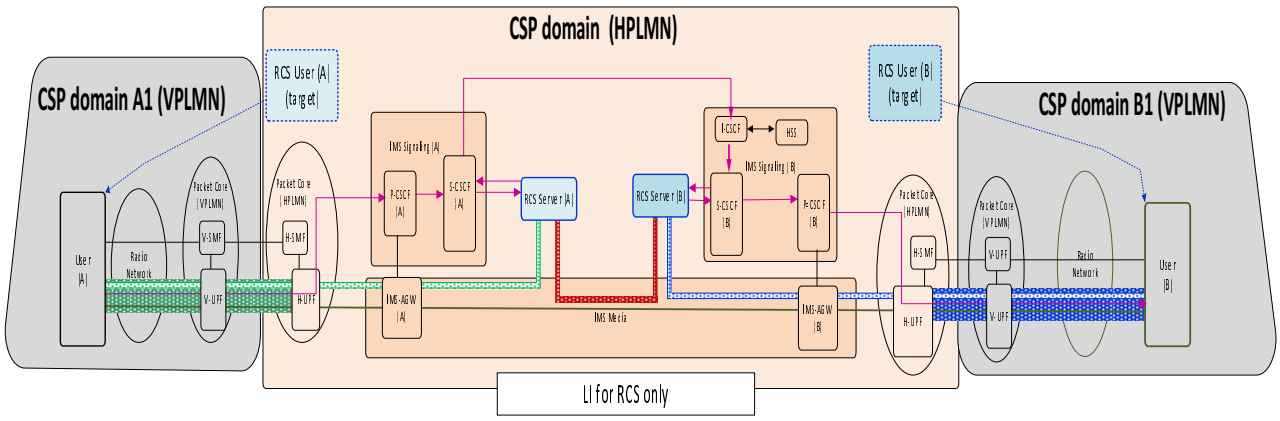


Figure 4.2.3.2.4-1: Roaming case (HR) – LI for RCS only

The RCS Server (A) provides the LI functions when User (A) is a target as an RCS user in the HPLMN and RCS Server (B) provides the LI functions when the User (B) is a target as an RCS user in HPLMN. The RCS Servers provide both IRI-POI and CC-POI functions. Both RCS Servers are in the HPLMN. There is no RCS related LI in the VPLMNs.

RCS LI and IMS LI

Figure 4.2.3.2.4-2 illustrates a case of roaming (HR) where both User (A) and User (B) are targets in the HPLMN for RCS and other IMS-based services (e.g. voice). In addition, User (A) is a target for other IMS-based services (e.g. voice) in the CSP domain A1 (VPLMN) and User (B) is a target for other IMS-based services (e.g. voice) in the CSP domain B1 (VPLMN). In this illustration, LI is applicable for RCS and other IMS-based services (e.g. voice).

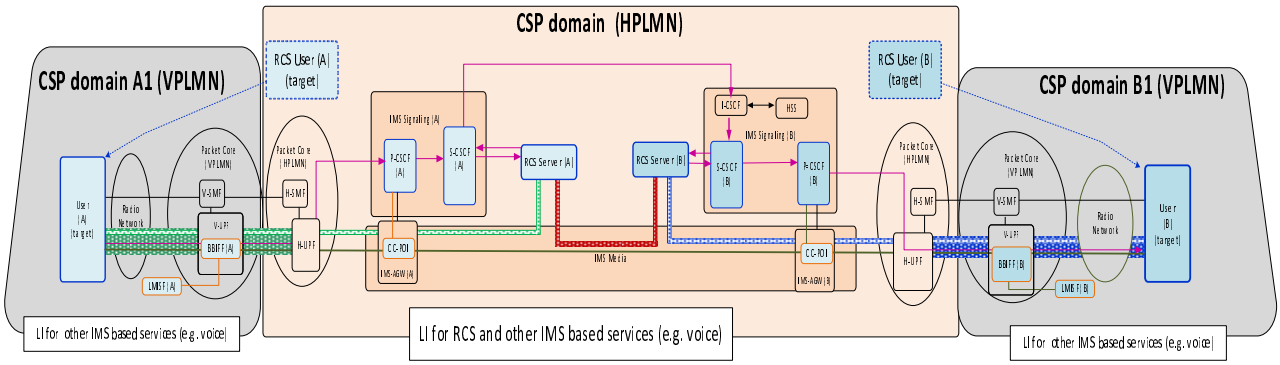


Figure 4.2.3.2.4-2: Roaming case (HR) - LI for RCS and other IMS-based services

For RCS LI, the RCS Server (A) provides the LI functions when User (A) is a target as an RCS user in the HPLMN and RCS Server (B) provides the LI functions when the User (B) is a target as an RCS user in HPLMN. RCS Servers are in the HPLMN. The RCS Servers provide both IRI-POI and CC-POI functions for RCS.

For LI of other IMS-based services (in the HPLMN), S-CSCF (A), P-CSCF (A) and IMS-AGW (A) provide the LI functions when the outbound User (A) is a target for other IMS-based services and S-CSCF (B), P-CSCF (B) and IMS-AGW (B) provide the LI functions when the outbound roaming User (B) is a target for other IMS-based services. The S-CSCFs provide the IRI-POI functions. The P-CSCFs provide the CC-TF functions and the IMS-AGWs provide the CC-POI functions.

For LI of other IMS-based services (in the VPLMNs), the BBIFFs and LMISFs, introduced as a part of N9HR, provide the LI functions when inbound roaming User (A) is a target for other IMS-based services in CSP domain A1 (VPLMN) and for inbound roaming User (B) is a target for other IMS-based services in CSP domain B1 (VPLMN). The BBIFFs and LMISFs in the VPLMNs provide the IRI-POI and CC-POI functions.

4.2.3.3 RCS users in two CSP domains

4.2.3.3.1 General

In this case, the two RCS users are in two different CSP domains. Accordingly, the two RCS Servers are in two different CSP domains.

The roaming cases where the roaming users can be the subscribers of respective HPLMNs are not illustrated here since the RCS LI does not apply to VPLMNs as illustrated in clause 4.2.3.2. For other aspects of LI in the VPLMN, the illustrations in clause 4.2.3.2.3 (LBO) and 4.2.3.2.4 (HR) apply here as well.

The case of target non-local ID is illustrated along with the LI for other IMS-based services (e.g. voice). The LI for other IMS-based services is shown for a comparison purpose.

4.2.3.3.2 Target is a subscriber of CSP domain that has the RCS Server

The figure 4.2.3.3.2-1 illustrates the LI for RCS and other IMS-based services when User (A) and User (B) are subscribers of two CSP domains (CSP domain A and CSP domain B). The illustration includes both RCS LI and LI for other IMS-based services.

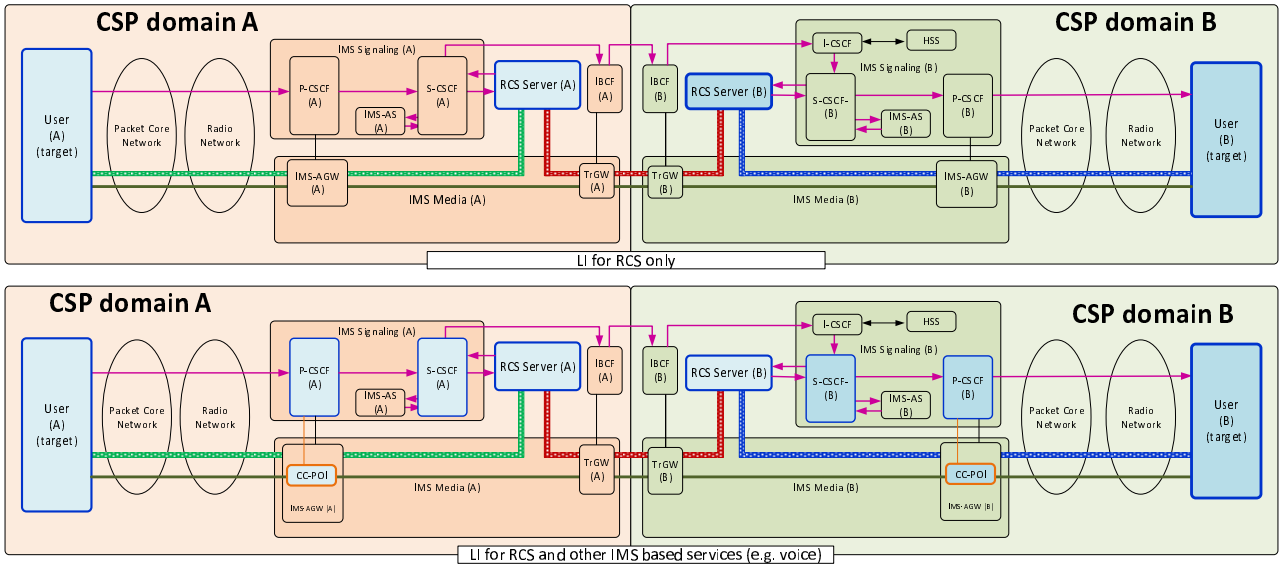


Figure 4.2.3.3.2-1: The two RCS users are in different CSP domains

The top diagram illustrates the RCS LI and bottom diagram illustrates RCS LI along with the other IMS LI.

For RCS LI, the RCS Server (A) provides the LI functions when User (A) is a target for RCS in CSP domain A and RCS Server (B) provides the LI functions when the User (B) is a target for RCS in CSP domain (B). The RCS Servers provide both IRI-POI and CC-POI functions.

For LI of other IMS-based services, the LI functions are provided by S-CSCFs, P-CSCFs, IMS-AGWs in the respective CSP domains when the users are targets for the other IMS-based services. The S-CSCFs provide the IRI-POI functions, P-CSCFs provide the CC-TF functions and the IMS-AGWs provide the CC-POI functions.

4.2.3.3.3 The case target non-local ID

The figure 4.2.3.3.3-1 illustrates the LI for RCS and other IMS-based services when User (A) and User (B) are target non-local IDs in other CSP domains. The illustration includes both RCS LI and LI for other IMS-based services.

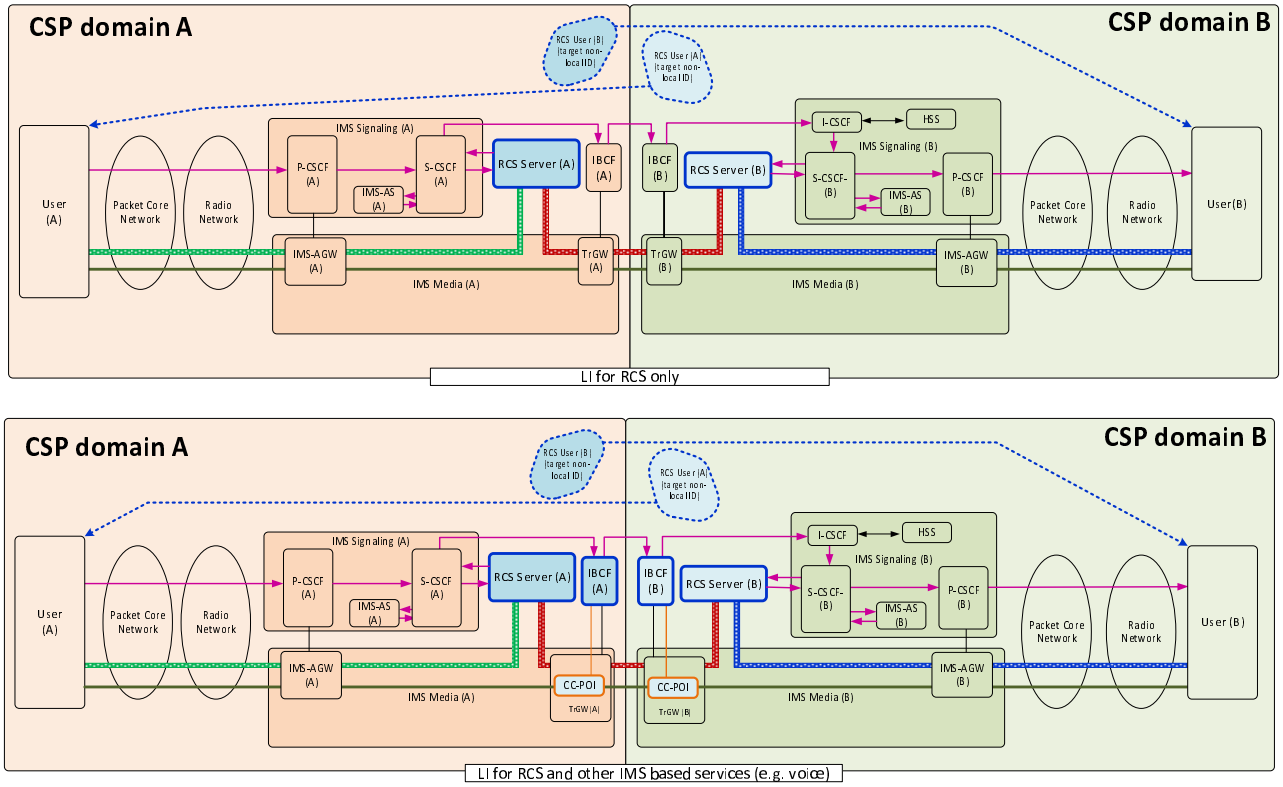


Figure 4.2.3.3.3-1: Two RCS users in different CSP domains (target non-local ID)

The top diagram illustrates the RCS LI and bottom diagram illustrates RCS LI along with the other IMS LI.

For RCS LI, the RCS Server (A) provides the LI functions when any user in CSP domain A is on an RCS session with User (B), a target non-local ID in CSP domain A. Likewise, the RCS Server (B) provides the LI functions when any user in CSP domain B receives an incoming RCS session from User (A), a target non-local ID in CSP domain (B). The RCS Servers provide both IRI-POI and CC-POI functions for RCS.

For LI of other IMS-based services, IBCF (A) and TrGW (A) provide the LI functions when any user in CSP domain A is on an IMS session with User (B), a target non-local ID in CSP domain A. Likewise, the IBCF (B) and TrGW (B) provide the LI functions when any user in CSP domain B receives an incoming IMS session from User (A), a target non-local ID in CSP domain (B). The IBCFs provide the IRI-POI and the CC-TF functions and the TrGWs provide the CC-POI functions.

4.2.4 Third party RCS service providers

4.2.4.1 General

In the case where the RCS service is provided by a third party RCS service provider, the RCS Server resides within that third RCS service provider domain. The RCS users may or may not be a subscriber of a CSP. In the former case (i.e. when the RCS user is a subscriber of CSP), the signalling and RCS media pass through the CSP domain whereas in the latter case, RCS signalling and RCS media pass through data network that is not related to any CSPs (e.g. Wi-Fi).

As in the case where the RCS Servers are in the CSP domain, there are two main cases when the RCS is provided by the third party RCS service provider:

- RCS Servers in one third party RCS service provider domain.
- RCS Servers in different third party RCS service provider domains.

When RCS user is in a CSP domain, there are two cases:

- RCS users in the same CSP domain.
- RCS users in different CSP domains.

It is possible that the third party RCS service provider may not have IMS NFs deployed. Instead, the SIP messages may pass through proprietary SIP message handling proxies. The illustrations shown in the present document, however, assume as if the third party RCS service provider domain also has the IMS NFs since the RCS Server is viewed as an IMS-based application in a general sense. However, the focus here (i.e. within the third party RCS service provider domain) is just on the RCS LI and applicability of LI for IMS other than the RCS (from a third party RCS service provider perspective) is outside the scope of the present document.

When RCS user is in a CSP domain with third party RCS service provider being used for RCS, two scenarios are considered:

- CSP domain does not have IMS.
- CSP domain has IMS.

When the CSP domain does not have IMS, RCS signalling and RCS media pass through the CSP domain over the packet data connection. This is similar to the case where RCS users are not subscribers of a CSP.

When the CSP domain has IMS, the present document illustrates the following deployment option:

- RCS Signalling and RCS media pass through the CSP domain over the packet data connection. In this case CSP IMS is not on the signalling and media path of an RCS session. This is referred to as deployment option 1.

When CSP domain has the IMS, another deployment option (referred to as deployment option 2) where the RCS Signalling and RCS media pass through the IMS present within the CSP domain is for further study.

To illustrate the point of signalling and media split (at the Ingress end) and combined (at the Egress end) within the third party RCS service provider domain, NFs named in the present document as Border GWs are shown at the Ingress and Egress end of the third party RCS service provider domain. The details how exactly such a split is done is outside the scope of present document.

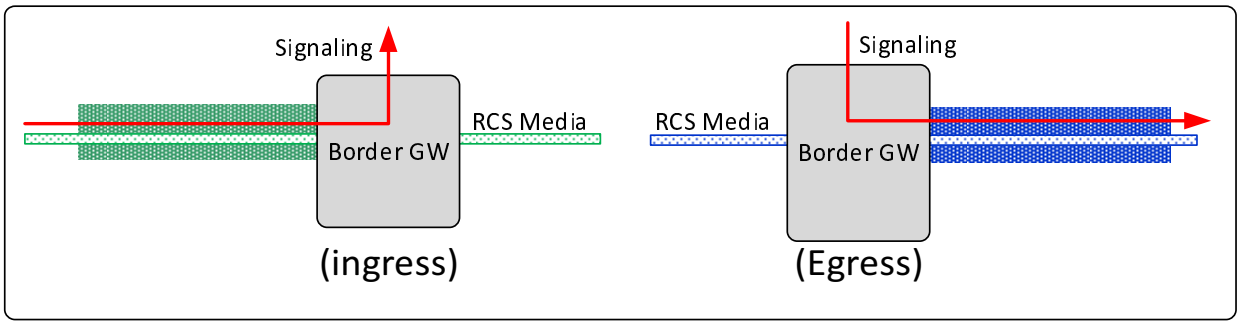


Figure 4.2.4.1-1: Conceptual Border Gateways that splits/combines signalling and media

4.2.4.2 One third party RCS service provider

4.2.4.2.1 General

This clause illustrates the case when RCS Servers are within the same third party RCS service provider domain with various scenarios depending on where the RCS users are.

Scenario 1: RCS users are not in CSP domain.

Scenario 2: RCS users are in the same CSP domain that does not have IMS.

Scenario 3: RCS users are in the same CSP domain that has IMS.

Scenario 4: RCS users are in the different CSP domains that do not have IMS.

Scenario 5: RCS users are in the different CSP domains that have IMS.

Not all the scenarios listed above may be deployed in practice, The present document only provides the illustrations in case those scenarios are deployed.

4.2.4.2.2 Scenario 1: RCS users are not in CSP domain

When RCS is provided by a third party RCS service provider, the RCS users are not necessarily to be the subscribers of any CSPs. The use-case for this can be RCS users are connected to the third party RCS service provider domain via Wi-Fi access.

Figure 4.2.4.2.2-1 illustrates RCS LI topology when RCS User (A) and RCS User (B) are targets for RCS in the third party RCS service provider domain.

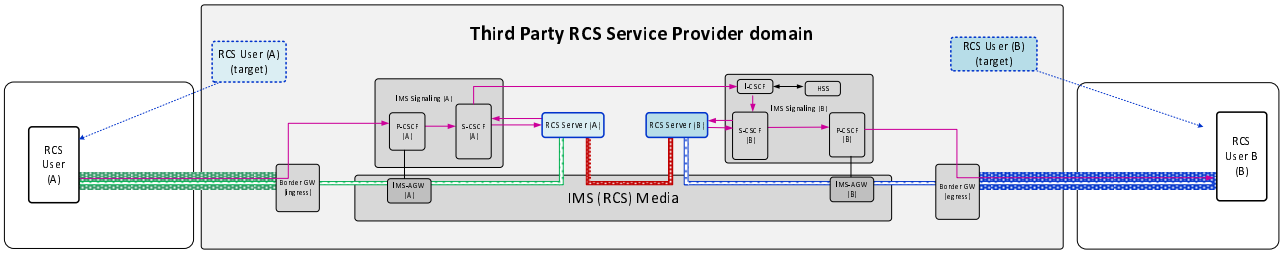


Figure 4.2.4.2.2-1: Non-CSP RCS users, third party RCS service provider

The RCS Server (A) present in the third party RCS service provider domain provides the LI functions when RCS User (A) is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the RCS User (B) is a target for in the third party RCS service provider domain.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.4.2.3 Scenario 2: RCS users are in the same CSP domain that does not have IMS

From an overall topology perspective, this case is similar to the topology shown in clause 4.2.4.2.2. One difference here is that RCS users being the subscribers of a CSP can also be outbound roaming.

Non-roaming case

RCS User (A) and RCS User (B) are the subscribers of the same CSP domain that does not have IMS. The CSP domain provides just the access when those users establish RCS sessions and as such both signalling and media pass through the CSP domain over the packet data connection.

Figure 4.2.4.2.3-1 illustrates the RCS LI for the non-roaming case when User (A) and User (B) are targets for RCS in the third party RCS service provider domain.

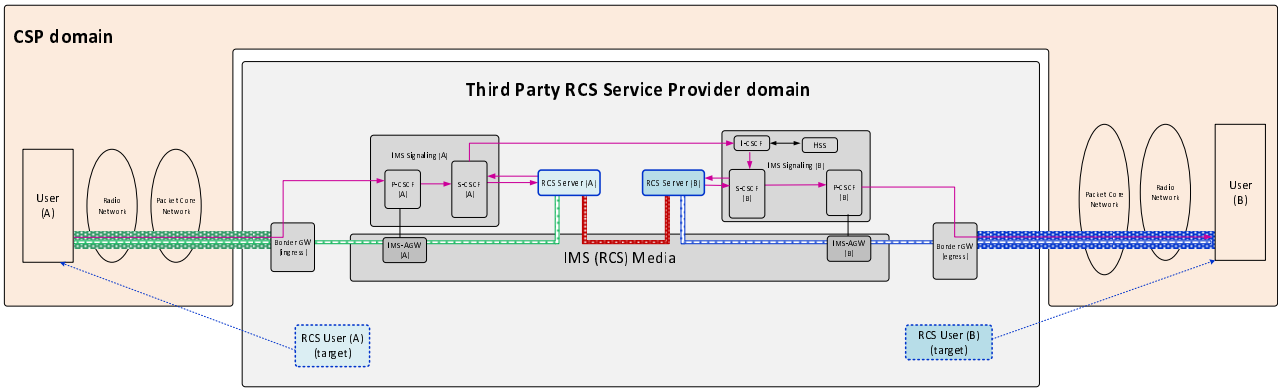


Figure 4.2.4.2.3-1: Non-roaming case, no IMS in CSP domain

The RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) in is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the User (B) is a target for RCS in the third party RCS service provider domain.

The RCS Servers provide both IRI-POI and CC-POI functions.

Roaming LBO

RCS User (A) and RCS User (B) are the subscribers of the same home CSP that does not support IMS. RCS User (A) is roaming in CSP domain A1 (VPLMN) and RCS User (B) is roaming in CSP domain B1 (VPLMN). The CSP domains (VPLMNs) just provide the access when those users establish RCS sessions and as such both signalling and media do not pass through the HPLMN with LBO, instead pass through the VPLMNs over the packet data connection.

Figure 4.2.4.2.3-2 illustrates the RCS LI for the roaming case (with LBO) when User (A) and User (B) are targets for RCS in the HPLMN.

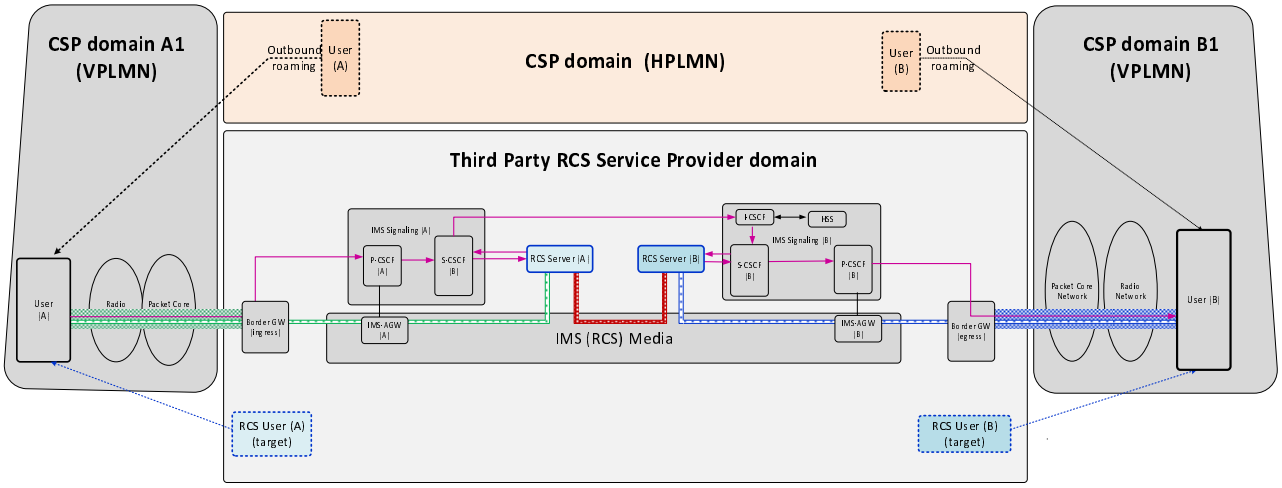


Figure 4.2.4.2.3-2: Roaming case (LBO), no IMS in CSP domain

The RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) in is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party service provider domain provides the LI functions when the User (B) is a target for RCS in the third party RCS service provider domain.

The RCS Servers provide both IRI-POI and CC-POI functions.

Roaming HR

RCS User (A) and RCS User (B) are the subscribers (shown as User (A) and User (B)) of the same home CSP that does not support IMS. User (A) is roaming in CSP domain A1 (VPLMN) and User (B) is roaming in CSP domain B1

(VPLMN). The, CSP domains (VPLMN and HPLMN) just provide the access when those users establish RCS sessions and as such both signalling.

Figure 4.2.4.2.3-3 illustrates the RCS LI for the roaming case (with HR) when RCS User (A) and RCS User (B) are targets for RCS in the HPLMN.

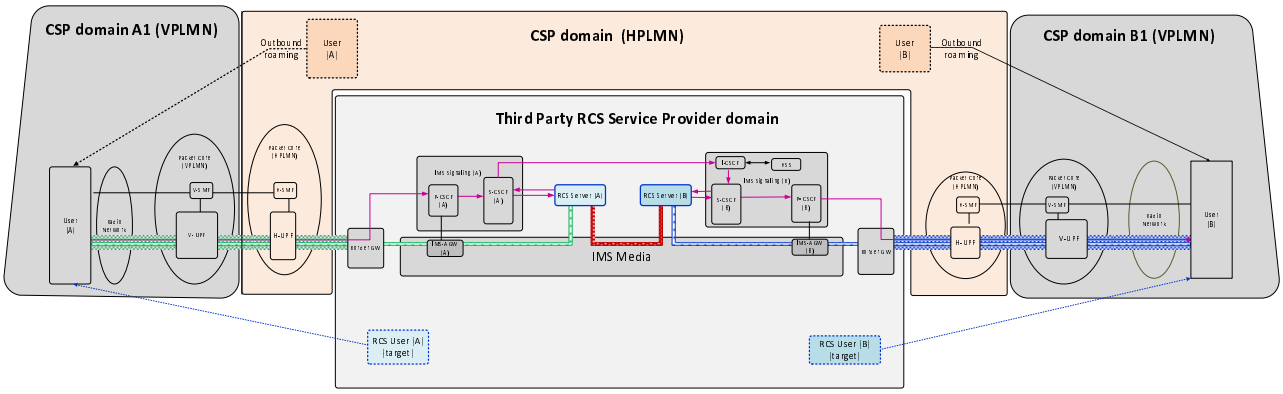


Figure 4.2.4.2.3-3: Roaming case (HR), no IMS in CSP domain

The RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the User (B) is a target for RCS in the third party RCS service provider domain.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.4.2.4 Scenario 3: RCS users are in the same CSP domain that has IMS

4.2.4.2.4.1 General

In the illustrations shown here RCS users are the subscribers in the CSP domain that has IMS.

Illustrations include the deployment option 1 mentioned in clause 4.2.4.1 with non-roaming, roaming with LBO and roaming with HR. LI for other IMS-based services is also illustrated for a comparison purpose to grasp the scope of RCS LI.

4.2.4.2.4.2 Non-roaming

Since the RCS Servers used for the RCS sessions are in third party RCS service provider domain, in this deployment option, CSP domain provides just the access when those users establish RCS sessions and as such both signalling and media pass through the CSP domain over the packet data connection.

The signalling and media for non-RCS related IMS sessions do not pass through the third party RCS service provider domain.

RCS LI

Figure 4.2.4.2.4.2-1 illustrates the CSP deployment option 1 when User (A) and User (B) are targets for RCS in the third party service provider domain. In the illustration, LI applies only for RCS.

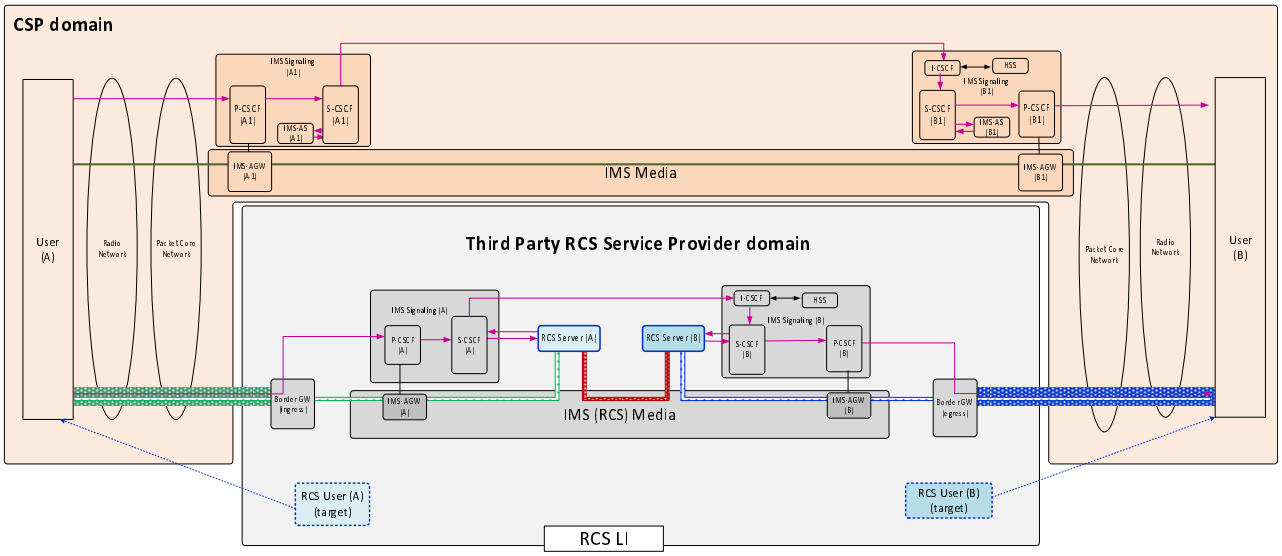


Figure 4.2.4.2.4.2-1: Non-roaming case, CSP with IMS - LI for RCS only

The RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the User (B) is a target for RCS in the third party RCS service provider domain.

The RCS Servers provide both IRI-POI and CC-POI functions.

RCS LI and IMS LI

Figure 4.2.4.2.4.2-2 illustrates the CSP deployment option 1 when User (A) and User (B) are targets for RCS in the third party service provider domain. The User (A) and User (B) are targets for other IMS-based services (e.g. voice) in the CSP domain.

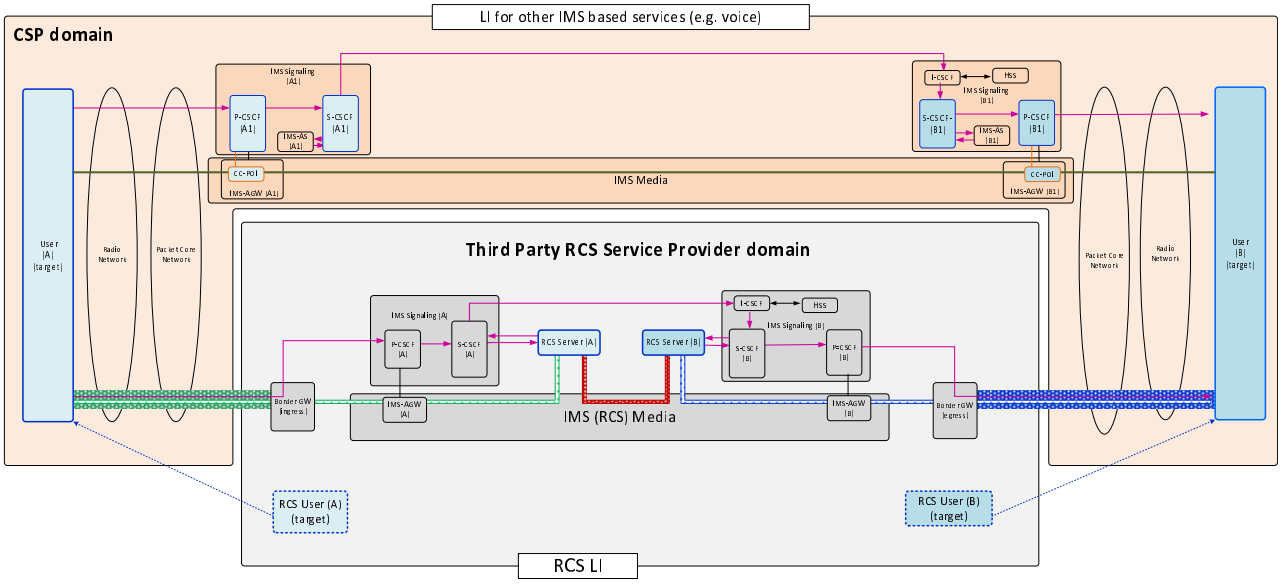


Figure 4.2.4.2.4.2-2: Non-roaming case, CSP with IMS - LI for RCS and other IMS based services

For RCS LI, the RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the User (B) is a target for in the third party RCS service provider domain. The RCS Servers provide both IRI-POI and CC-POI functions.

For LI of other IMS-based services, the S-CSCF (A1), P-CSCF (A1), IMS-AGW (A1) present in the CSP domain provide the LI functions when the User (A) is a target for other IMS-based services in the CSP domain and the S-CSCF

(B1), P-CSCF (B1), IMS-AGW (B1) present in the CSP domain provide the LI functions when the User (B) is a target for other IMS-based services in the CSP domain. The S-CSCFs provide the IRI-POI functions, the P-CSCFs provide the CC-TF functions and the IMS-AGWs provide the CC-POIs functions.

4.2.4.2.4.3 Roaming (LBO)

In these illustrations, User (A) and User (B) are the subscribers of the same home CSP. User (A) is roaming in CSP domain A2 (VPLMN) and User (B) is roaming in CSP domain B2 (VPLMN).

Since the RCS Servers used for the RCS sessions are in third party RCS service provider domain, in this deployment option, CSP domains just provide the access when those users establish RCS sessions and as such both signalling and media do not pass through the HPLMN, instead pass through the VPLMNs over the packet data connection.

The signalling and media for non-RCS related IMS sessions do not pass through the third party RCS service provider domain.

RCS LI

Figure 4.2.4.2.4.3-1 illustrates the CSP deployment option 1 when User (A) and User (B) roaming in CSP domain A2 (VPLMN) and CSP domain B2 (VPLMN) respectively, are targets for RCS in the third party service provider domain.

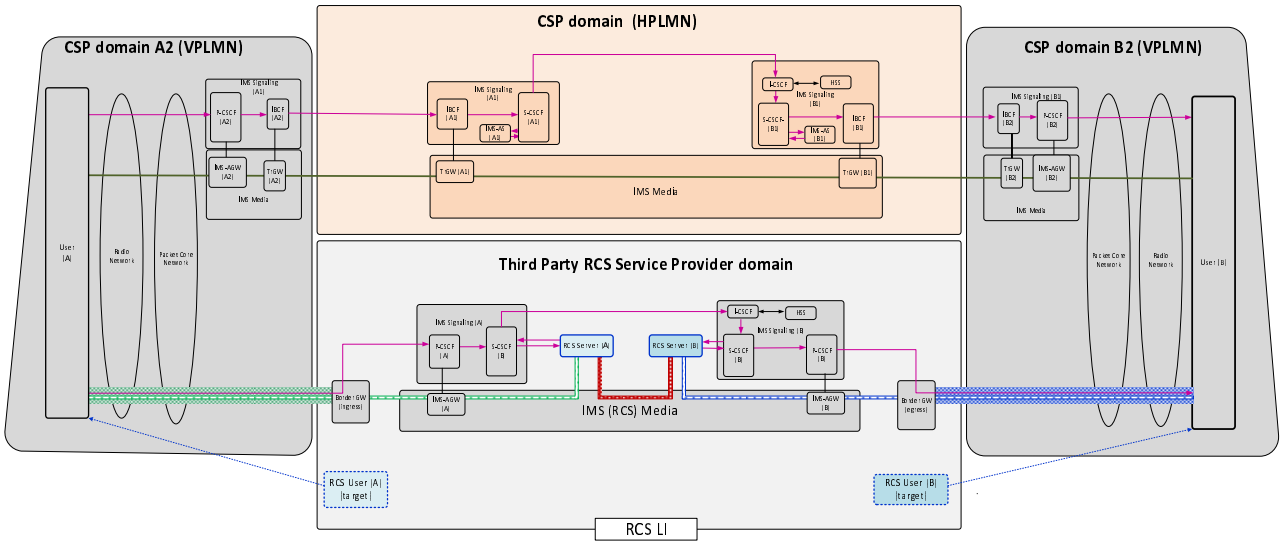


Figure 4.2.4.2.4.3-1: Roaming case (LBO), CSP with IMS – LI for RCS only

The RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) is a target for in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the User (B) is a target in the third party RCS service provider domain.

The RCS Servers provide both IRI-POI and CC-POI functions.

RCS LI and IMS LI

Figure 4.2.4.2.4.3-2 illustrates the CSP deployment option 1 when User (A) and User (B) roaming in CSP domain A2 (VPLMN) and CSP domain B2 (VPLMN) respectively, are targets for RCS in the third party service provider domain. The User (A) and User (B) are targets for other IMS-based services (e.g. voice) in the CSP domain (HPLMN) and the respective VPLMNs.

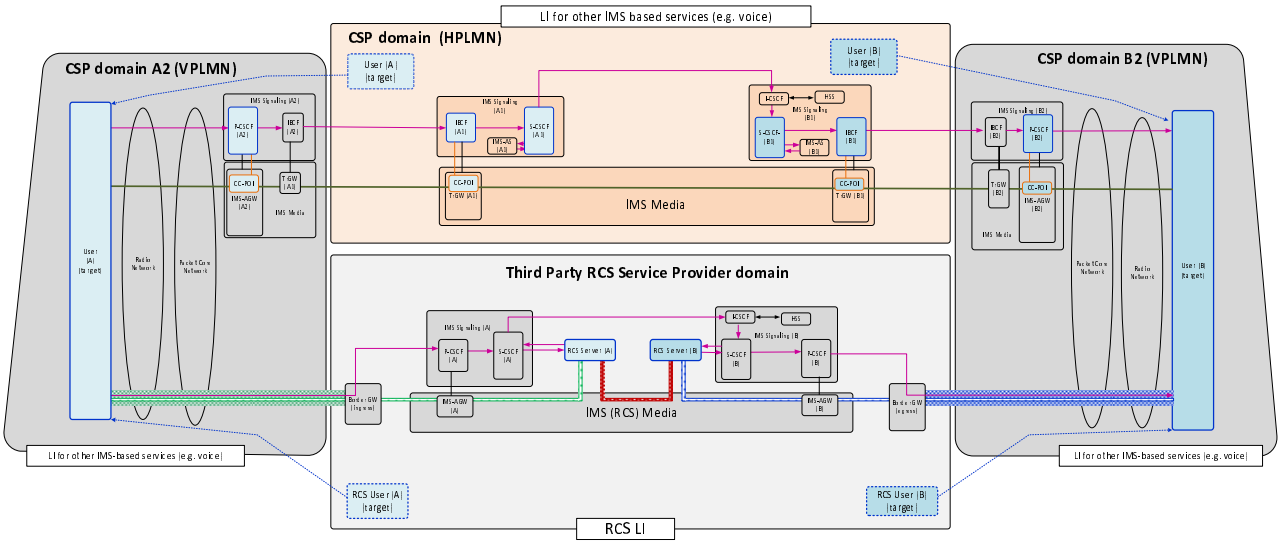


Figure 4.2.4.2.4.3-2: Roaming case (LBO), CSP with IMS – LI for RCS and other IMS-based services

For RCS LI, the RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the User (B) is a target for RCS in the third party RCS service provider domain. The RCS servers provide both IRI-POI and CC-POI functions.

For LI of other IMS-based service in the HPLMN, the S-CSCF (A1), IBCF (A1), TrGW (A1) present in the CSP domain (HPLMN) provide the LI functions when the User (A) is a target for other IMS-based services in the CSP domain (HPLMN) and the S-CSCF (B1), IBCF (B1), TrGW (B1) present in the CSP domain (HPLMN) provide the LI functions when the User (B) is a target for other IMS-based services in the CSP domain (HPLMN). The S-CSCFs provide the IRI-POI functions, the IBCFs provide the CC-TF functions and the TrGWs provide the CC-POIs functions.

For LI of other IMS-based services in the VPLMNs, the P-CSCF (A2), IMS-AGW (A2) present in the CSP domain A2 (VPLMN) provide the LI functions when the User (A) is a target for other IMS-based services in the CSP domain A2 (VPLMN) and the P-CSCF (B2), IMS-AGW (B2) present in the CSP domain B2 (VPLMN) provide the LI functions when the User (B) is a target for other IMS-based services in the CSP domain B2 (VPLMN). The P-CSCFs provide the IRI-POI and CC-TF functions and the IMS-AGWs provide the CC-POIs functions.

4.2.4.2.4.4 Roaming (HR)

In these illustrations, User (A) and User (B) are the subscribers of the same home CSP. User (A) is roaming in CSP domain A2 (VPLMN) and User (B) is roaming in CSP domain B2 (VPLMN).

Since the RCS Servers used for the RCS sessions are in third party RCS service provider domain, in this deployment option, CSPs provide just the access when those users establish RCS sessions and as such both signalling and media pass through the CSP domains (VPLMN and HPLMN) over the packet data connection.

The signalling and media for non-RCS related IMS sessions do not pass through the third party RCS service provider domain.

RCS LI

Figure 4.2.4.2.4.4-1 illustrates the CSP deployment option 1 when User (A) and User (B) roaming in CSP domain A2 (VPLMN) and CSP domain B2 (VPLMN) respectively, are targets for RCS in the third party service provider domain.

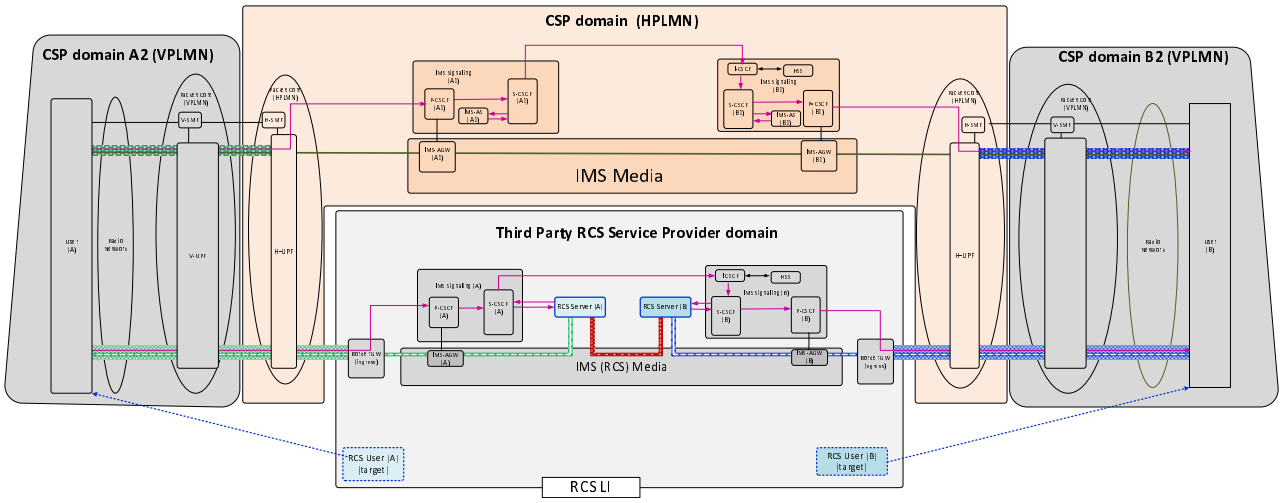


Figure 4.2.4.2.4.4-1: Roaming case (HR), CSP with IMS – LI for RCS only

The RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the User (B) is a target for RCS in the third party RCS service provider domain. The RCS Servers provide both IRI-POI and CC-POI functions.

RCS LI and IMS LI

Figure 4.2.4.2.4.4-2 illustrates the CSP deployment option 1 when User (A) and User (B) roaming in CSP domain A2 (VPLMN) and CSP domain B2 (VPLMN) respectively, are targets for RCS in the third party service provider domain. The User (A) and User (B) are targets for other IMS-based services (e.g. voice) in the CSP domain (HPLMN) and the respective VPLMNs.

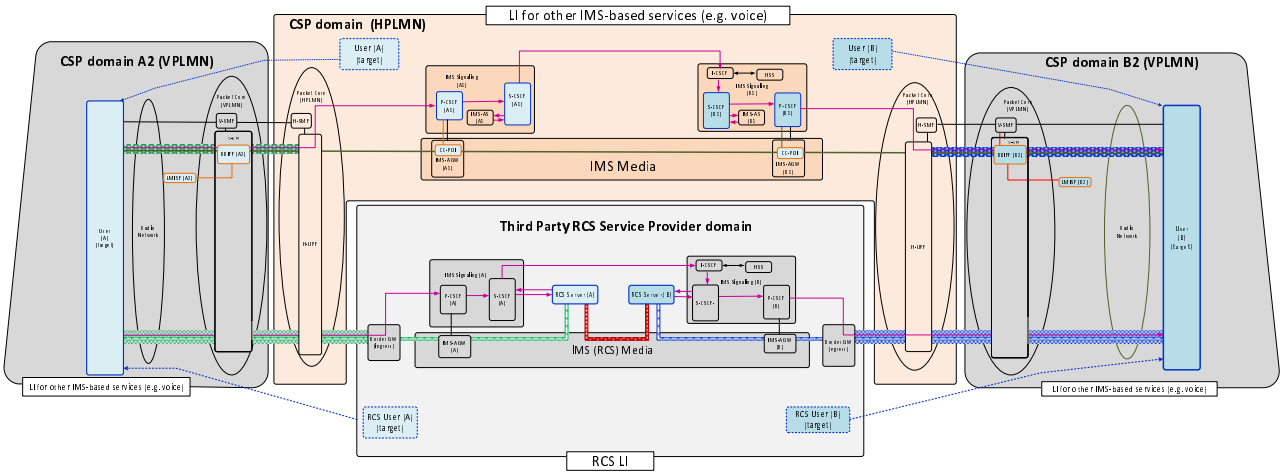


Figure 4.2.4.2.4.4-2: Roaming case (HR), CSP with IMS – LI for RCS and other IMS-based services

For RCS LI, the RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the User (B) is a target for RCS in the third party RCS service provider domain. The RCS servers provide both IRI-POI and CC-POI functions.

For LI of other IMS-based services in the HPLMN, the S-CSCF (A1), P-CSCF(A1), IMS-AGW (A1) present in the CSP domain (HPLMN) provide the LI functions when the User (A) is a target for other IMS-based services in the CSP domain (HPLMN) and the S-CSCF (B1), P-CSCF (B1), IMS-AGW (B1) present in the CSP domain (HPLMN) provide the LI functions when the User (B) is a target for other IMS-based services in the CSP domain (HPLMN). The S-CSCFs provide the IRI-POI functions, the P-CSCFs provide the CC-TF functions and the IMS-AGWs provide the CC-POIs functions.

For LI of other IMS-based services (in the VPLMNs), the BBIFFs and LMISFs, introduced as a part of N9HR, provide the LI functions for inbound roaming User (A) is a target for other IMS-based services in CSP domain A2 (VPLMN)

and for inbound roaming User (B) is a target for other IMS-based services in CSP domain B2 (VPLMN). The BBIFFs and LMISFs in the VPLMNs provide the IRI-POI and CC-POI functions.

4.2.4.2.5 Scenario 4: RCS users are different CSP domains that do not have IMS

This clause illustrates the case when RCS Servers are in the same third party RCS service provider domain and the RCS users are different CSP domains that do not have IMS.

Only non-roaming scenario is illustrated here. The roaming scenarios are similar to the illustrations shown in clause 4.2.4.2.3 except that two CSP domains are involved here.

User (A) is a subscriber in CSP domain A1 and User (B) is a subscriber in the CSP domain B1. The CSP domains provide just the access when those users establish RCS sessions and as such both signalling and media pass through the CSP domain over the packet data connection.

Figure 4.2.4.2.5-1 illustrates the case when RCS User (A) and RCS User (B) are targets in the third party RCS service provider domain.

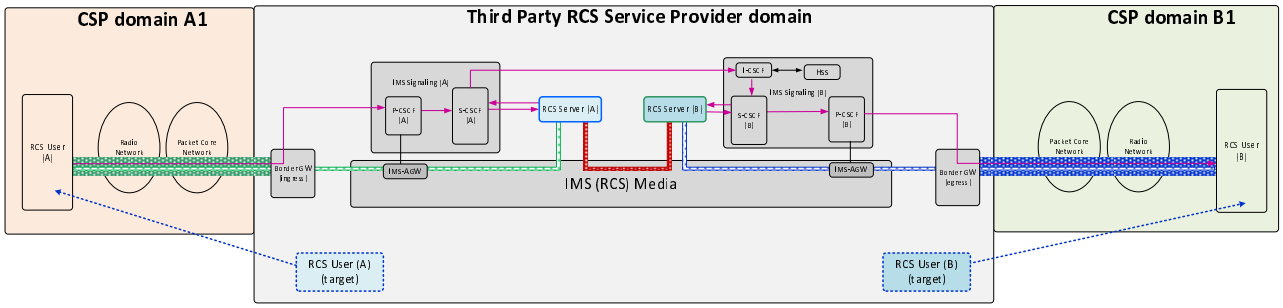


Figure 4.2.4.2.5-1: RCS users in different CSP domains that do not have IMS

The RCS Server (A) present in the third party RCS service provider domain provides the LI functions when RCS User (A) is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the RCS User (B) is a target for RCS in the third party RCS service provider domain.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.4.2.6 Scenario 5: RCS users are in different CSP domains that have IMS

This clause illustrates the case when RCS Servers are in the same third party RCS service provider domain and the RCS users are different CSP domains that have IMS.

The illustrations focus RCS LI and non-roaming scenarios. Roaming scenarios and LI for other IMS-based services are similar to the illustrations to the case 3 (clause 4.2.4.2.4).

User (A) is a subscriber in the CSP domain A1 and User (B) is a subscriber in the CSP domain B1. Since the RCS Servers used for the RCS sessions are in third party RCS service provider domain, in this deployment option, both CSPs provide just the access when those users establish RCS sessions and as such both signalling and media pass through the CSP domains over the packet data connection.

The signalling and media for non-RCS related IMS sessions do not pass through the third party RCS service provider domain.

Figure 4.2.4.2.6-1 illustrates the CSP deployment option 1 when User (A) and User (B) are targets for RCS in the third party service provider domain.

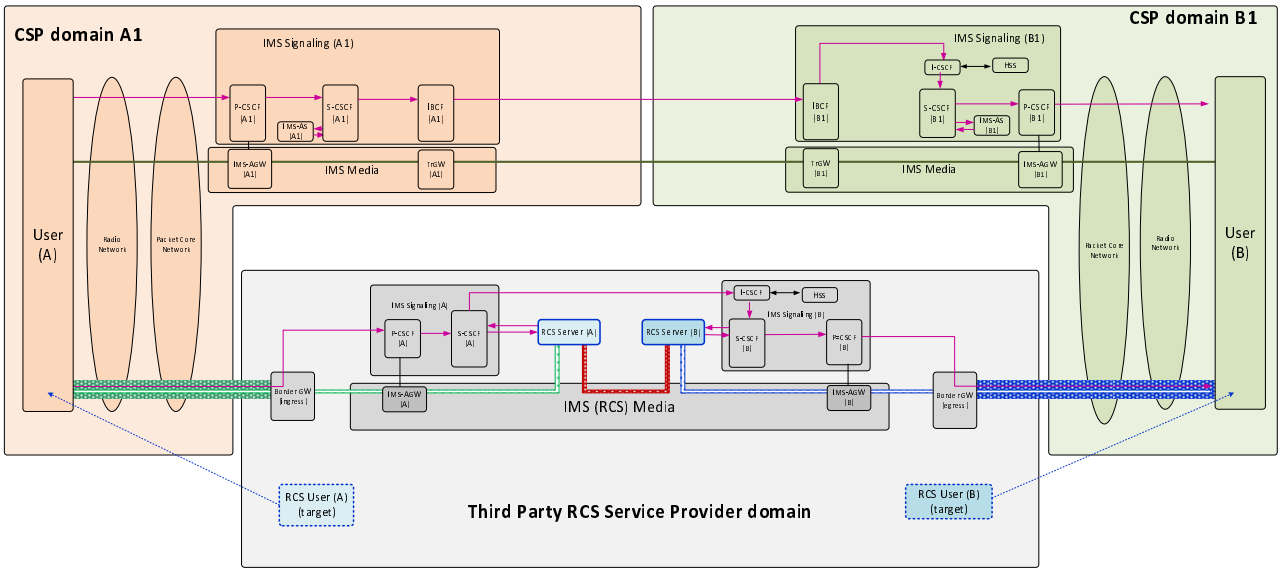


Figure 4.2.4.2.6-1: RCS users in different CSP domains that have IMS

The RCS Server (A) present in the third party RCS service provider domain provides the LI functions when User (A) is a target for RCS in the third party RCS service provider domain and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the User (B) is a target for RCS in the third party RCS service provider domain.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.4.3 Different third party RCS service providers

4.2.4.3.1 General

This clause illustrates the case when RCS Servers are in different third party RCS service provider domains with various cases depending on where the RCS users are.

Scenario 1: RCS users are not in CSP domain.

Scenario 2: RCS users are in the same CSP domain that does not have IMS.

Scenario 3: RCS users are in the same CSP domain that has IMS.

Scenario 4: RCS users are in the different CSP domains that do not have IMS.

Scenario 5: RCS users are in the different CSP domains that have IMS.

Not all the scenarios listed above may be deployed in practice, The present document only provides the illustrations in case those scenarios are deployed.

The illustrations do not show the roaming cases as they are similar to the illustrations shown in clause 4.2.4.2. The illustrations do not show the LI for other IMS-based services as they are also similar to the illustrations shown in clause 4.2.4.2.

4.2.4.3.2 Scenario 1: RCS users are not in CSP domain

Figure 4.2.4.3.2-1 illustrates RCS LI topology when RCS User (A) is a target for RCS in third party RCS provider domain A and RCS User (B) is a target for RCS in the third party RCS service provider B. and RCS User (B) are targets for RCS in the third party RCS service provider domain.

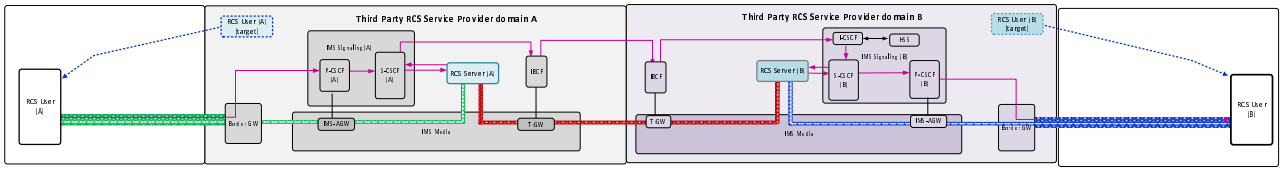


Figure 4.2.4.3.2-1: Non-CSP RCS users

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when RCS User (A) is a target for RCS in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain provides the LI functions when the RCS User (B) is a target for RCS in the third party RCS service provider domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.4.3.2-2 illustrates RCS LI topology when RCS User (A) is a target non-local ID for RCS in third party RCS provider domain B and RCS User (B) is a target non-local ID for RCS in the third party RCS service provider A.

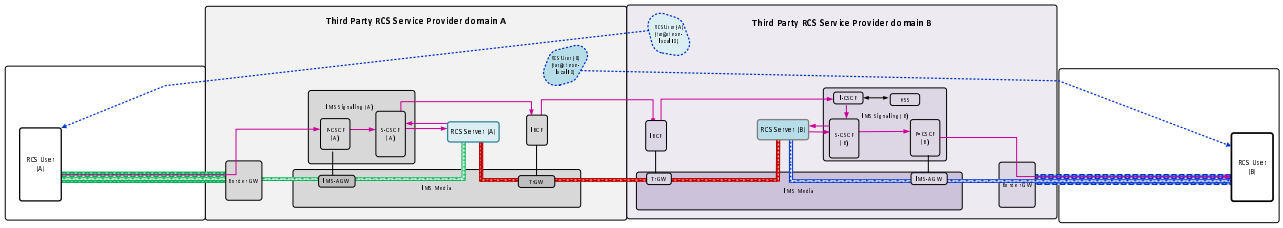


Figure 4.2.4.3.2-2: Non-CSP users (target non-local ID)

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when RCS User (B) is a target in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the RCS User (A) is a target in the third party RCS service provider domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.4.3.3 Scenario 2: RCS users are in the same CSP domain that does not have IMS

In the illustrations shown in this clause, RCS service is provided by different third party RCS service providers and the RCS users are in the same CSP domain that does not have IMS.

RCS User (A) and RCS User (B) are the subscribers of the same CSP domain that does not have IMS. The CSP domain provides just the access when those users establish RCS sessions and as such both signalling and media pass through the CSP domain over the packet data connection.

Figure 4.2.4.3.3-1 illustrates the case where RCS User (A) is a target for RCS in the third party RCS service provider domain A and RCS User (B) is a target for RCS in the third party RCS service provider domain B.

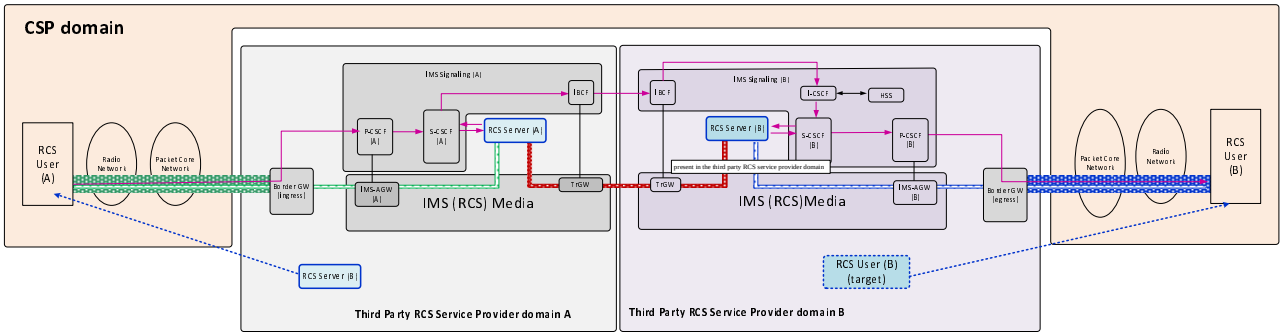


Figure 4.2.4.3.3-1: RCS users in the same CSP domain that does not have IMS

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when RCS User (A) in is a target for in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the RCS User (B) is a target for RCS in the third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.4.3.3-2 illustrates the case where RCS User (A) is a target non-local ID for RCS in the third party RCS service provider domain B and RCS User (B) is a target non-local ID for RCS in the third party RCS service provider domain A.

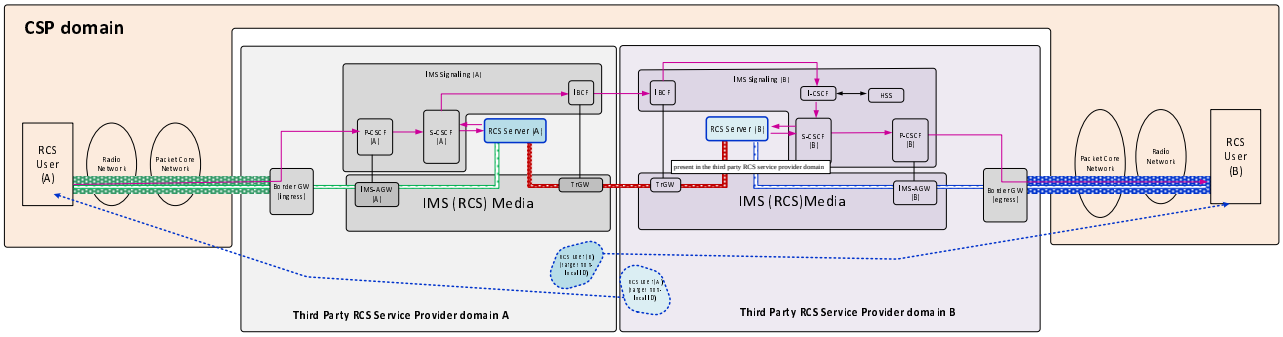


Figure 4.2.4.3.3-2: RCS users in the same CSP domain that does not have IMS (target non-local ID)

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when RCS User (B) is a target non-local ID in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the RCS User (A) is a target non-local ID in the third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.4.3.4 Scenario 3: RCS users are in the same CSP domain that has IMS

This clause illustrates the case when RCS Servers are in different third party RCS service provider domains and the RCS users are the same CSP domain that has IMS.

The illustrations focus RCS LI and non-roaming scenarios.

In the illustrations shown in this clause, User (A) and User (B) are the subscribers in the same CSP domain. Since the RCS Servers used for the RCS sessions are in third party RCS service provider domains, in this deployment option, the CSP domain provides just the access when those users establish RCS sessions and as such both signalling and media pass through the CSP domain over the packet data connection.

The signalling and media for non-RCS related IMS sessions do not pass through the third party RCS service provider domain.

Figure 4.2.4.3.4-1 illustrates the CSP deployment option 1 where User (A) is a target for RCS in RCS third party domain A and User (B) is a target for RCS in the RCS third party domain B.

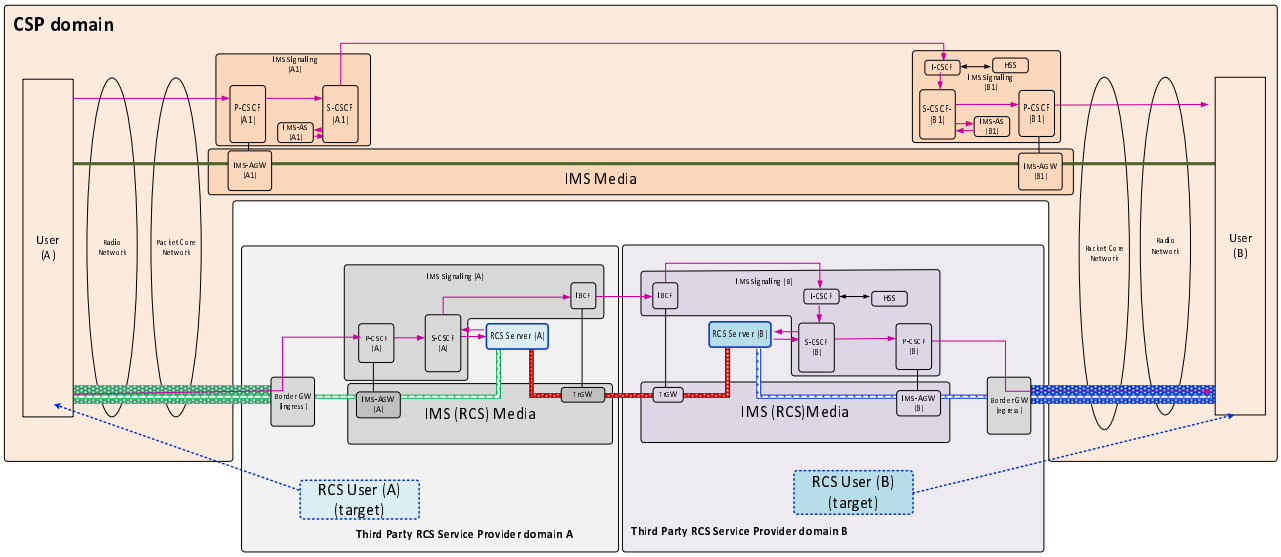


Figure 4.2.4.3.4-1: RCS users in the same CSP domain that has IMS

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target for in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the User (B) is a target for in the third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.4.3.4-2 illustrates the CSP deployment option 1 where User (A) is a target non-local ID for RCS in RCS third party domain B and User (B) is a target non-local ID for RCS in the RCS third party domain A.

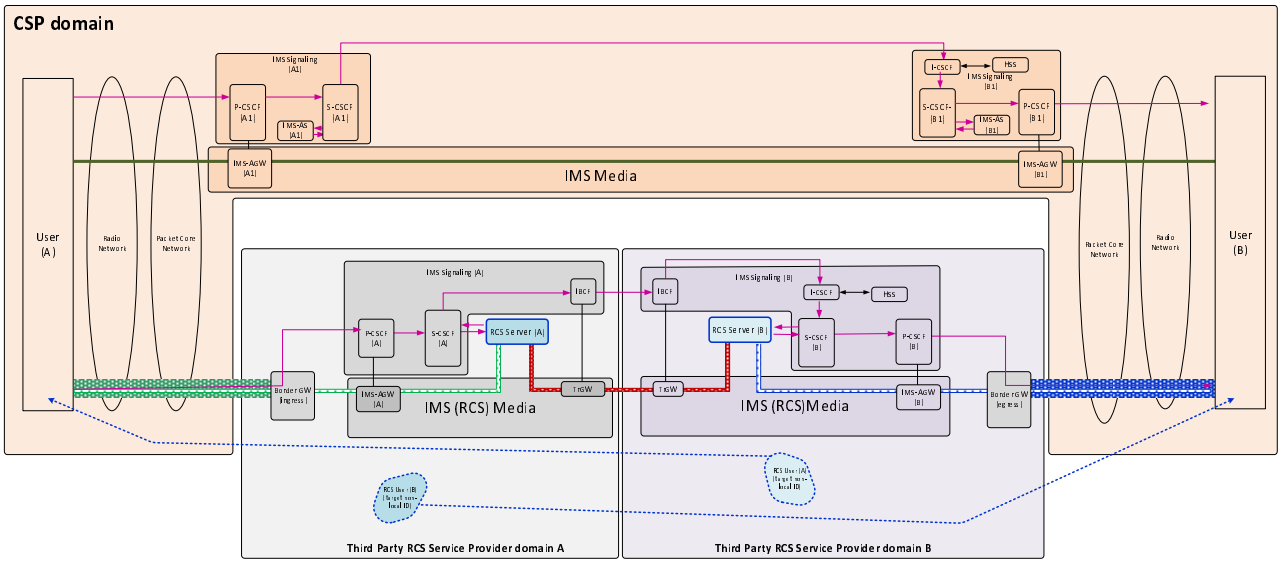


Figure 4.2.4.3.4-2: RCS users in the same CSP domain that has IMS (target non-local ID)

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (B) is a target non-local ID for RCS in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the User (A) is a target non-local ID for RCS in the third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.4.3.5 Scenario 4: RCS users are different CSP domains that do not have IMS

In the illustrations shown in this clause, RCS service is provided by different third party RCS service providers and the RCS users are in two CSP domains that do not have IMS.

RCS User (A) is a subscriber in CSP domain A1) and RCS User (B) is a subscriber in the CSP domain B1. The CSP domains provide just the access when those users establish RCS sessions and as such both signalling and media pass through the CSP domains over the packet data connections.

Figure 4.2.4.3.5-1 illustrates the case where RCS User (A) is a target for RCS in the third party RCS service provider domain A and RCS User (B) is a target for RCS in the third party RCS service provider domain B.

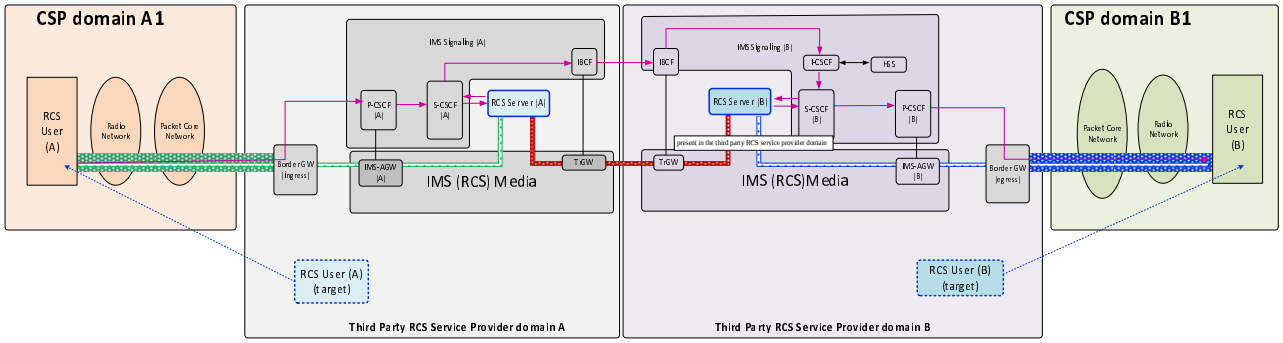


Figure 4.2.4.3.5-1: RCS users in the different CSP domains that do not have IMS

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when RCS User (A) is a target for RCS in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the RCS User (B) is a target for RCS in the third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.4.3.5-2 illustrates the case where RCS User (A) is a target non-local ID for RCS in the third party RCS service provider domain B and RCS User (B) is a target non-local ID for RCS in the third party RCS service provider domain A.

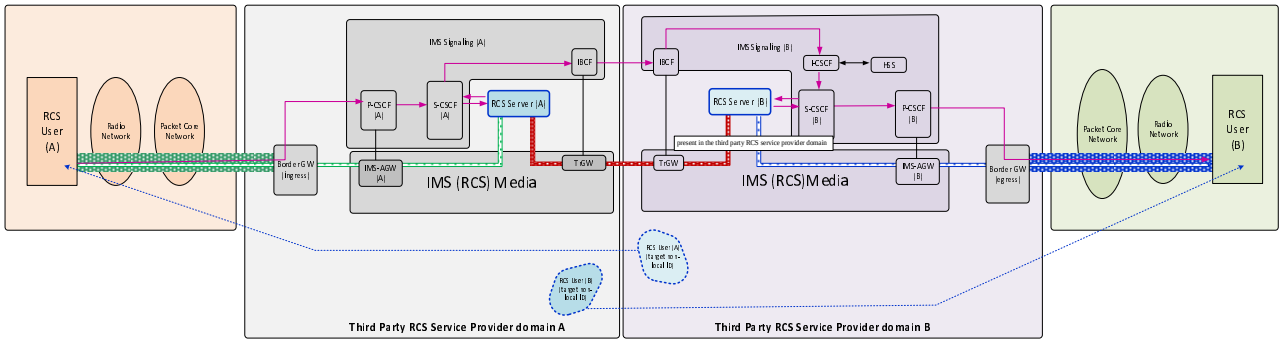


Figure 4.2.4.3.5-2: RCS users in the different CSP domains that do not have IMS (target non-local ID)

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when RCS User (B) is a target non-local ID for RCS in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the RCS User (A) is a target non-local ID for RCS in the third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.4.3.6 Scenario 5: RCS users are in the different CSP domains that have IMS

This clause illustrates the case when RCS Servers are in different third party RCS service provider domains and the RCS users are different CSP domains that have IMS.

The illustrations focus RCS LI and non-roaming scenarios.

In the illustrations show in the clause, User (A) is a subscriber of CSP domain A1 and User (B) is a subscriber of CSP domain B1. Since the RCS Servers used for the RCS sessions are in third party RCS service provider domains, in this deployment option, both CSP domains provide just the access when those users establish RCS sessions and as such both signalling and media pass through the CSP domains over the packet data connection.

The signalling and media for non-RCS related IMS sessions do not pass through the third party RCS service provider domains.

Figure 4.2.4.3.6-1 illustrates the CSP deployment option 1 where RCS User (A) is a target for RCS in the third party RCS service provider domain A and RCS User (B) is a target for RCS in the third party RCS service provider domain B:

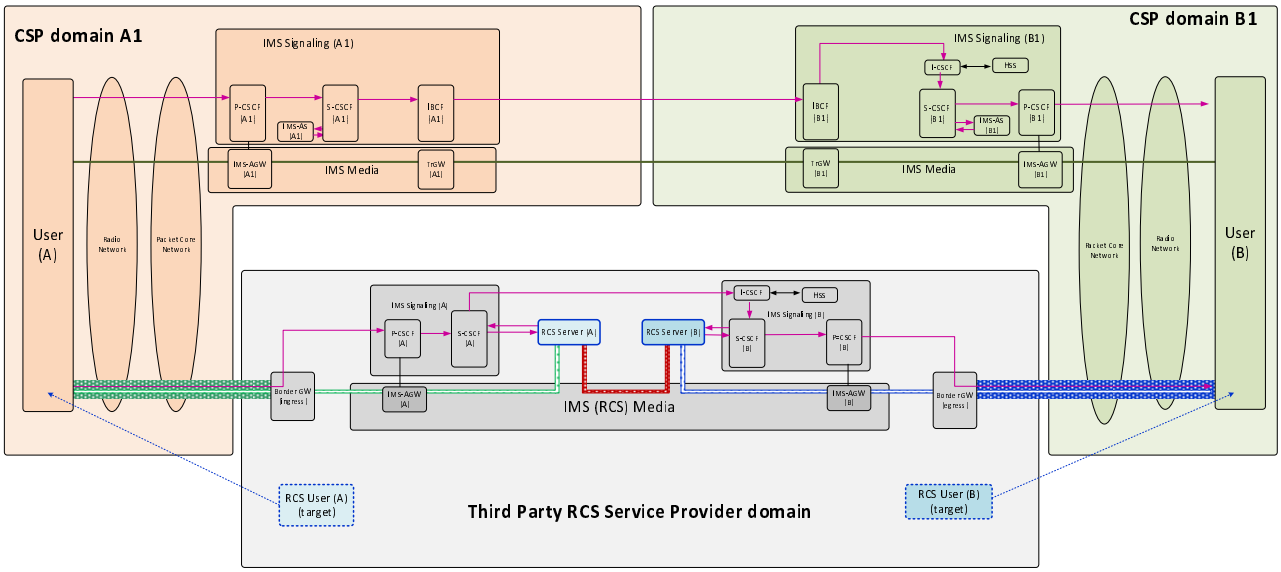


Figure 4.2.4.3.6-1: RCS users in the different CSP domains that have IMS

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target for RCS in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the User (B) is a target for RCS in the third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.4.3.6-2 illustrates the case where RCS User (A) is a target non-local ID for RCS in the third party RCS service provider domain B and RCS User (B) is a target non-local ID for RCS in the third party RCS service provider domain A.

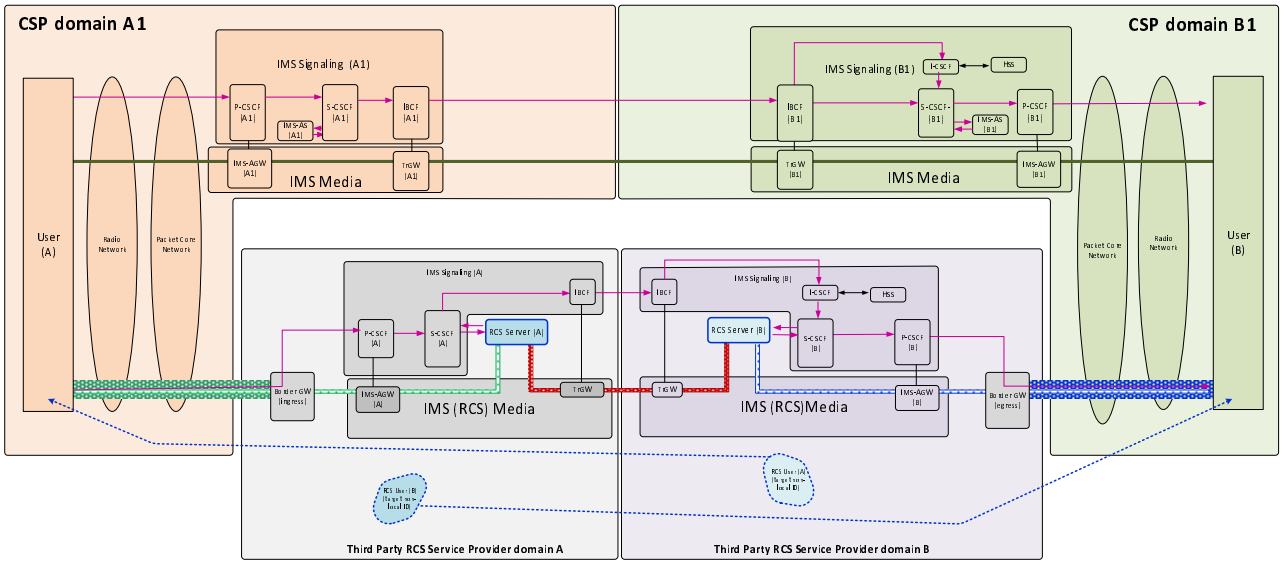


Figure 4.2.4.3.6-2: RCS users in the different CSP domains that have IMS (target non-local ID)

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (B) is a target non-local ID for RCS in the third party RCS service provider domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the User (A) is a target non-local ID for RCS in the third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.5 Hybrid cases

4.2.5.1 General

Clause 4.2.5 illustrates the hybrid cases where one RCS Server is in third party service provider domain and the other RCS Server is in the CSP domain.

The illustrations focus RCS LI and non-roaming scenarios.

4.2.5.2 Originating RCS Server in third party, terminating RCS Server in CSP domain

4.2.5.2.1 General

When the originating RCS server is in third party RCS service provider, the following cases apply to the originating end:

Scenario 1: Originating RCS user is not in CSP domain.

Scenario 2: Originating RCS user is in CSP domain that does not have IMS.

Scenario 3: Originating RCS user is in CSP domain that has IMS.

Not all scenarios listed above may be deployed in practice. The present document includes the scenarios, to illustrate, in case, the scenarios are deployed.

4.2.5.2.2 Scenario 1: Originating RCS user not in CSP domain

Figure 4.2.5.2.2-1 illustrates the case when RCS User (A) is a target in the third party RCS provider domain A and the User B is a target for RCS in the CSP domain B.

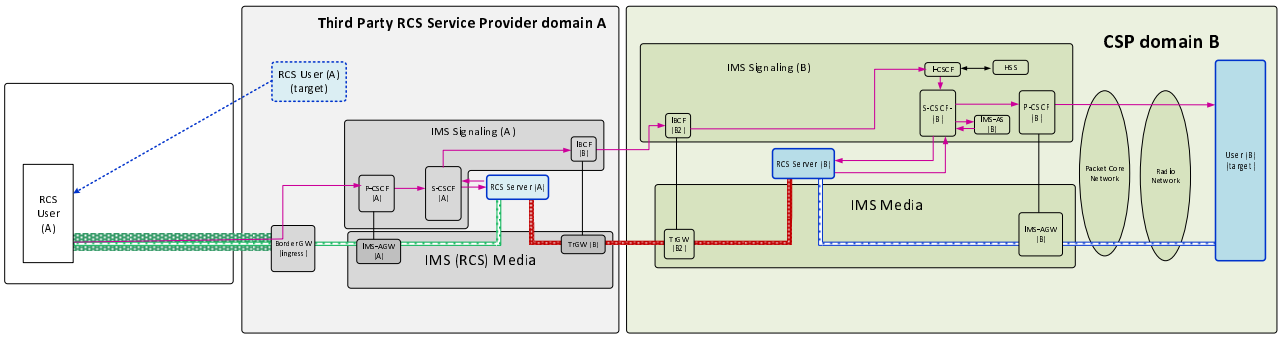


Figure 4.2.5.2.2-1: RCS server (third party) at the originating end, non-CSP user

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when RCS User (A) is a target in the third party RCS service provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the RCS User (B) is a target for RCS in CSP domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.5.2.2-2 illustrates the case when User (B) is a target non-local ID in the third party RCS provider domain A and the RCS User A is a target for RCS non-local ID in the CSP domain B.

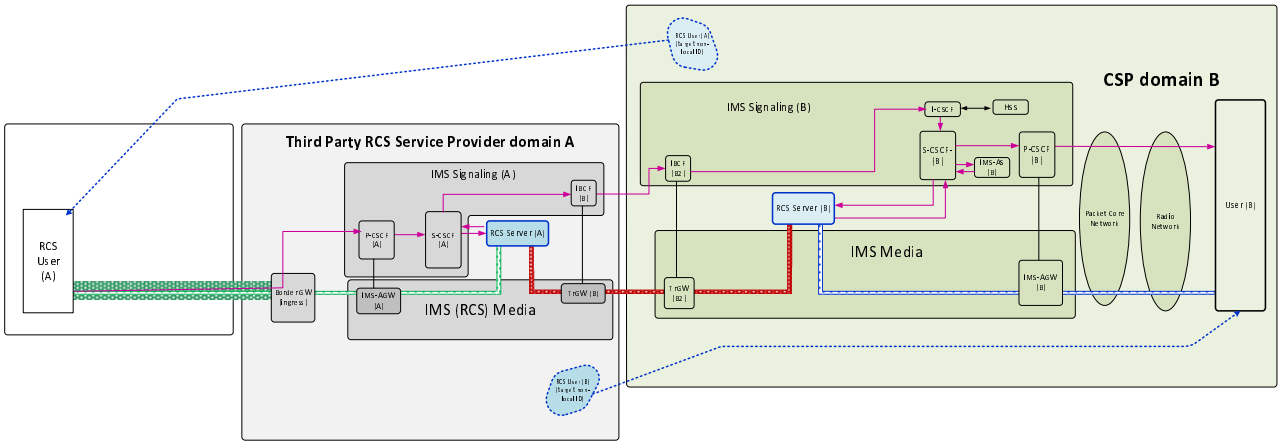


Figure 4.2.5.2.2-2: RCS server (third party) at the originating end, non-CSP user (target non-local ID)

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (B) is a target non-local ID in the third party RCS service provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the RCS User (A) is a target non-local ID for in the CSP domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.5.2.3 Scenario 2: Originating RCS user in CSP domain that does not have IMS

In the illustrations shown in this clause, originating RCS Server is in third party RCS service provider domain and the terminating RCS server is in CSP domain. Originating RCS user is in CSP domain that does not have IMS.

In the illustrations, RCS User (A) is a subscriber in CSP domain A1 that does not have IMS and User (B) is a subscriber in CSP domain B. The CSP domain A1 provides just the access when the RCS User (A) establishes RCS sessions and as such both signalling and media pass through the CSP domain A1 over the packet data connection.

Figure 4.2.5.2.3-1 illustrates the case where User (B) is a target for in CSP domain B and RCS User (A) is a target in third party RCS provider domain A.

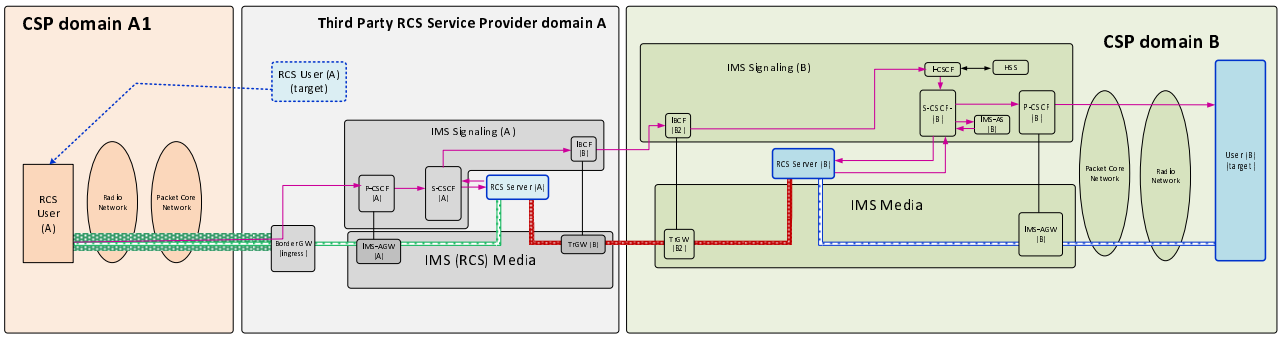


Figure 4.2.5.2.3-1: RCS server (third party) at the originating end, no IMS

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when RCS User (A) in is a target in the third party RCS service provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (B) is a target for RCS in the CSP domain B for RCS.

The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.5.2.3-2 illustrates the case where RCS User (A) is a target non-local ID for RCS in the CSP domain B and User B is a target non-local ID in third party RCS provider domain A.

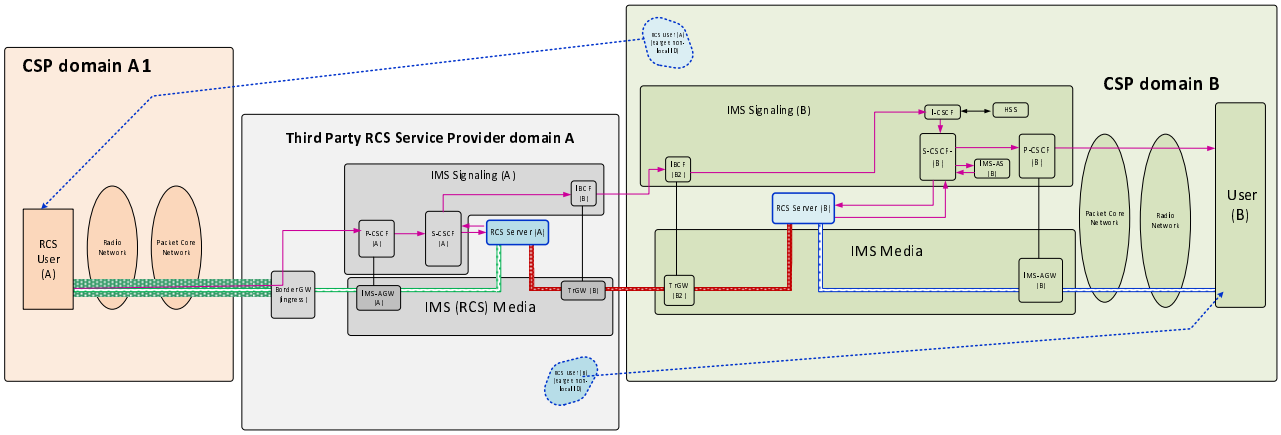


Figure 4.2.5.2.3-2: RCS server (third party) at the originating end, no IMS (target non-local ID)

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (B) in is a target non-local ID in the third party RCS service provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the RCS User (A) is a target non-local ID for RCS in the CSP domain B for RCS. The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.5.2.4 Scenario 3: Originating RCS user in CSP domain that has IMS

In the illustrations shown in this clause, originating RCS Server is in third party RCS service provider domain and the terminating RCS server is in CSP domain. Originating RCS user is in CSP domain that has IMS.

In the illustrations shown in this clause, the User (A) is a subscriber in CSP domain A1 and User (B) is a subscriber in CSP domain B. At the originating end, since the RCS Server (A) used for the RCS sessions is in third party RCS service provider domain A, in this deployment option, the CSP domain A1 provides just the access to the RCS users and as such both signalling and media pass through the CSP domain A1 over the packet data connection.

The signalling and media for non-RCS related IMS sessions do not pass through the third party RCS service provider domain A.

Figure 4.2.5.2.4-1 illustrates the deployment option 1 for the case where User (A) in CSP domain A1 is a target in the third party RCS provider domain A and the User (B) is a target for RCS in CSP domain B.

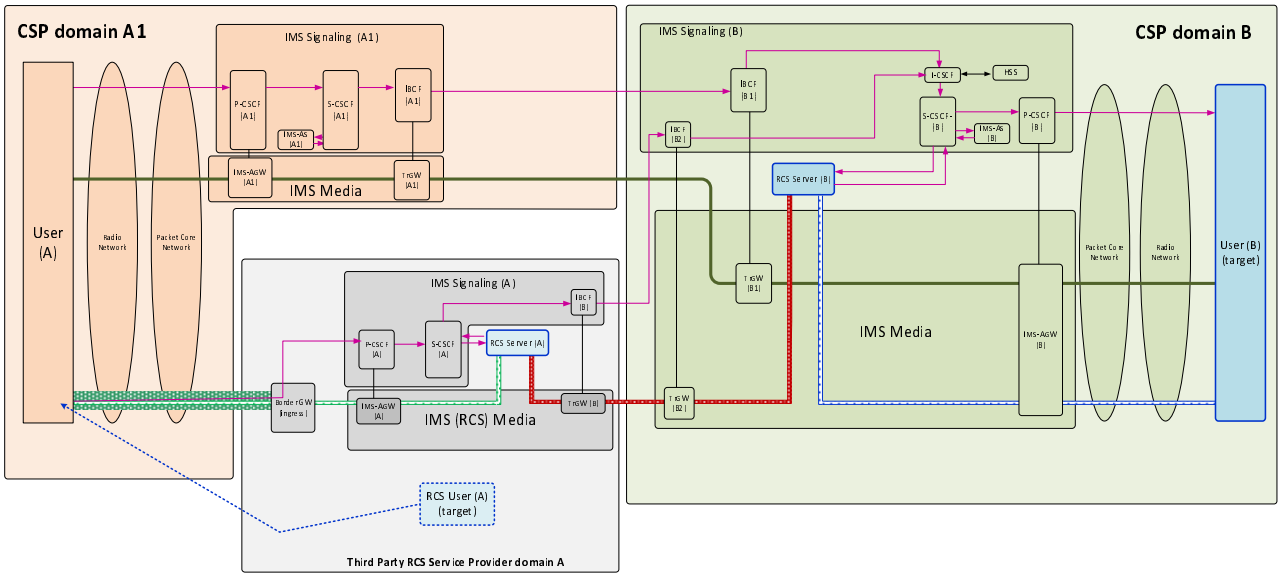


Figure 4.2.5.2.4-1: RCS server (third party) at the originating end, CSP has IMS

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target in the third party RCS service provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (B) is a target for RCS in the CSP domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.5.2.4-2 illustrates the deployment option 1 for the case where User A in CSP domain A1 is a target non-local ID for RCS in CSP domain B and the User (B) is a target non-local ID in the third party RCS provider domain A.

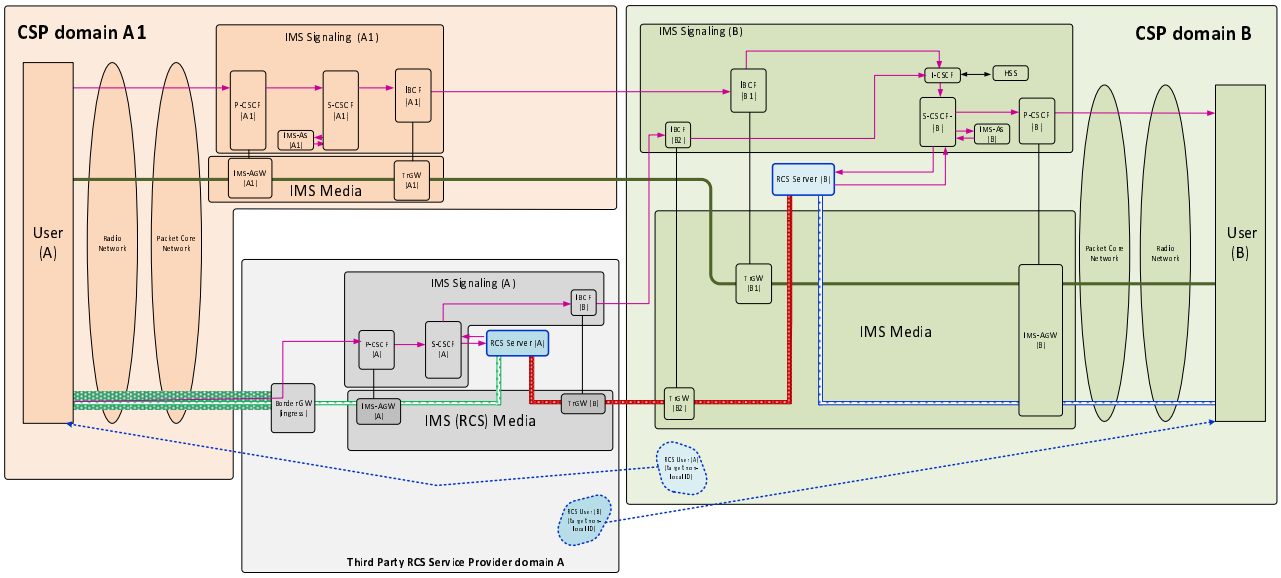


Figure 4.2.5.2.4-2: RCS server (third party) at the originating end, CSP has IMS (target non-local ID)

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (B) is a target non-local ID in the third party RCS service provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (A) is a target non-local ID for RCS in the CSP domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.5.3 Terminating RCS Server in third party, originating RCS Server in CSP domain

4.2.5.3.1 General

When the terminating RCS server is in third party RCS service provider domain, the following cases apply to the originating end:

Scenario 1: Terminating RCS user is not in CSP domain.

Scenario 2: Terminating RCS user is in CSP domain that does not have IMS.

Scenario 3: Terminating RCS user is in CSP domain that has IMS.

Not all scenarios listed above may be deployed in practice. Not all the scenarios listed above may be deployed in practice. The present document includes the scenarios, to illustrate, in case, the scenarios are deployed.

4.2.5.3.2 Scenario 1: Terminating RCS user not in CSP domain

Figure 4.2.5.3.2-1 illustrates the case when RCS User (B) is a target in the third party RCS provider domain B and the User A is a target for RCS in the CSP domain A. RCS User (B) is not a CSP user.

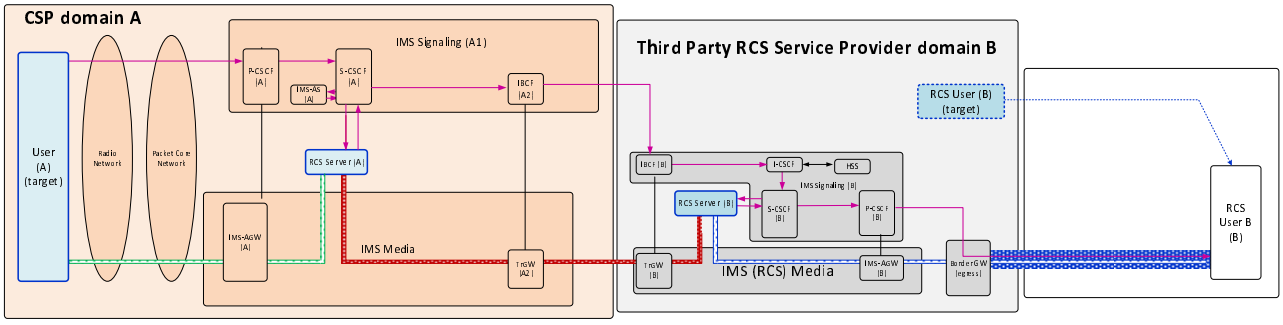


Figure 4.2.5.3.2-1: RCS server (third party) at the terminating end, non-CSP user

The RCS Server (A) present in the CSP domain A provides the LI functions when User (A) is a target for RCS in the CSP domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the RCS User (B) is a target in third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.5.3.2-2 illustrates the case when RCS User (B) is a target non-local ID for RCS in CSP domain A and User A is a target non-local ID in the third party RCS provider domain B.

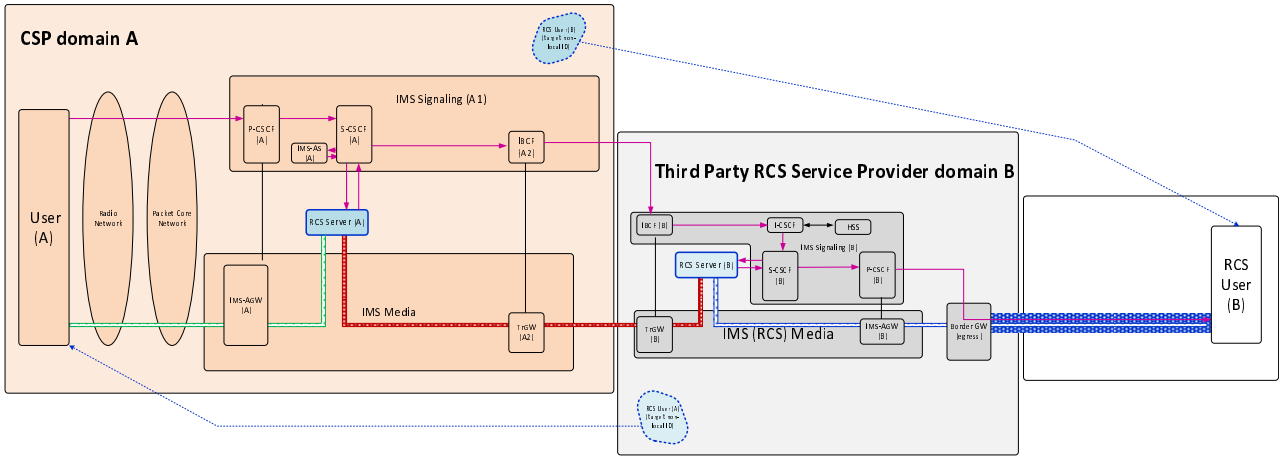


Figure 4.2.5.3.2-2: RCS server (third party) at the terminating end, non-CSP user (target non-local ID)

The RCS Server (A) present in the CSP domain A provides the LI functions when RCS User (B) is a target non-local ID for RCS in the CSP domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the User (A) is a target non-local ID for RCS in third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.5.3.3 Scenario 2: Terminating RCS user in CSP domain that does not have IMS

In the illustrations shown in the clause, User (A) is a subscriber in CSP domain A and RCS User (B) is a subscriber in CSP domain B1 that does not have IMS. The CSP domain B1 provides just the access to RCS User (B) when RCS User (B) is on an RCS session. As such both signalling and media pass through the CSP domain B1 over the packet data connection for RCS User (B).

Figure 4.2.5.3.3-1 illustrates the case when RCS User (B) is a target in the third party RCS provider domain B and the User A is a target for RCS in the CSP domain A. The RCS User (B) is in CSP domain that does not have IMS.

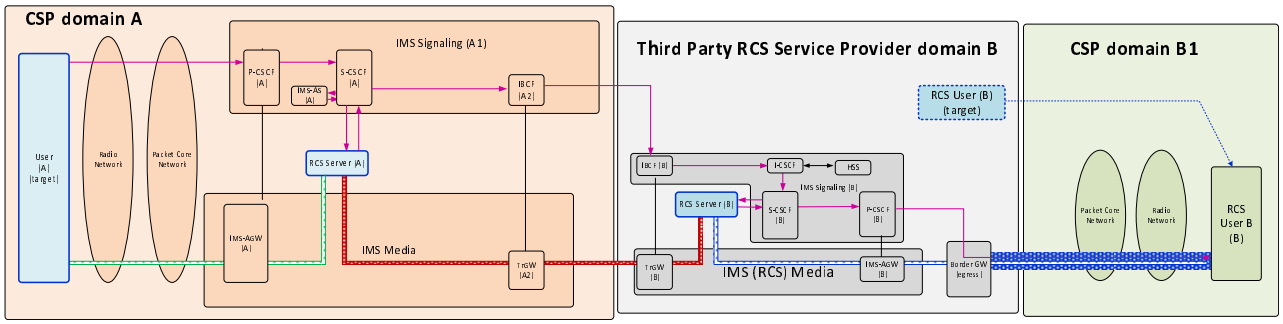


Figure 4.2.5.3.3-1: RCS server (third party) at the terminating end, no IMS

The RCS Server (A) present in the CSP domain A provides the LI functions when User (A) is a target for in the CSP domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the RCS User (B) is a target in third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.5.3.3-2 illustrates the case when RCS User (B) is a target non-local ID for RCS in CSP domain A and the User A is a target non-local ID in third party RCS provider domain B. RCS User (B) is in CSP domain that does not have IMS.

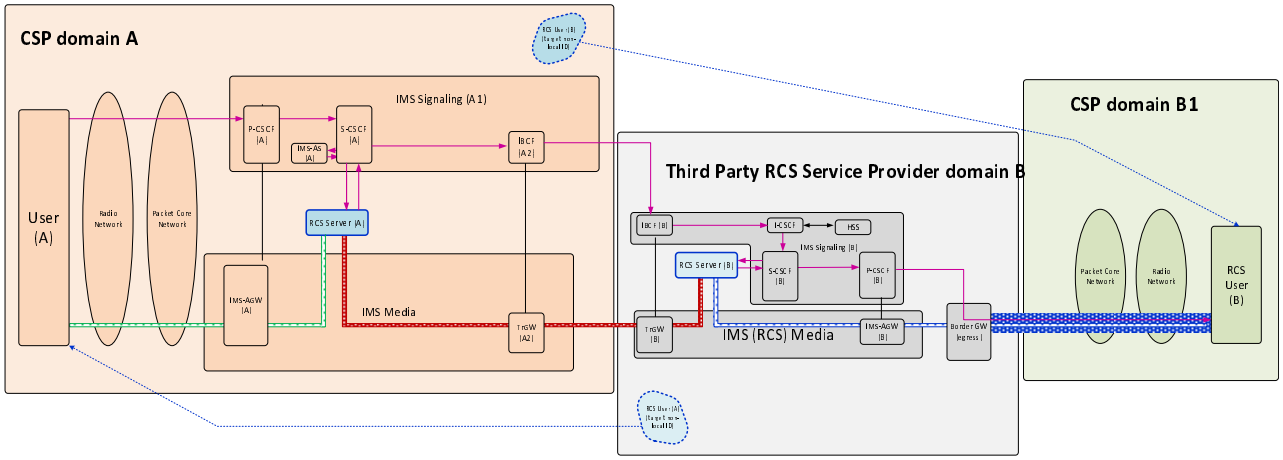


Figure 4.2.5.3.3-2: RCS server (third party) at the terminating end, no IMS (target non-local ID)

The RCS Server (A) present in the CSP domain A provides the LI functions when RCS User (B) is a target non-local ID for RCS in the CSP domain A and RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when the User (A) is a target non-local ID in third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.5.3.4 Scenario 3: Terminating RCS user in CSP domain that has IMS

In the illustrations shown in this clause, originating RCS Server is in CSP domain and terminating RCS Server is in third party RCS service provider domain. Terminating RCS user is in CSP domain that has IMS.

In the illustrations shown in this clause, User (A) is a subscriber in CSP domain A and User (B) is a subscriber in CSP domain B1. At the terminating end, since the RCS Server (B) used for the RCS sessions is in third party RCS service provider domain B, in this deployment option, CSP domain B1 provides just the access to User (B) and as such both signalling and media pass through the CSP domain B1 over the packet data connection.

The signalling and media for non-RCS related IMS sessions do not pass through the third party RCS service provider domain B.

Figure 4.2.5.3.4-1 illustrates the deployment option 1 for the case when User (A) is a target for RCS in the CSP domain A and RCS User (B) present in CSP domain B1 is a target in third party RCS provider domain B.

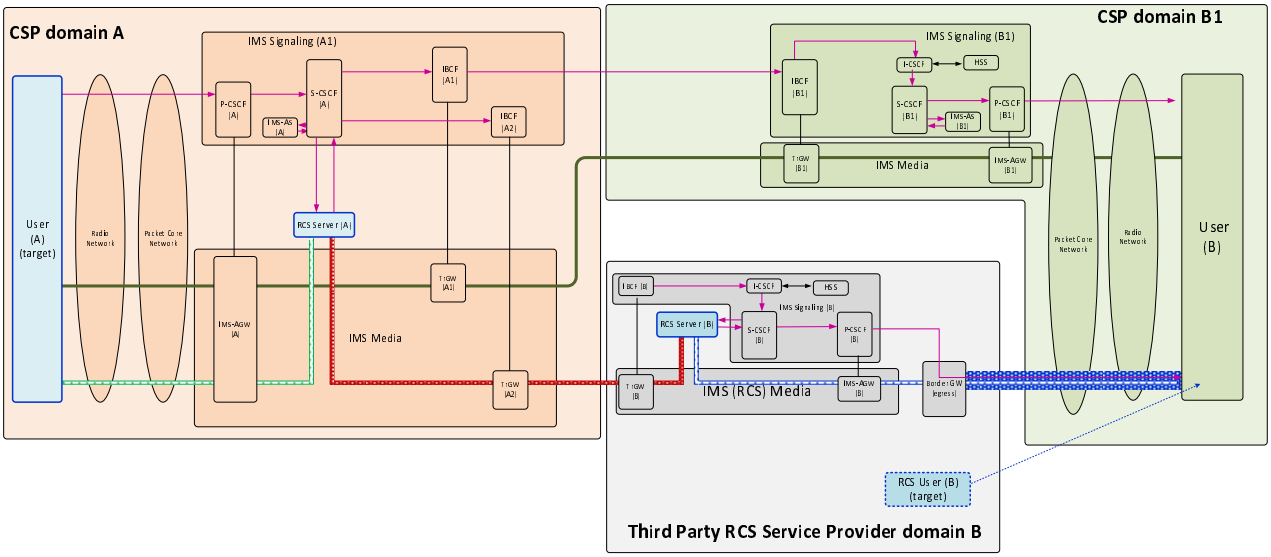


Figure 4.2.5.3.4-1: RCS server (third party) at the terminating end, CSP has IMS

RCS Server (A) present in the CSP domain A provides the LI functions when the User (A) is a target for RCS in the CSP domain A. The RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when User (B) is a target in the third party RCS service provider domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

Target non-local ID

Figure 4.2.5.3.4-2 illustrates the deployment option 1 for the case when User (B) present in CSP domain B1 is a target non-local ID for RCS in the CSP domain A and User (A) present in CSP domain A is a target non-local ID in third party RCS provider domain B.

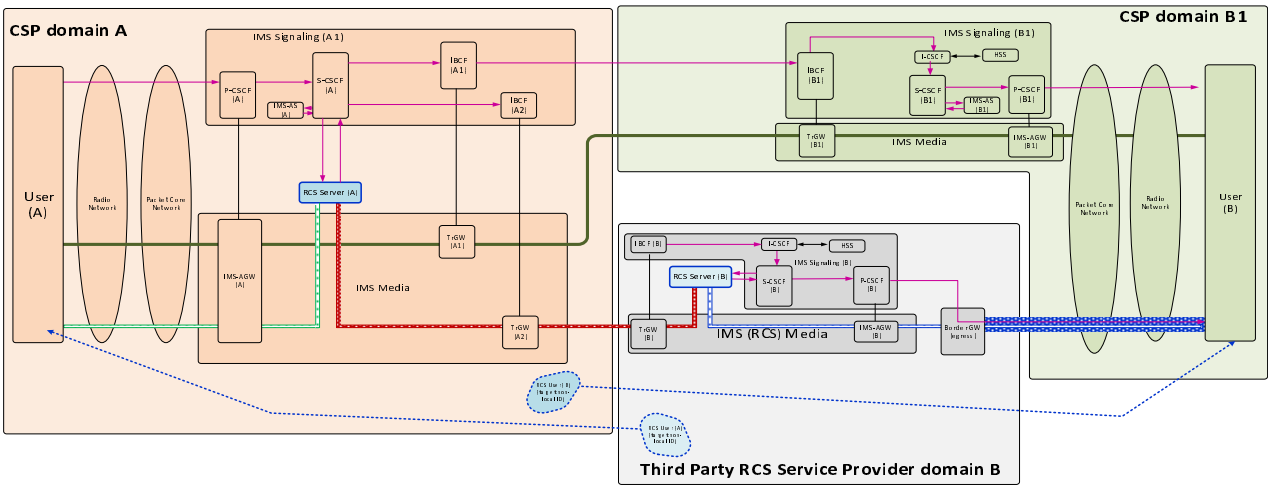


Figure 4.2.5.3.4-2: RCS server (third party) at the terminating end, CSP has IMS (target non-local ID)

The RCS Server (A) present in the CSP domain A provides the LI functions when the User (B) is a target non-local ID for RCS in the CSP domain A. The RCS Server (B) present in the third party RCS service provider domain B provides the LI functions when User (A) is a target non-local ID in the third party RCS service provider domain B.

The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.6 RCS interworking with SMS

4.2.6.1 General

The term Converged IP Messaging (CPM) is used when two users in a message exchange use different messaging format (e.g. RCS and SMS).

When the terminating user is not registered for the RCS or when the terminating network domain does not support, the RCS message originated from an RCS user may be delivered to the terminating user as an SMS message. The network function that performs the interworking is referred to as Interworking Function (IWF). The present document refers to the same as CPM Interworking Function which, in part, includes the RCS Server and the IP-SM-GW.

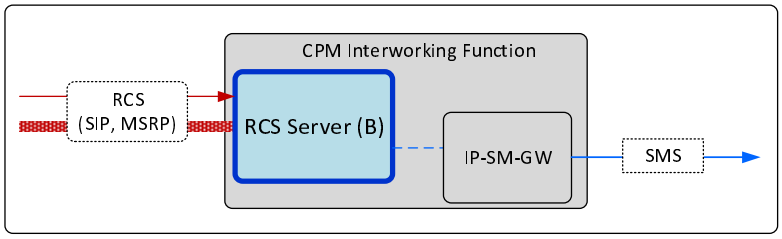


Figure 4.2.6.1-1: CPM Interworking Function for RCS to SMS interworking

For RCS to SMS interworking, the CPM Interworking Function may reside in the terminating network domain (e.g. when the terminating user is not registered for RCS) or in the originating network domain (e.g. when the terminating network does not support the RCS).

With the understanding that SMS is a CSP offered service, terminating device needs to have the SMS capability and therefore, it is presumed that the terminating user is served in a CSP domain. The RCS for the originating user may either be provided by the originating CSP domain or by a third party RCS service provider domain.

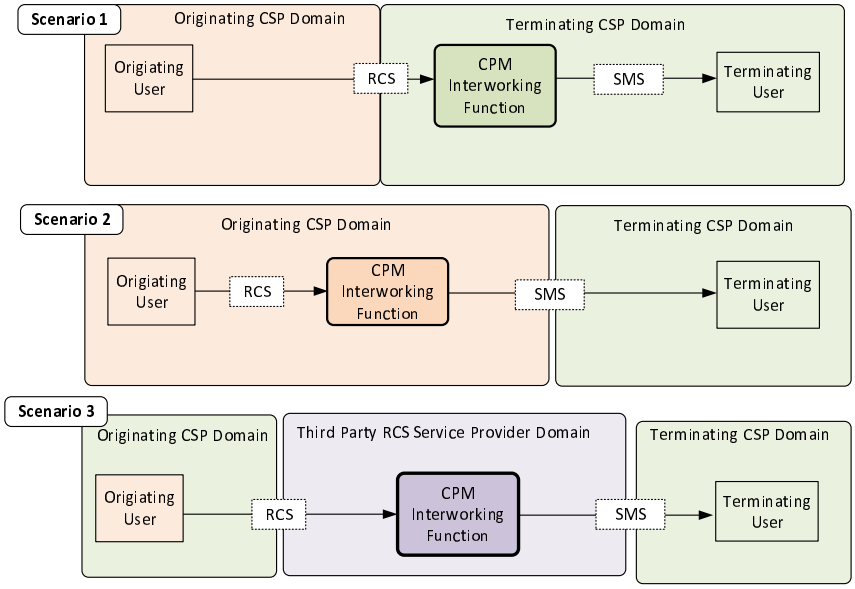


Figure 4.2.6.1-2: Scenarios of CPM Interworking Function for RCS to SMS interworking

The present document illustrates two cases of SMS delivery:

- SMS over IP.
- SMS over NAS.

For RCS to SMS interworking to happen, the destination should be addressed using a SIP URI, or TEL URI, MSISDN or an E.164 number. When the destination is a SIP URI, the SMS is delivered over IMS. When the destination is MSISDN or E.164 number, the SMS is delivered over NAS.

The RCS topology scenarios illustrated in clause 4.2.3 and clause 4.2.5.2 are applicable for the originating end of the RCS handling.

The subsequent clauses illustrate the topology scenarios for RCS to SMS interworking (i.e. destination is the SMS user). The illustrations of SMS to RCS interworking (if possible) are for further study from this TR perspective.

4.2.6.2 CPM interworking function at the terminating end

4.2.6.2.1 General

This clause illustrates the scenarios of RCS to SMS interworking (i.e. CPM Interworking Function) at the terminating end when SMS is delivered either over IP or over NAS to the destination.

The scenarios illustrate the following:

- Originating RCS server is in the CSP domain.
- Originating RCS server is in third party RCS service provider domain.

The illustration starts with the assumption that two users involved are in two different domains. An illustration is shown at the end, in case two users happens to be in the same CSP domain. The exact use of this case is further study.

When the originating RCS server is in third party RCS service provider domain, the case 2 illustrated in clause 4.2.5.2 (hybrid case) are considered. However, other cases can also be applicable.

Illustrative examples of LI for messaging to support the interception of SMS to the terminating target are in TR 33.929-3 [12].

4.2.6.2.2 Originating RCS server is in a CSP domain

4.2.6.2.2.1 SMS over IP

Figure 4.2.6.2.2.1-1 illustrates a case of SMS over IP where User (A) is a target in the CSP domain A and User (B) is a target in the CSP domain B. Both users are targets for RCS in the respective domain.

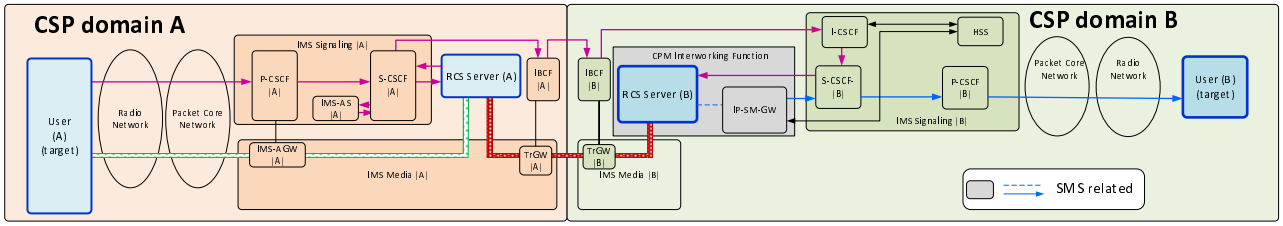


Figure 4.2.6.2.2.1-1: RCS to SMS interworking at the terminating end (different CSP domains)

For RCS LI, the RCS Server (A) present in the CSP domain A provides the LI functions when User (A) is a target for RCS in CSP domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (B) is a target for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case the User (B) is a target for messaging, the SMS related LI functions are provided by the S-CSCF (B) present in the CSP domain B. The details of LI for messaging for SMS over IP with IRI-POI at S-CSCF are illustrated in TR 33.929-3 [12].

NOTE: Since the terminating user (User (B) in figure 4.2.6.2.2.1-1) does not receive the RCS service, the scope of the RCS LI when the terminating user is a target for RCS in the terminating domain (CSP domain in figure 4.2.6.2.2.1-1) may be limited (e.g. reporting only the incoming SIP messages). Alternatively, the RCS Server in the terminating end may not provide the LI functions when the terminating user is a target for RCS.

The above NOTE is applicable to the illustrations shown in the subsequent clauses as well.

The case of target non-local ID

Figure 4.2.6.2.2.1-2 illustrates a case of SMS over IP where User (A) is a target non-local ID in the CSP domain B and User (B) is a target non-local ID in the CSP domain A.

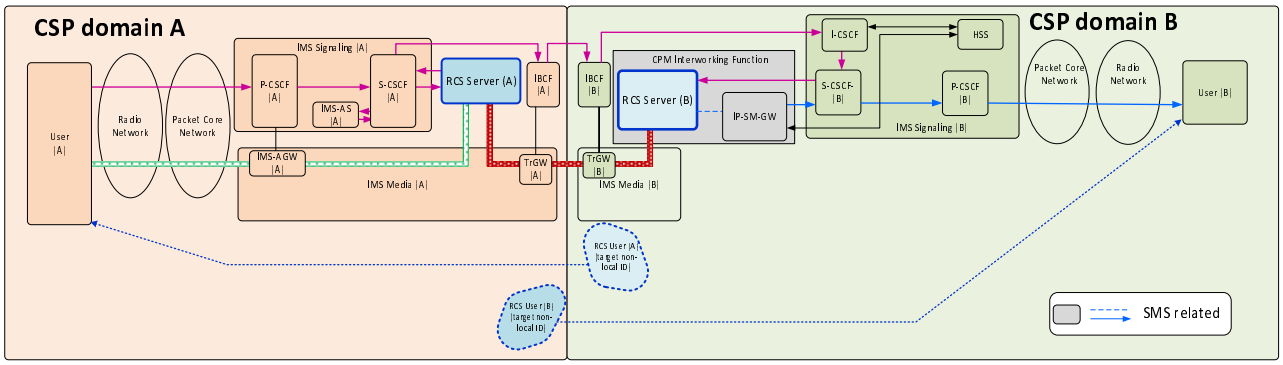


Figure 4.2.6.2.2.1-2: RCS to SMS interworking at the terminating end (target non-local ID)

RCS Server (A) present in the CSP domain A provides the LI functions when User (B) is a target non-local ID for RCS in CSP domain A and RCS Server (B) provides the LI functions when the User (A) is a target non-local ID for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case the User (A) is a target non-local ID for messaging in CSP domain B, the SMS related LI functions are provided by the S-CSCF (B) present in the CSP domain B.

4.2.6.2.2.2 SMS over NAS

Figure 4.2.6.2.2.2-1 illustrates a case of SMS over NAS where User (A) is a target for RCS in the CSP domain A and User (B) is a target for RCS in the CSP domain B.

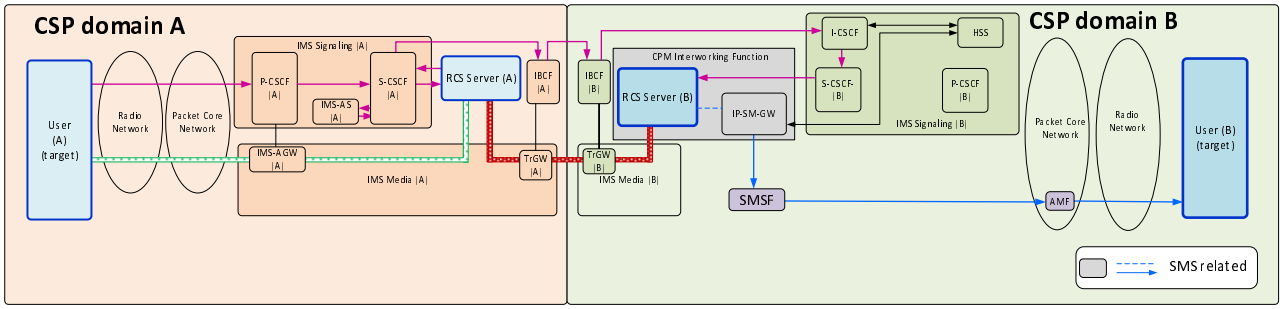


Figure 4.2.6.2.2.2-1: RCS to SMS interworking at the terminating end (different CSP domains)

For RCS LI, the RCS Server (A) present in the CSP domain A provides the LI functions when User (A) is a target for RCS in CSP domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (B) is a target for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case the User (B) is a target for messaging in CSP domain B, the SMS related LI functions are provided by the SMSF present in the CSP domain B. The details of LI for messaging for SMS over NAS with IRI-POI at SMSF are illustrated in TR 33.929-3 [12].

The case of target non-local ID

Figure 4.2.6.2.2.2-2 illustrates a case of SMS over NAS where User (A) is a target non-local ID in the CSP domain B for RCS and User (B) is a target non-local ID for RCS in the CSP domain A.

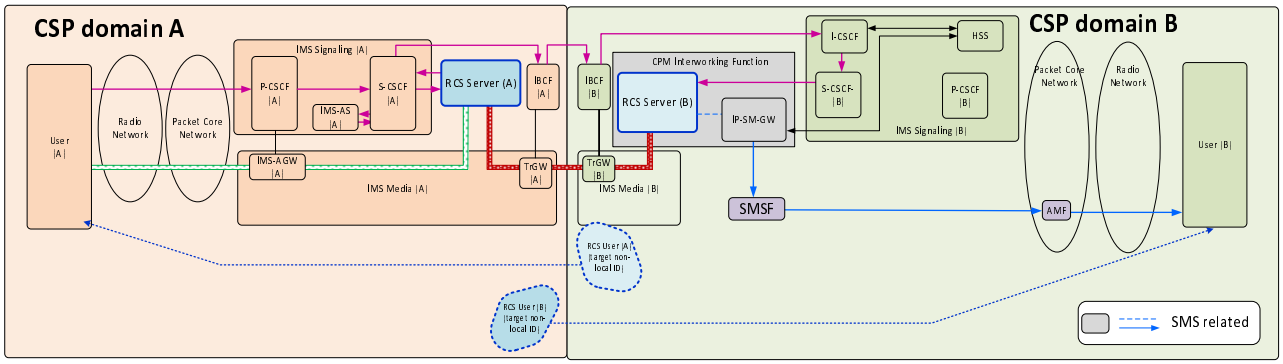


Figure 4.2.6.2.2.2-2: RCS to SMS interworking at the terminating end (target non-local ID)

RCS Server (A) present in the CSP domain A provides the LI functions when User (B) is a target non-local ID for RCS in CSP domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (A) is a target non-local ID for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case the User (A) is a target non-local ID for messaging in CSP domain B, the SMS related LI functions are provided by the SMSF present in the CSP domain B.

4.2.6.2.3 Originating RCS server is in third party RCS provider domain

4.2.6.2.3.1 SMS over IP

Figure 4.2.6.2.3.1-1 illustrates a case of SMS over IP where User (A) served by CSP domain A1 is a target in the third party RCS provider domain A and User (B) is a target in the CSP domain B.

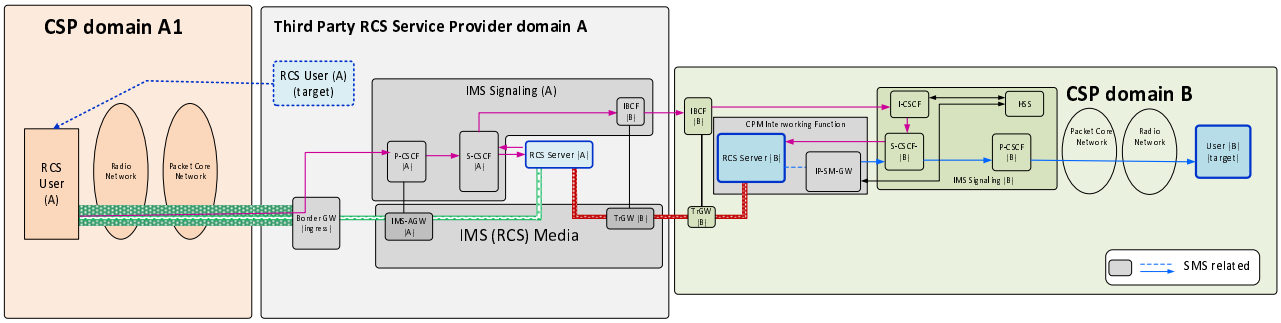


Figure 4.2.6.2.3.1-1: RCS to SMS interworking at the terminating end

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target for RCS in third party RCS service provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (B) is a target for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case the User (B) is a target for messaging in CSP domain B, the SMS related LI functions are provided by the S-CSCF (B) present in the CSP domain B.

The case of target non-local ID

Figure 4.2.6.2.3.1-2 illustrates a case of SMS over IP where User (A) is a target non-local ID in the CSP domain B for RCS and User (B) is a target non-local ID for RCS in the third party RCS provider domain A.

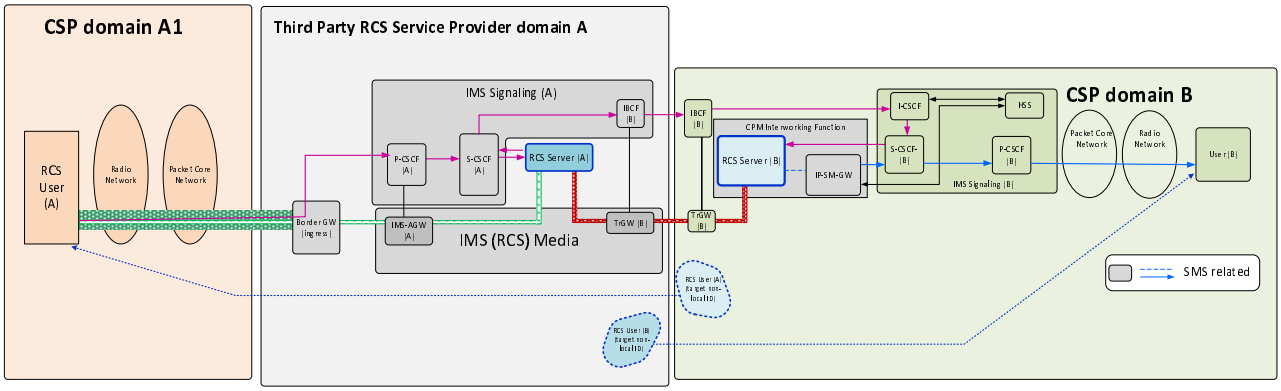


Figure 4.2.6.2.3.1-2: RCS to SMS interworking at the terminating end (target non-local ID)

RCS Server (A) present in the third party RCS provider domain A provides the LI functions when User (B) is a target non-local ID for RCS in third party RCS provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (A) is a target non-local ID for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case the User (A) is a target non-local ID for messaging in CSP domain B, the SMS related LI functions are provided by the S-CSCF (B) present in the CSP domain B.

4.2.6.2.3.2 SMS over NAS

Figure 4.2.6.2.3.2-1 illustrates a case of SMS over NAS where User (A) served by CSP domain A1 is a target in the third party RCS provider domain A and User (B) is a target in the CSP domain B.

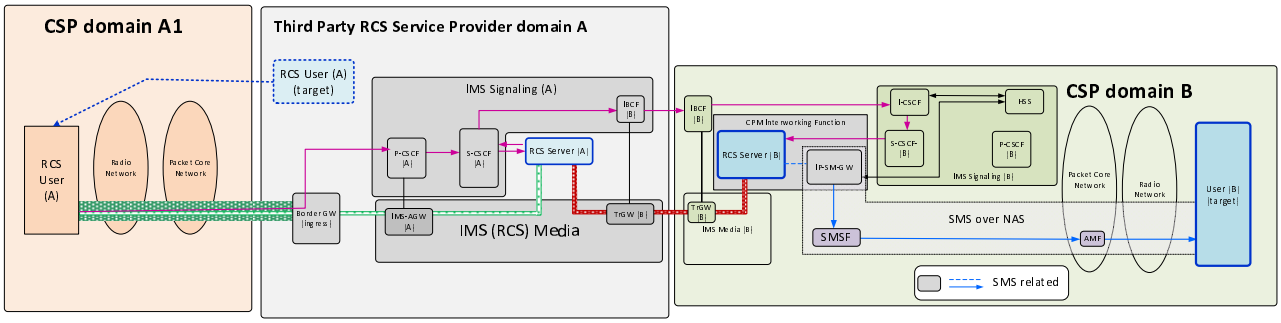


Figure 4.2.6.2.3.2-1: RCS to SMS interworking at the terminating end

For RCS LI, the RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target for RCS in third party RCS service provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (B) is a target for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case the User (B) is a target for messaging in CSP domain B, the SMS related LI functions are provided by the SMSF present in the CSP domain B.

The case of target non-local ID

Figure 4.2.6.2.3.2-2 illustrates a case where User (A) is a target non-local ID in the CSP domain B for RCS and User (B) is a target non-local ID for RCS in the third party RCS provider domain A.

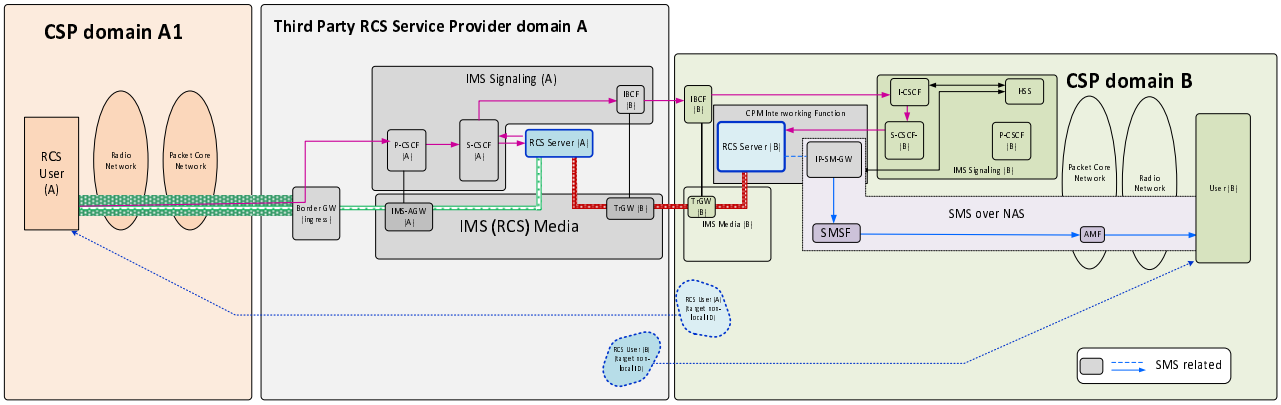


Figure 4.2.6.2.3.2-2: RCS to SMS interworking at the terminating end (target non-local ID)

RCS Server (A) present in the third party RCS provider domain A provides the LI functions when User (B) is a target non-local ID for RCS in third party RCS provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (A) is a target non-local ID for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case the User (A) is a target non-local ID for messaging in CSP domain B, the SMS related LI functions are provided by the SMSF present in the CSP domain B.

4.2.6.2.4 Two users in the same CSP domain

4.2.6.2.4.1 SMS over IP

Figure 4.2.6.2.4.1-1 illustrates a case of SMS over IP where both User (A) and User (B) are in the same CSP domain and happens to be the targets.

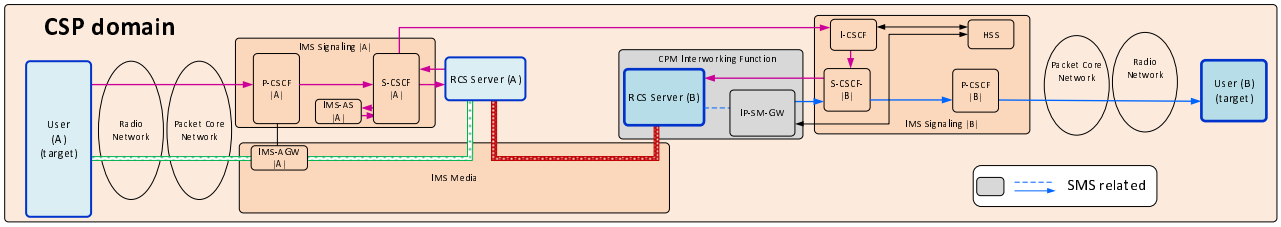


Figure 4.2.6.2.4.1-1: RCS to SMS interworking at the terminating end (same CSP)

For RCS LI, the RCS Server (A) provides the LI functions when User (A) is a target for RCS and RCS Server (B) provides the LI functions when the User (B) is a target for RCS. The RCS Servers provide both IRI-POI and CC-POI functions.

In case the User (B) is a target for messaging, the SMS related LI functions are provided by the S-CSCF (B).

4.2.6.2.4.2 SMS over NAS

Figure 4.2.6.2.4.2-1 illustrates a case of SMS over NAS for an IMS capable UE where both User (A) and User (B) are in the same CSP domain and happens to be the targets.

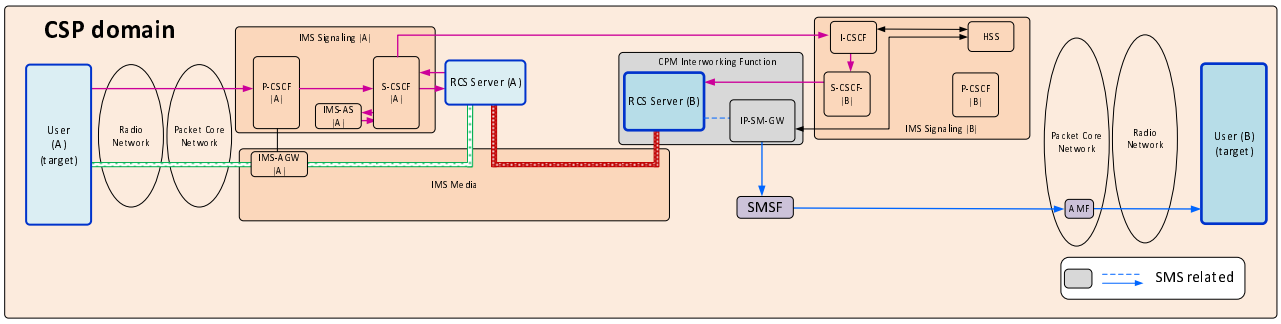


Figure 4.2.6.2.4.2-1: RCS to SMS interworking at the terminating end (same CSP)

For RCS LI, the RCS Server (A) provides the LI functions when User (A) is a target for RCS and RCS Server (B) provides the LI functions when the User (B) is a target for RCS. The RCS Servers provide both IRI-POI and CC-POI functions.

4.2.6.3 CPM interworking function at the originating end

4.2.6.3.1 General

This clause illustrates the scenarios of RCS to SMS interworking (i.e. CPM Interworking Function) at the originating end when SMS is delivered either over IP or over NAS to the destination.

The scenarios illustrate the following:

- Originating RCS server is in the CSP domain.
- Originating RCS server is in third party RCS service provider domain.

When the originating RCS server is in third party RCS service provider domain, the case 2 illustrated in clause 4.2.5.2 (hybrid case) are considered. However, case 3 can be applicable too.

Note that the use-case for this scenario is when originating domain determines that the terminating domain does not support RCS. Therefore, the illustrations do not consider single CSP domain case.

4.2.6.3.2 Originating user is in CSP domain

4.2.6.3.2.1 SMS over IP

Figure 4.2.6.3.2.1-1 illustrates a case of SMS over IP where User (A) is a target for RCS in the CSP domain A and User (B) is a target for messaging in the CSP domain B.

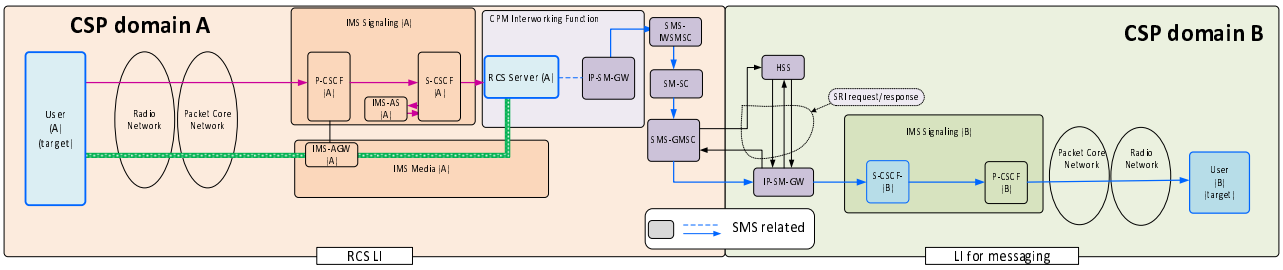


Figure 4.2.6.3.2.1-1: RCS to SMS interworking at the originating end

For RCS LI, the RCS Server (A) present in the CSP domain A provides the LI functions when User (A) is a target for RCS in CSP domain A. The RCS-Server (A) provides the IRI-POI and CC-POI functions. Since the terminating domain does not support RCS, there is no LI for RCS in the terminating domain.

For SMS LI, the S-CSCF (B) present in the CSP domain B provides the LI functions when the User (B) is a target for messaging in CSP domain B. The S-CSCF (B) provides the IRI-POI function.

4.2.6.3.2.2 SMS over NAS for IMS capable UE

Figure 4.2.6.3.2.2-1 illustrates a case of SMS over NAS for IMS capable UE where User (A) is a target for RCS in the CSP domain A and User (B) is a target for messaging in the CSP domain B.

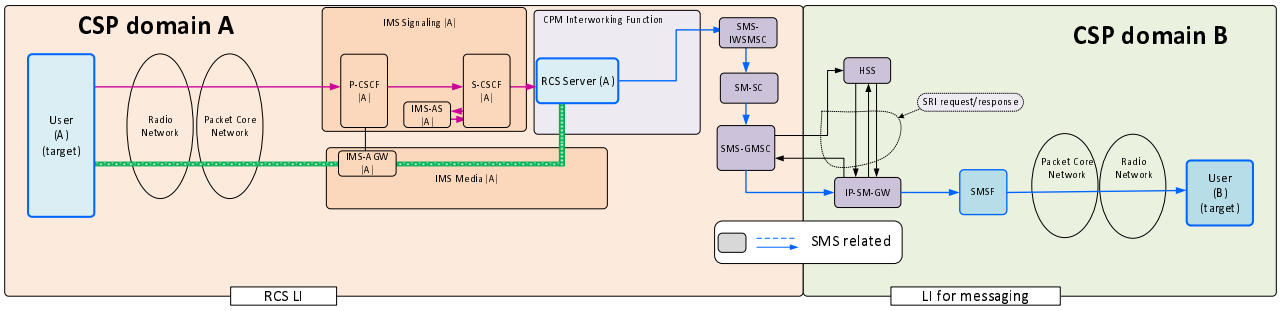


Figure 4.2.6.3.2.2-1: RCS to SMS interworking at the originating end

For RCS LI, the RCS Server (A) present in the CSP domain A provides the LI functions when User (A) is a target for RCS in CSP domain A. The RCS-Server (A) provides the IRI-POI and CC-POI functions. Since the terminating domain does not support RCS, there is no LI for RCS in the terminating domain.

For SMS LI, the SMSF present in the CSP domain B provides the LI functions when the User (B) is a target for messaging in CSP domain B. The SMSF provides the IRI-POI function.

4.2.6.3.2.3 SMS over NAS non-IP capable UE

Figure 4.2.6.3.2.3-1 illustrates a case of SMS over NAS for non-IP capable UE where User (A) is a target for RCS in the CSP domain A and User (B) is a target for messaging in the CSP domain B.

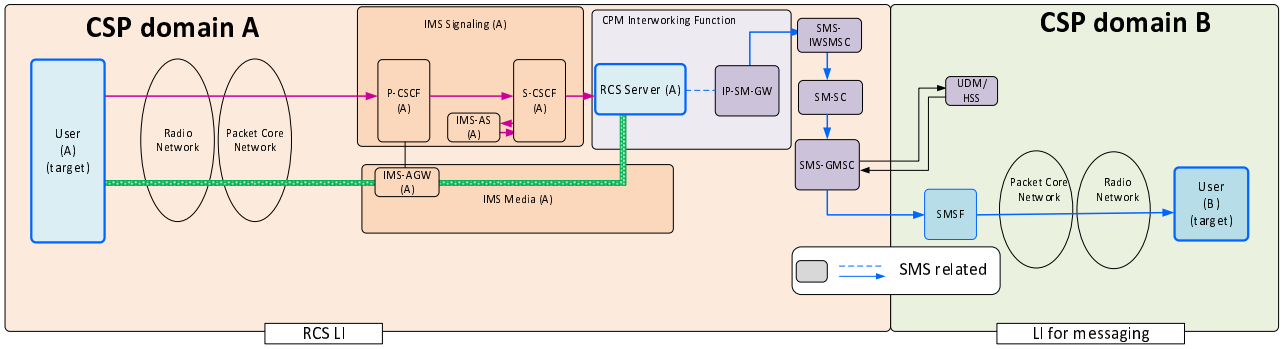


Figure 4.2.6.3.2.3-1: RCS to SMS interworking at the originating end

For RCS LI, the RCS Server (A) present in the CSP domain A provides the LI functions when User (A) is a target for RCS in CSP domain A. The RCS-Server (A) provides the IRI-POI and CC-POI functions. Since the terminating domain does not support RCS, there is no LI for RCS in the terminating domain.

For SMS LI, the SMSF present in the CSP domain B provides the LI functions when the User (B) is a target for messaging in CSP domain B. The SMSF provides the IRI-POI function.

4.2.6.3.3 Originating RCS server is in third party RCS provider domain

4.2.6.3.3.1 SMS over IP

Figure 4.2.6.3.3.1-1 illustrates a case of SMS over IP where User (A) served by CSP domain A1 is a target in the third party RCS provider domain A and User (B) is a target for messaging in CSP domain B.

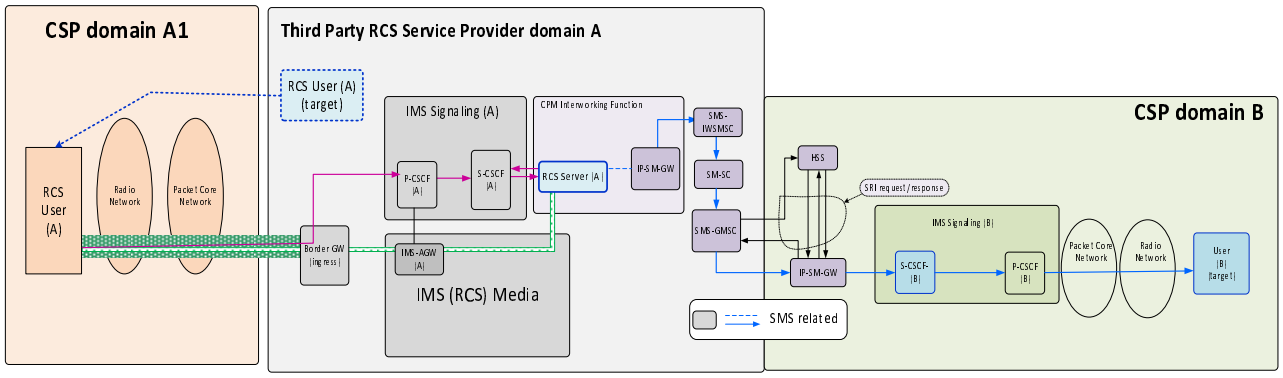


Figure 4.2.6.3.3.1-1: RCS to SMS interworking at the originating end

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target for RCS in third party RCS service provider domain A. Since the terminating domain does not support RCS, there is no LI for RCS in the terminating domain.

For SMS LI, the S-CSCF (B) present in the CSP domain B provides the LI functions when the User (B) is a target for messaging in CSP domain B. The S-CSCF (B) provides the IRI-POI function.

4.2.6.3.3.2 SMS over NAS for IP capable UE

Figure 4.2.6.3.3.2-1 illustrates a case of SMS over NAS where User (A) served by CSP domain A1 is a target in the third party RCS provider domain A and User (B) is a target for messaging in CSP domain B.

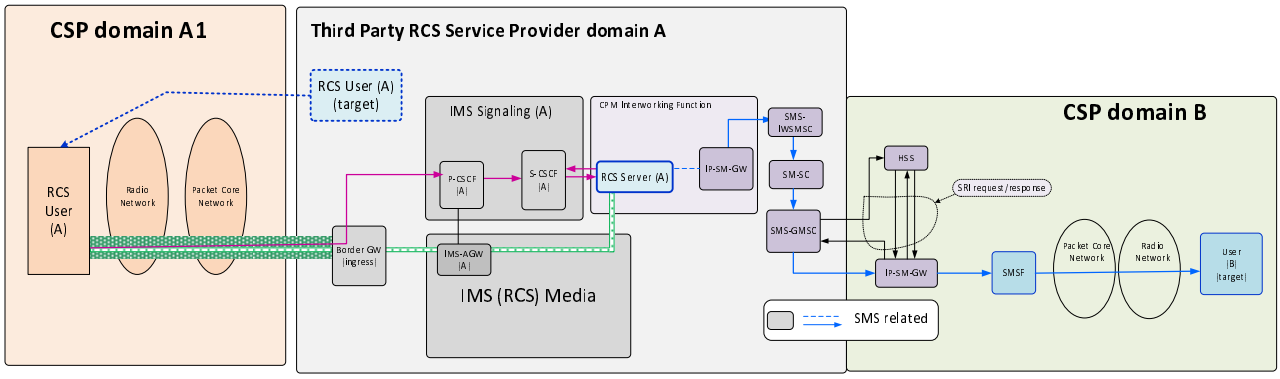


Figure 4.2.6.3.3.2-1: RCS to SMS interworking at the originating end

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target for RCS in third party RCS service provider domain A. Since the terminating domain does not support RCS, there is no LI for RCS in the terminating domain.

For SMS LI, the SMSF present in the CSP domain B provides the LI functions when the User (B) is a target for messaging in CSP domain B. The SMSF provides the IRI-POI function.

4.2.6.3.3.3 SMS over NAS for non-IP capable UE

Figure 4.2.6.3.3.3-1 illustrates a case of SMS over NAS for non-IP capable UE where User (A) served by CSP domain A1 is a target in the third party RCS provider domain A and User (B) is a target for messaging in CSP domain B.

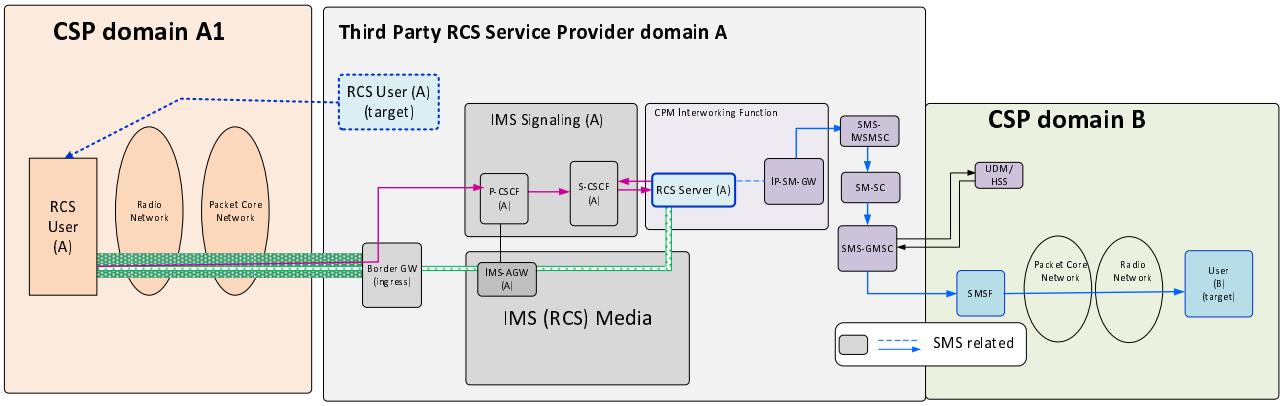


Figure 4.2.6.3.3.3-1: RCS to SMS interworking at the originating end

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target for RCS in third party RCS service provider domain A. Since the terminating domain does not support RCS, there is no LI for RCS in the terminating domain.

For SMS LI, the SMSF present in the CSP domain B provides the LI functions when the User (B) is a target for messaging in CSP domain B. The SMSF provides the IRI-POI function.

4.2.7 RCS interworking with MMS

4.2.7.1 General

The RCS message may be delivered as an MMS message to the destination when the RCS message can be delivered neither as an RCS message nor as an SMS message, but can be delivered as MMS.

As described in clause 4.2.6.1, the network function that performs the interworking from RCS to MMS and vice-versa is also referred to as Interworking Function (IWF). The present document refers to the same as CPM Interworking Function which, in part, includes the RCS Server and the MMS Proxy/Relay Server.

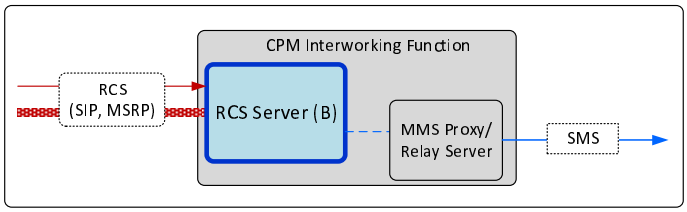


Figure 4.2.7.1-1: CPM Interworking Function for RCS to MMS interworking

For RCS to MMS interworking, the CPM Interworking Function may reside in the terminating network domain (e.g. when the terminating user is not registered for RCS) or in the originating network domain (e.g. when the terminating network does not support the RCS).

The third party RCS service provider, by definition, is an RCS service provider and therefore, the present document limits the illustrations to the case where the terminating user is served in a CSP domain. The RCS for the originating user may either be provided by the originating CSP domain or by a third party RCS service provider domain.

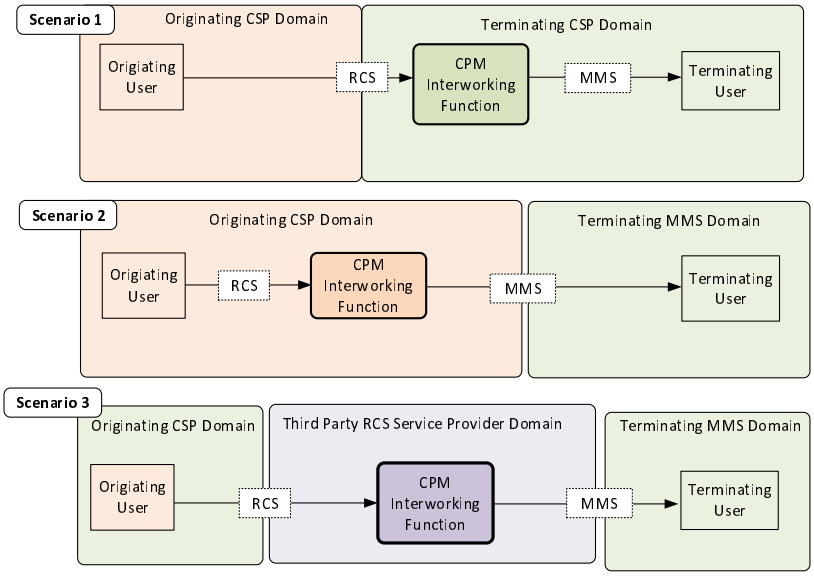


Figure 4.2.7.1-2: Scenarios of CPM Interworking Function for RCS to MMS interworking

For RCS to MMS interworking to happen, the destination should be addressed using a MSISDN or an E.164 number, both from an RCS perspective are in IMPU format.

The RCS topology scenarios illustrated in clause 4.2.3 and clause 4.2.5.2 are applicable for the originating end of the RCS handling.

The subsequent clauses illustrate the topology scenarios for RCS to MMS interworking (i.e. destination is the MMS user). The illustrations of MMS to RCS interworking (if possible) are for further study from this TR perspective.

4.2.7.2 CPM interworking function at the terminating end

4.2.7.2.1 General

This clause illustrates the scenarios of RCS to MMS interworking at the terminating end.

The scenarios illustrate the following:

- Originating RCS server is in the CSP domain.
- Originating RCS server is in third party RCS service provider domain.

The illustration starts with the assumption that two users involved are in two different domains. An illustration is shown at the end, in case two users happens to be in the same CSP domain. The exact use of this case is further study.

When the originating RCS server is in third party RCS service provider domain, the case 2 illustrated in clause 4.2.5.2 (hybrid case) are considered. However, other cases can be applicable too.

Illustrative examples of LI for messaging to support the interception of MMS to the terminating target are in TR 33.929-8 [16].

In all the illustrations, since the terminating user does not receive the RCS service, the scope of the RCS LI when the terminating user is a target for RCS may be limited (e.g. reporting only the incoming SIP messages). Alternatively, the RCS Server in the terminating end may not provide the LI functions when the terminating user is a target for RCS. Likewise, for the case of target non-local ID, the LI functions provided in the terminating domain may be limited when the originating user is a target non-local ID for RCS in the terminating domain.

4.2.7.2.2 Originating RCS user is in CSP domain

Figure 4.2.7.2.2-1 illustrates a case of RCS to MMS interworking where User (A) is a target for RCS in the CSP domain A and User (B) is a target for RCS and messaging in the CSP domain B.

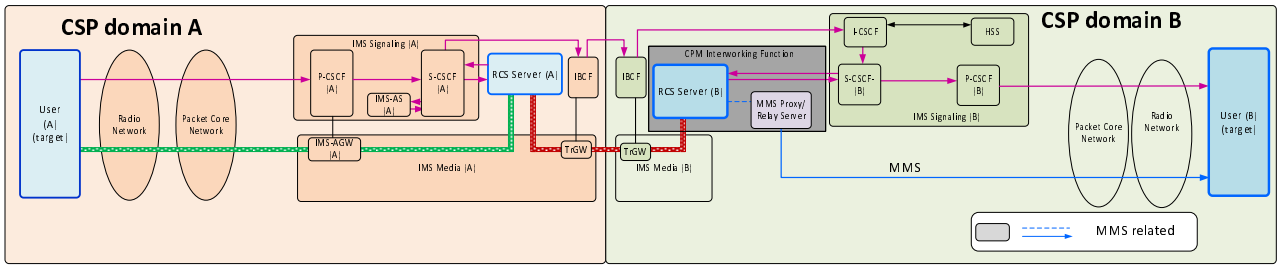


Figure 4.2.7.2.2-1: RCS to MMS interworking at the terminating end

For RCS LI, the RCS Server (A) present in the CSP domain A provides the LI functions when User (A) is a target for RCS in CSP domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (B) is a target for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case User (B) is also a target for messaging, the MMS Proxy/Relay Server present in the CSP domain B provides the LI functions. The MMS Proxy/Relay Server provides the IRI-POI and CC-POI functions as illustrated in TR 33.929-8 [16].

NOTE: Since the terminating user (User (B) in figure 4.2.7.2.2-1) does not receive the RCS service, the scope of the RCS LI when the terminating user is a target for RCS in the terminating domain (CSP domain in figure 4.2.7.2.2-1) may be limited (e.g. reporting only the incoming SIP messages). Alternatively, the RCS Server in the terminating end may not provide the LI functions when the terminating user is a target for RCS.

The above NOTE is applicable to the illustrations shown in the subsequent clauses as well.

The case of target non-local ID

Figure 4.2.7.2.2-2 illustrates a case of RCS to MMS interworking where User (A) is a target non-local ID for RCS in the CSP domain B and User (B) is a target non-local ID for RCS in the CSP domain A.

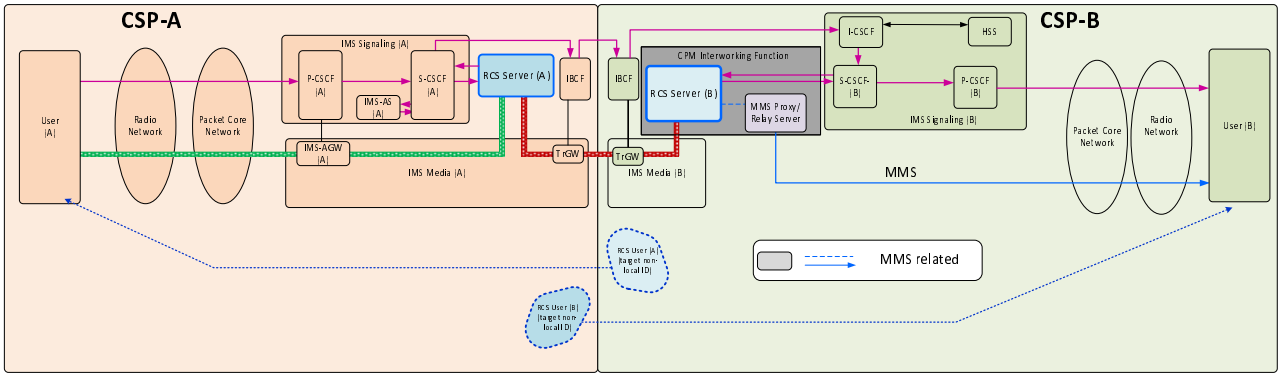


Figure 4.2.7.2.2-2: RCS to MMS interworking at the terminating end (target non-local ID)

RCS Server (A) present in the CSP domain A provides the LI functions when User (B) is a target non-local ID for RCS in CSP domain A. The RCS Server (A) provide both IRI-POI and CC-POI functions. RCS Server (B) present in the CSP domain B provides the LI functions when the User (A) is a target non-local ID in CSP domain B.

In case User (A) is target non-local ID for messaging, the MMS Proxy/Relay Server present in the CSP domain B provides the LI functions.

4.2.7.2.3 Originating RCS server is in third party RCS provider domain

Figure 4.2.7.2.3-1 illustrates a case of RCS to MMS interworking case when User (A) served by CSP domain A1 is a target in the third party RCS provider domain A and User (B) is a target for RCS in the CSP domain B.

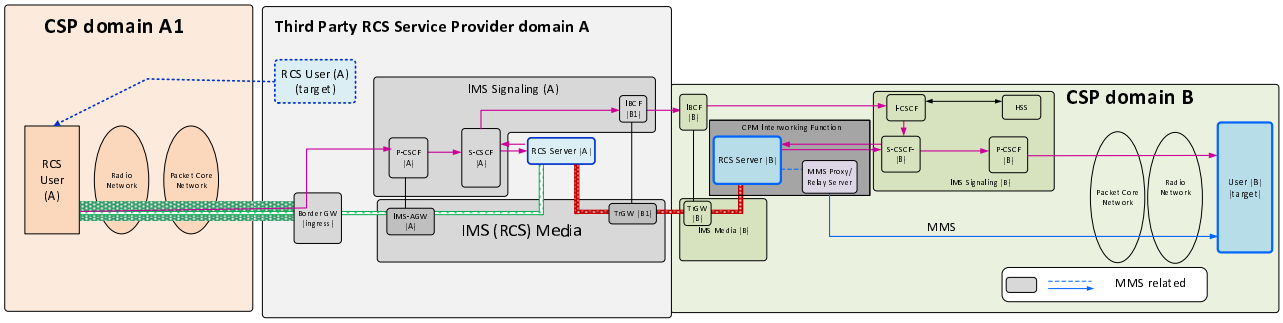


Figure 4.2.7.2.3-1: RCS to MMS interworking at the terminating end

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target for RCS in third party RCS service provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (B) is a target for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case User (B) is also a target for messaging, the MMS Proxy/Relay Server present in the CSP domain B provides the LI functions. The MMS Proxy/Relay Server provides the IRI-POI and CC-POI functions as illustrated in TR 33.929-8 [16].

The case of target non-local ID

Figure 4.2.7.2.3-2 illustrates a case of RCS to MMS interworking where User (A) is a target non-local ID in the CSP domain B for RCS and User (B) is a target non-local ID for RCS in the third party RCS provider domain A.

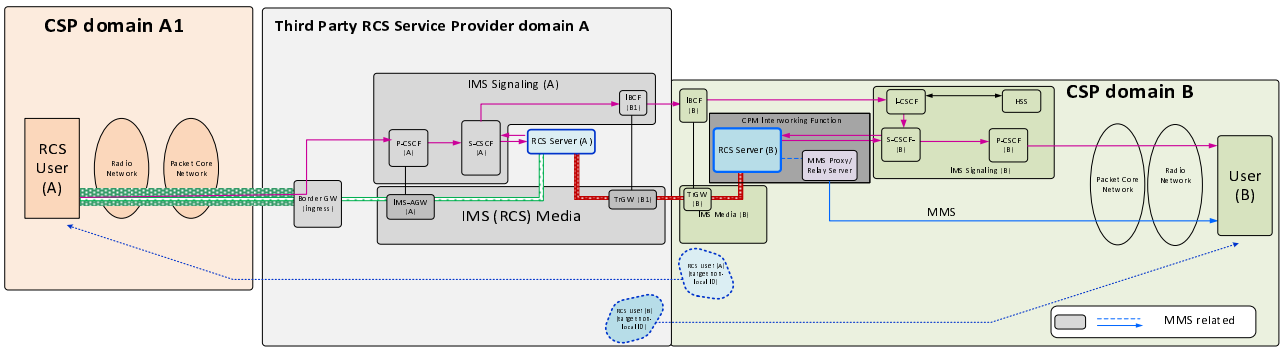


Figure 4.2.7.2.3-2: RCS to MMS interworking at the terminating end (target non-local ID)

RCS Server (A) present in the third party RCS provider domain A provides the LI functions when User (B) is a target non-local ID for RCS in third party RCS provider domain A and RCS Server (B) present in the CSP domain B provides the LI functions when the User (A) is a target non-local ID for RCS in CSP domain B. The RCS Servers provide both IRI-POI and CC-POI functions.

In case User (A) is target non-local ID for messaging, the MMS Proxy/Relay Server present in the CSP domain B provides the LI functions.

4.2.7.3 CPM interworking function at the originating end

4.2.7.3.1 General

This clause illustrates the scenarios of RCS to MMS interworking at the originating end.

The scenarios illustrate the following:

- Originating RCS server is in the CSP domain.
- Originating RCS server is in third party RCS service provider domain.

When the originating RCS server is in third party RCS service provider domain, the case 2 illustrated in clause 4.2.5.2 (hybrid case) are considered. However, other cases can be applicable too.

Since the RCS to MMS interworking happens in the originating domain, the terminating domain is shown as MMS domain rather than a CSP domain. The illustrative examples of MMS LI are shown in TR 33.929-8 [16].

4.2.7.3.2 Originating RCS user is in CSP domain

Figure 4.2.7.3.2-1 illustrates a case of RCS to MMS interworking where User (A) is a target for RCS in the CSP domain A and User (B) is a target for messaging (MMS) and messaging in the MMS domain B.

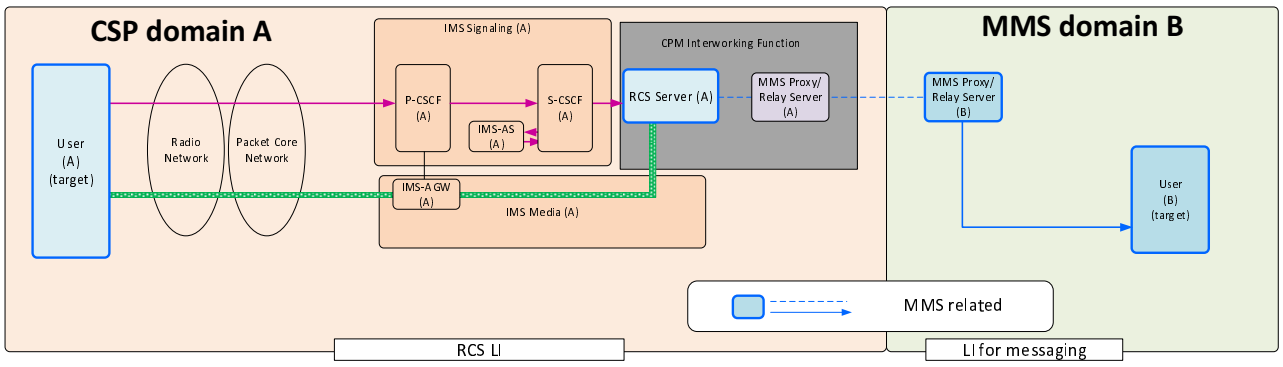


Figure 4.2.7.3.2-1: RCS to MMS interworking at the originating end

For RCS LI, the RCS Server (A) present in the CSP domain A provides the LI functions when User (A) is a target for RCS in CSP domain A. The RCS Server provides both IRI-POI and CC-POI functions. Since the terminating domain does not support, there is no LI for RCS in the terminating domain.

For MMS LI, the MMS Proxy/Relay Server (B) present in CSP domain B provides the LI functions when User (B) is a target for messaging in MMS domain B. MMS Proxy/Relay Server provides the IRI-POI and CC-POI functions.

4.2.7.3.3 Originating RCS server is in third party RCS provider domain

Figure 4.2.7.3.3-1 illustrates a case of RCS to MMS interworking when User (A) served by CSP domain A1 is a target in the third party RCS provider domain A and User (B) is a target for messaging in the CSP domain B.

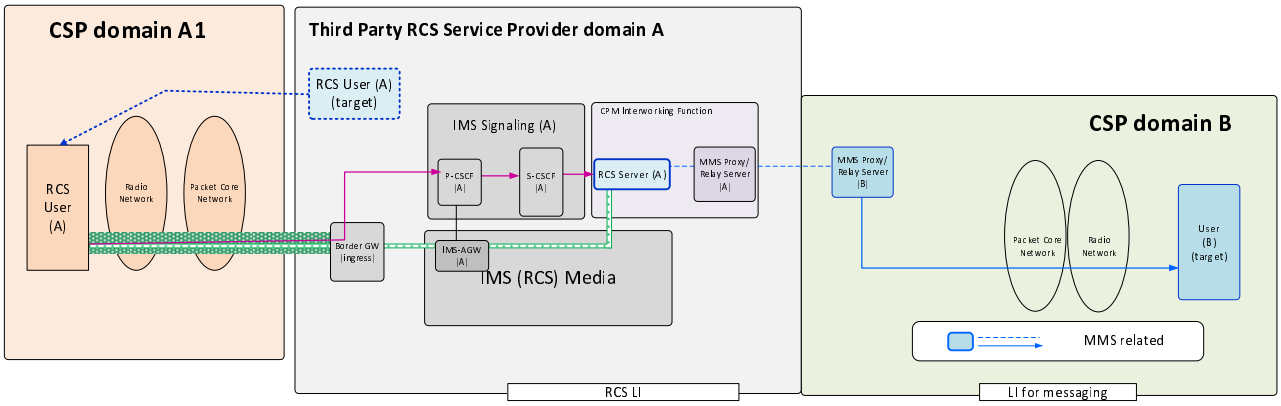


Figure 4.2.7.3.3-1: RCS to MMS interworking at the originating end

The RCS Server (A) present in the third party RCS service provider domain A provides the LI functions when User (A) is a target for RCS in third party RCS service provider domain A. The RCS Server provides both IRI-POI and CC-POI functions.

For MMS LI, the MMS Proxy/Relay Server (B) present in CSP domain B provides the LI functions when User (B) is a target for messaging in MMS domain B. MMS Proxy/Relay Server provides the IRI-POI and CC-POI functions.

4.2.8 File transfer via content server

4.2.8.1 General

The subsequent clauses illustrate the topology scenarios related to file transfer via content server between the two RCS users. To transfer a file the sender uploads the file to a server and then message (via the RCS servers) the recipient (s) with the location of the file at the content server which would allow the recipient (s) to retrieve the file from the content server.

The content server that stores the uploaded file is referred to as HTTP Content Server. Before the file is downloaded by the recipient present in a different domain, the file may be retrieved and stored at a local server referred to as s File Localization Function. In the illustrations, it is assumed that the HTTP Content Server is in the same domain as the originating RCS Server while the File Localization Function is in the same domain as the terminating RCS Server. When the two users are in the same domain (and hence, the two servers), the need for a File Localization Function does not arise.

When the information regarding the uploaded file location (an URL) is sent to the recipient using the SIP messages, those SIP messages transfer through the RCS Server.

The RCS servers may be present in one CSP domain or different CSP domains (clause 4.2.3) , one third party RCS provider domain or different RCS provider domains (clause 4.2.3), or the hybrid cases (clause 4.2.4). A few of those scenarios are repeated in the subsequent clauses with the addition of HTTP Content Server and the File Localization Function:

- File transfer between two users in one domain.
- File transfer between two users in two domains with a File Localization Function.
- File transfer between two users in two domains without a File Localization Function.

The case of RCS servers and Content Servers in CSP domain and third party RCS service provider domain are considered. When two domains are involved the case of target non-local ID is also considered.

NOTE: Whether the RCS to SMS/MMS interworking would apply for the file transfer related procedures is for further study.

4.2.8.2 File transfer between two users in one CSP domain

Figure 4.2.8.2-1 illustrates the topology for the case when two RCS Users and hence, the two RCS servers are in the same CSP domain. Since the two users are in the same CSP domain, the File Localization Function is not used here. In this case, User (A) is a target for RCS at RCS server (A) and User (B) is a target for RCS at RCS Server (B).

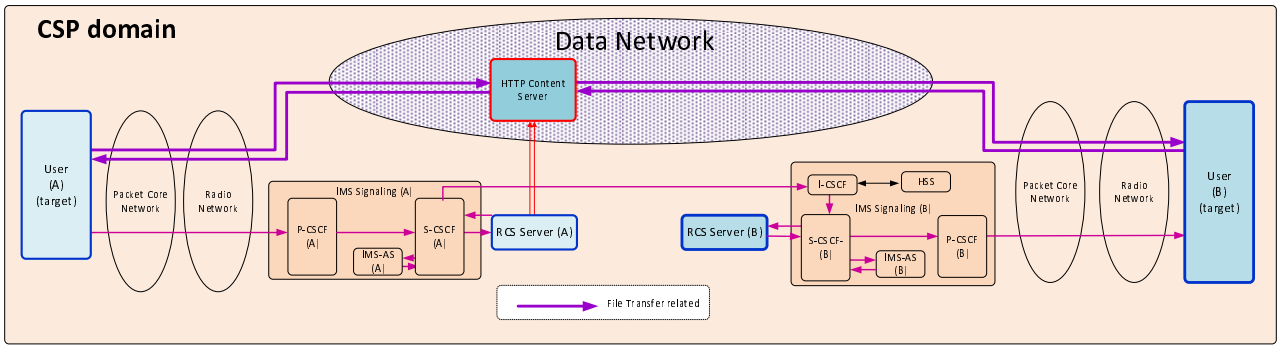


Figure 4.2.8.2-1: File transfer between users in the same CSP domain

The RCS Server (A) provides the LI functions when User (A) is a target for RCS and the RCS Server (B) providers the LI functions when User (B) is a target for RCS. For file transfer related functions, the RCS servers provide the IRI-POI functions.

When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/download related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL.

The HTTP Content Server may also provide the LI functions when the identity of the RCS User (A) received from RCS User (A) during file upload related authentication procedure is target for RCS. Likewise, HTTP Content Server may also provide the LI functions when the identity of the RCS User (B) received from RCS User (B) during file download related authentication procedure is target for RCS.

4.2.8.3 File transfer between two users in one third party RCS service provider domain

When the RCS server is in third party RCS service provider domain, all the topologies illustrated in clause 4.2.4 are applicable.

Only one of those topology cases is used for the illustration here.

Figure 4.2.8.3-1 illustrates the topology for the case when two RCS Users use the same third party RCS service provider. Therefore, in this illustration, the two RCS servers are in the same third party RCS service provider domain. Since the two users use the same third party RCS service provider domain, the File Localization Function is not used here. In this case, User (A), a subscriber of CSP domain A1 is a target for RCS in third party RCS service provider domain and User (B) a subscriber of CSP domain B1 is a target for RCS in third party RCS service provider.

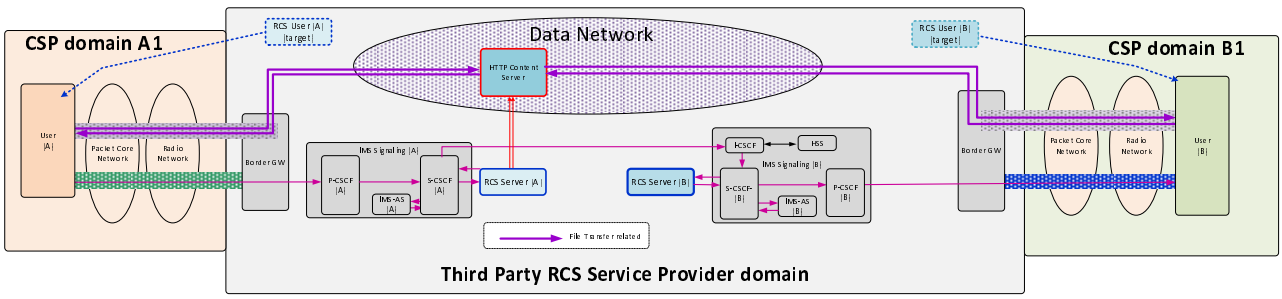


Figure 4.2.8.3-1: File transfer between users in the same third party domain

The RCS Server (A) provides the LI functions when User (A) is a target for RCS and the RCS Server (B) providers the LI functions when User (B) is a target for RCS. For file transfer related functions, the RCS servers provide the IRI-POI functions.

When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/download related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL.

The HTTP Content Server may also provide the LI functions when the identity of the RCS User (A) received from RCS User (A) during file upload related authentication procedure is target for RCS. Likewise, HTTP Content Server may also provide the LI functions when the identity of the RCS User (B) received from RCS User (B) during file download related authentication procedure is target for RCS.

4.2.8.4 File transfer between two users in two CSP domains with a File Localization Function

Figure 4.2.8.4-1 illustrates the topology for the case when two RCS Users and hence, the two RCS servers are in the different CSP domains. In this illustration, a File Localization Function is present in the terminating CSP domain. In this case, User (A) is a target for RCS at RCS server (A) and User (B) is a target for RCS at RCS Server (B).

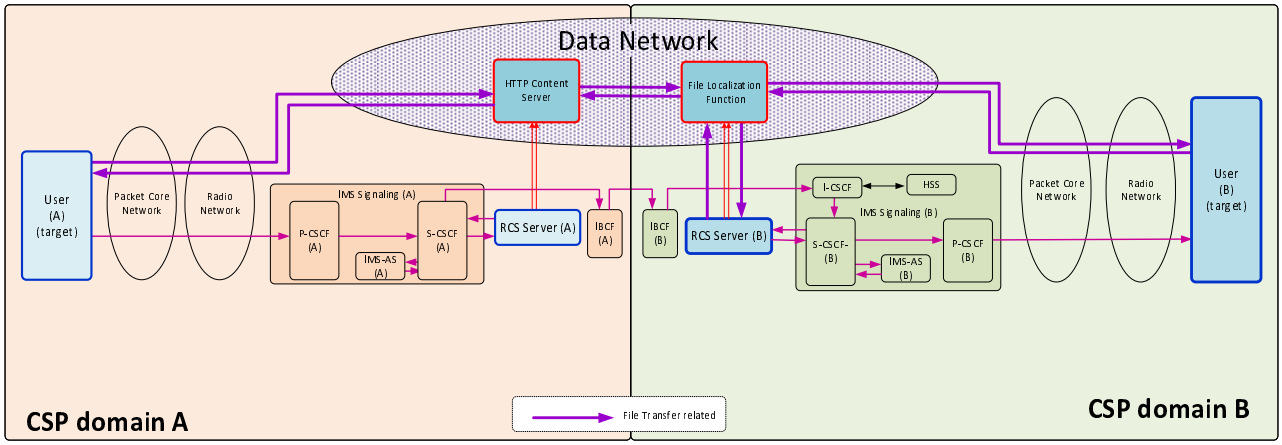


Figure 4.2.8.4-1: File transfer between users in the different CSP domains with file localization function

The RCS Server (A) provides the LI functions when User (A) is a target for RCS and the RCS Server (B) providers the LI functions when User (B) is a target for RCS. For file transfer related functions, the RCS servers provide the IRI-POI functions.

When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/retrieval related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL. Likewise, when the file is retrieved from the HTTP Content Server or when the downloaded by an RCS user, the File Localization Function provides the LI functions for the file retrieval/download related events. Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL even though the File Localization Function would store the file locally at a different URL.

The HTTP Content Server may also provide the LI functions when the identity of the RCS User (A) received from RCS User (A) during file upload related authentication procedure is target for RCS.

The File Localization Function may also provide the LI functions when the identity of the RCS User (B) received from RCS User (B) during file download related authentication procedure is target for RCS.

Target non-local ID

Figure 4.2.8.4-2 illustrates the topology for the case when two RCS Users and hence, the two RCS servers are in the different CSP domains. In this illustration, a File Localization Function is present in the terminating CSP domain. In

this case User (A) is a target non-local ID for RCS at RCS server (B) and User (B) is a target non-local ID for RCS at RCS Server (A).

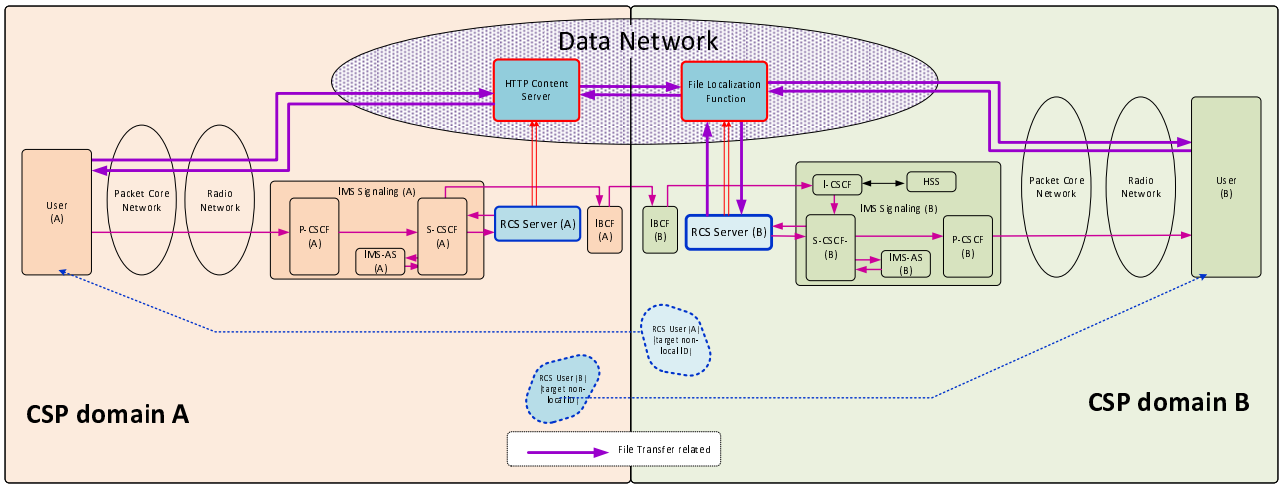


Figure 4.2.8.4-2: File transfer between users in the different CSP domains with file focalization function (target non-local ID)

The RCS Server (A) provides the LI functions when User (B) is a target non-local ID for RCS in CSP domain A and the RCS Server (B) providers the LI functions when User (A) is a target non-local ID for RCS in CSP domain B. For file transfer related functions, the RCS servers provide the IRI-POI functions. When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/retrieval related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL. Likewise, when the file is retrieved from the HTTP Content Server or when the downloaded by an RCS user, the File Localization Function provides the LI functions for the file retrieval/download related events. Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL even though the File Localization Function would store the file locally at a different URL.

4.2.8.5 File transfer between two users in two third party domains with a File Localization Function

When the RCS server is in third party RCS service provider domain, all the topologies illustrated in clause 4.2.4 are applicable. Hybrid cases illustrated in clause 4.2.5 are applicable as well.

Only one of those topology cases is used for the illustration here.

Figure 4.2.8.5-1 illustrates the topology for the case when two RCS Users use different third party RCS service providers. In this illustration, the two RCS servers are in the different third party RCS service provider domains. Furthermore, in this illustration, a File Localization Function is present in the terminating third party RCS service provider domain. In this case, User (A), a subscriber of CSP domain A1 is a target for RCS in third party RCS service provider domain A and User (B) a subscriber of CSP domain B1 is a target for RCS in third party RCS service provider B.

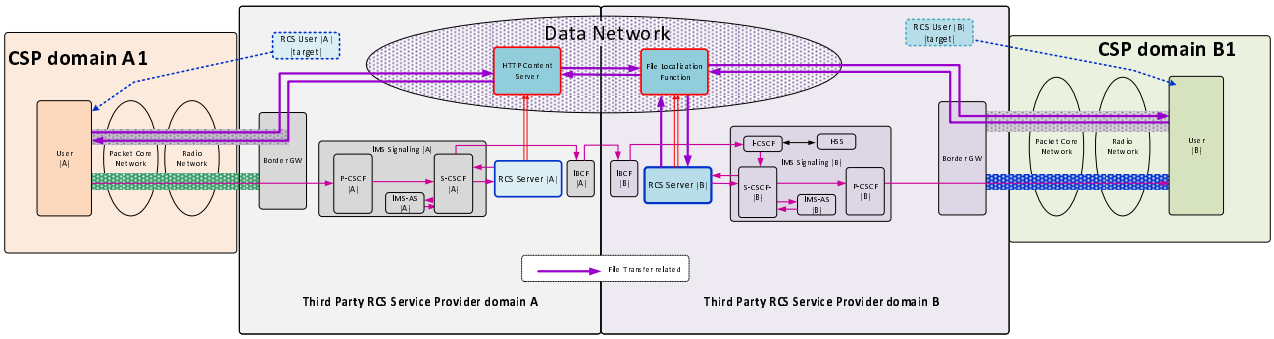


Figure 4.2.8.5-1: File transfer between users in the different third party domains with file localization function

The RCS Server (A) provides the LI functions when User (A) is a target for RCS in third party RCS provider domain A and the RCS Server (B) providers the LI functions when User (B) is a target for RCS third party RCS service provider B. For file transfer related functions, the RCS servers provide the IRI-POI functions.

When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/retrieval related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL. Likewise, when the file is retrieved from the HTTP Content Server or when the downloaded by an RCS user, the File Localization Function provides the LI functions for the file retrieval/download related events. Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL even though the File Localization Function would store the file locally at a different URL.

The HTTP Content Server may also provide the LI functions when the identity of the RCS User (A) received from RCS User (A) during file upload related authentication procedure is target for RCS. Likewise, File Localization Function may also provide the LI functions when the identity of the RCS User (B) received from RCS User (B) during file download related authentication procedure is target for RCS.

Target non-local ID

Figure 4.2.8.5-2 illustrates the topology for the case when two RCS Users use different third party RCS service providers. In this illustration, the two RCS servers are in the different third party RCS service provider domains. Furthermore, in this illustration, a File Localization Function is present in the third party RCS service provider domain B. In this case, User (A), a subscriber of CSP domain A1 is a target non-local ID for RCS in third party RCS service provider domain B and User (B) a subscriber of CSP domain B1 is a target non-local ID for RCS in third party RCS service provider B.

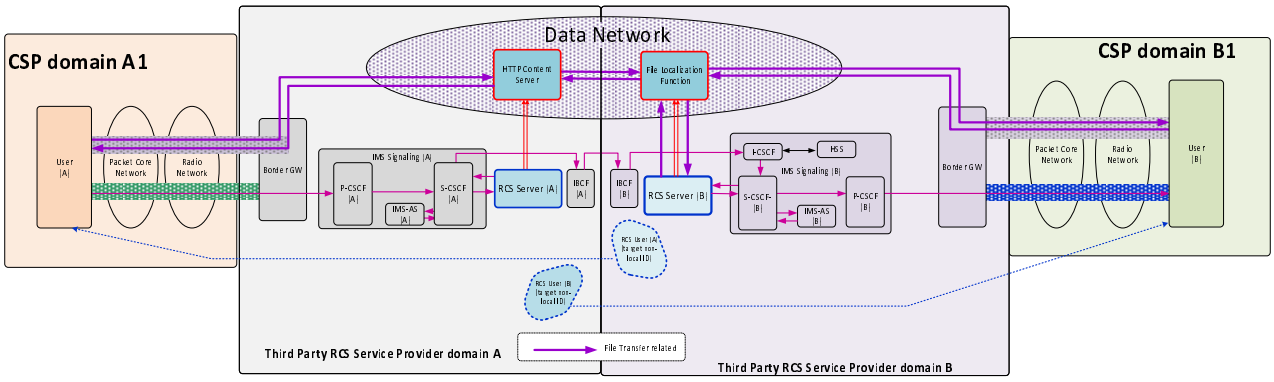


Figure 4.2.8.5-2: File transfer between users in the different third party domains with file focalization function (target non-local ID)

The RCS Server (A) provides the LI functions when User (B) is a target non-local ID for RCS in third party RCS service provider domain A and the RCS Server (B) providers the LI functions when User (A) is a target non-local ID for RCS in third party RCS service provider domain B. For file transfer related functions, the RCS servers provide the IRI-POI functions.

When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/retrieval related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL. Likewise, when the file is retrieved from the HTTP Content Server or when the downloaded by an RCS user, the File Localization Function provides the LI functions for the file retrieval/download related events. Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL even though the File Localization Function would store the file locally at a different URL.

4.2.8.6 File transfer between two users in two CSP domains without a File Localization Function

Figure 4.2.8.6-1 illustrates the topology for the case when two RCS Users and hence, the two RCS servers are in the different CSP domains. In this illustration, a File Localization Function is not present in the terminating CSP domain. In this case, User (A) is a target for RCS at RCS server (A) and User (B) is a target for RCS at RCS Server (B).

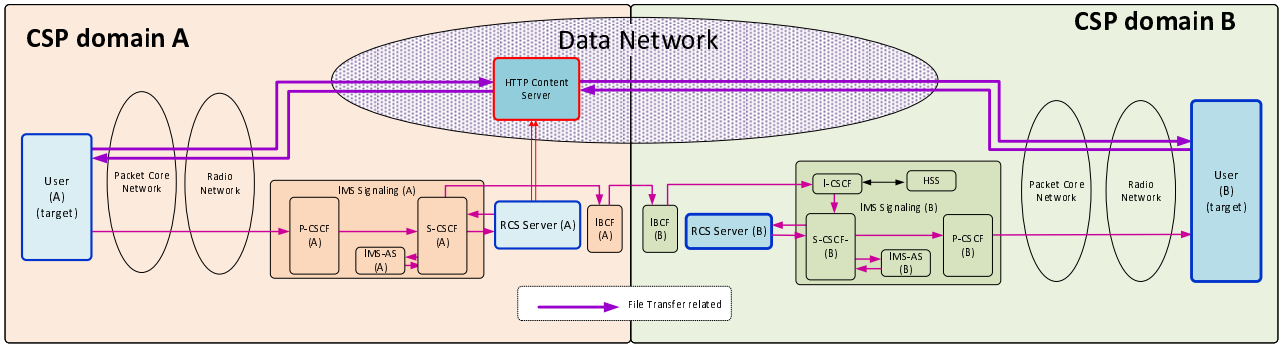


Figure 4.2.8.6-1: File transfer between users in the different CSP domains without a File Localization Function

The RCS Server (A) provides the LI functions when User (A) is a target for RCS and the RCS Server (B) providers the LI functions when User (B) is a target for RCS. For file transfer related functions, the RCS servers provide the IRI-POI functions.

When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/retrieval related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL.

The HTTP Content Server may also provide the LI functions when the identity of the RCS User (A) received from RCS User (A) during file upload related authentication procedure is target for RCS.

Target non-local ID

Figure 4.2.8.6-2 illustrates the topology for the case when two RCS Users and hence, the two RCS servers are in the different CSP domains. In this illustration, a File Localization Function is not present in the terminating CSP domain. In this case User (A) is a target non-local ID for RCS at RCS server (B) and User (B) is a target non-local ID for RCS at RCS Server (A).

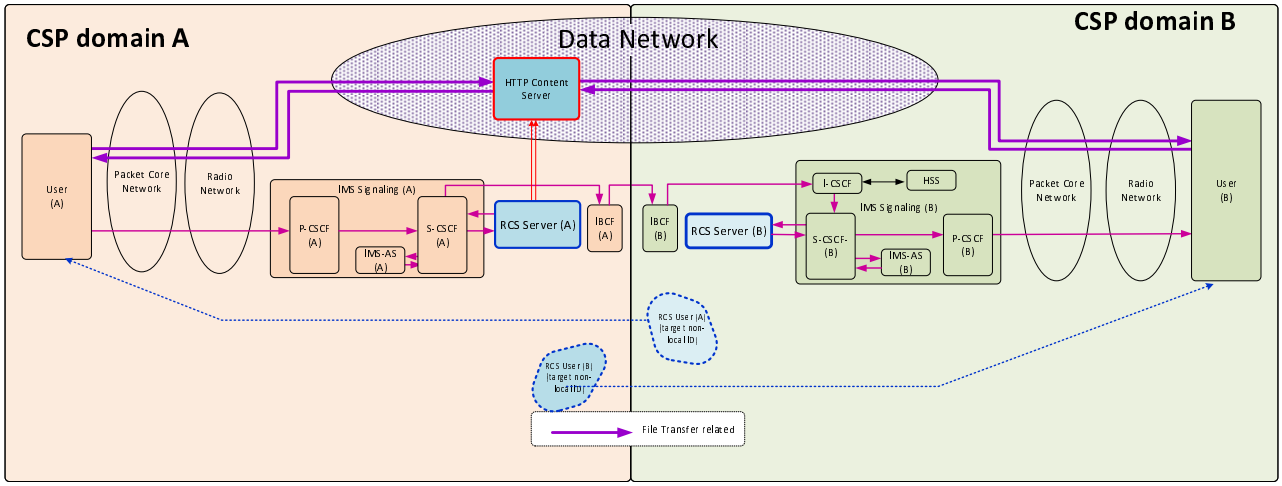


Figure 4.2.8.6-2: File transfer between users in the different CSP domains without a File Localization Function (target non-local ID)

The RCS Server (A) provides the LI functions when User (B) is a target for RCS in CSP domain A and the RCS Server (B) providers the LI functions when User (A) is a target for RCS in CSP domain B. For file transfer related functions, the RCS servers provide the IRI-POI functions.

When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/retrieval related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL.

4.2.8.7 File transfer between two users in two third party domains without a File Localization Function

When the RCS server is in third party RCS service provider domain, all the topologies illustrated in clause 4.2.4 are applicable. Hybrid cases illustrated in clause 4.2.5 are applicable as well.

Only one of those topology cases is used for the illustration here.

Figure 4.2.8.7-1 illustrates the topology for the case when two RCS Users use different third party RCS service providers. In this illustration, the two RCS servers are in the different third party RCS service provider domains. Furthermore, in this illustration, a File Localization Function is not present in the terminating third party RCS service provider domain. In this case, User (A), a subscriber of CSP domain A1 is a target for RCS in third party RCS service provider domain A and User (B) a subscriber of CSP domain B1 is a target for RCS in third party RCS service provider B.

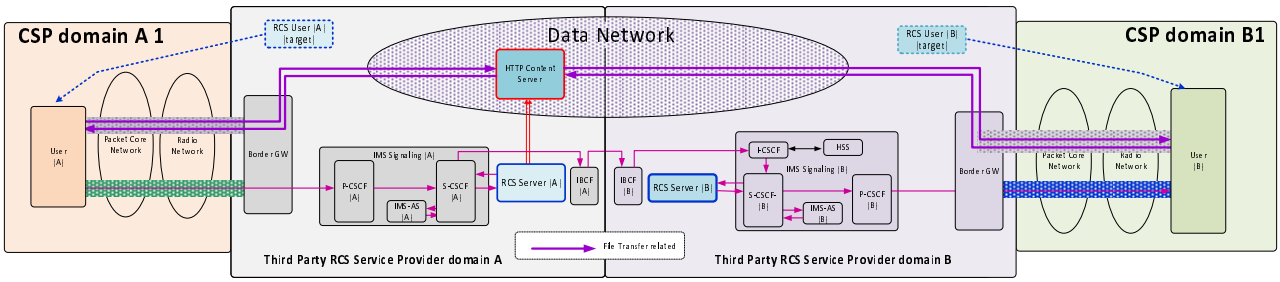


Figure 4.2.8.7-1: File transfer between users in the different third party domains without a File Localization Function

The RCS Server (A) provides the LI functions when User (A) is a target for RCS in third party RCS service provider domain A and the RCS Server (B) providers the LI functions when User (B) is a target for RCS in third party RCS service provider domain B. For file transfer related functions, the RCS servers provide the IRI-POI functions.

When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/retrieval related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include

the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL.

The HTTP Content Server may also provide the LI functions when the identity of the RCS User (A) received from RCS User (A) during file upload related authentication procedure is target for RCS.

Target non-local ID

Figure 4.2.8.7-2 illustrates the topology for the case when two RCS Users use different third party RCS service providers. In this illustration, the two RCS servers are in the different third party RCS service provider domains. Furthermore, in this illustration, a File Localization Function is not present in the third party RCS service provider domain B. In this case, User (A), a subscriber of CSP domain A1 is a target non-local ID for RCS in third party RCS service provider domain B and User (B) a subscriber of CSP domain B1 is a target non-local ID for RCS in third party RCS service provider B.

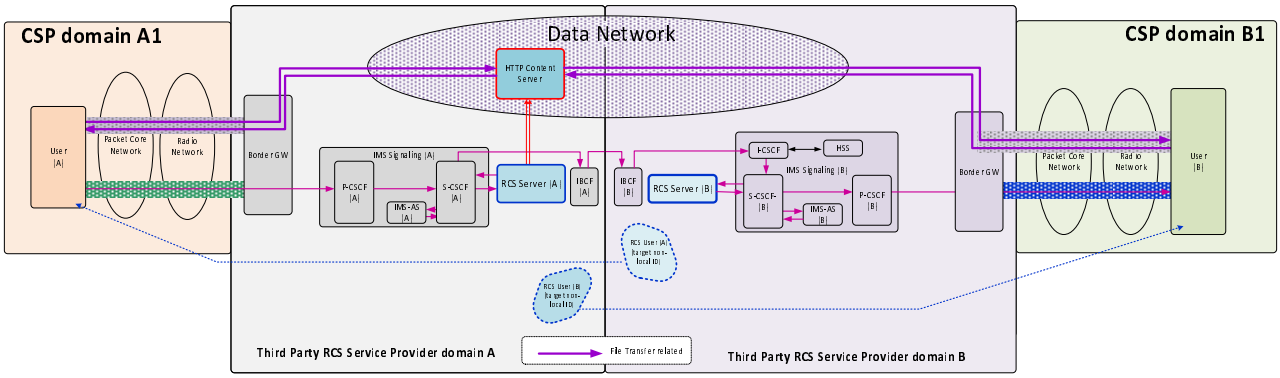


Figure 4.2.8.7-2: File transfer between users in the different third party domains without a File Localization Function (target non-local ID)

The RCS Server (A) provides the LI functions when User (B) is a target for RCS in third party RCS service provider domain A and the RCS Server (B) providers the LI functions when User (A) is a target for RCS in third party RCS service provider domain B. For file transfer related functions, the RCS servers provide the IRI-POI functions.

When the file is uploaded or retrieved, HTTP Content Server provides the LI functions for file upload/retrieval related events based on the trigger received from the IRI-TF and CC-TF present in the RCS Server (A). Those triggers include the location of the file (an URL) stored at the HTTP Content Server. Therefore, the LI functions provided for this case are purely based on the URL.

4.3 Steps of file transfer via content server

4.3.1 Background

4.3.1.1 General

The subsequent clauses illustrate the steps involved during the file transfer via content server procedure between the two RCS users.

To transfer a file the sender uploads the file to a server (referred to as the HTTP Content Server), receives the URL where the file was stored from the HTTP Content Server and sends the URL to recipient using a SIP message through the RCS Servers.

Using the received URL, the recipient downloads the file from the HTTP Content Server. When the recipient is a different domain, the file may be retrieved and stored at a local server referred to as s File Localization Function. The recipient downloads the file from the File Localization Function.

The transfer of the SIP message carrying the URL may happen independent of whether, or not, the RCS users are in an RCS session.

The illustrations shown in clause 4.2 provide various topology scenarios for the RCS LI. The illustrations shown in clause 4.2.8 provide a few scenarios that include the HTTP Content Server and File Localization Function. The illustrations shown in the following clauses mask out those topology details to avoid repetition in the illustrations. Focus here is more on the file-transfer related steps.

In all the illustration, it is assumed that the HTTP Content Server is in the same domain as the originating RCS Server while the File Localization Function is in the same domain as the terminating RCS Server. When the two users are in the same domain (and hence, the two servers), the need for a File Localization Function does not arise.

Furthermore, it is assumed that the RCS users are authenticated when a file is uploaded or retrieved for downloading. The HTTP Content Server would authenticate the File Localization Function before the latter can retrieve the file.

4.3.1.2 File transfer in the same domain

Figure 4.3.1.2-1 illustrates the steps involved during the file transfer via content server procedure when two RCS Users and hence, the two RCS Servers are in the same domain. Since the two RCS users are served by the same RCS domain, the File Localization Function is not used here. In the illustration the RCS User (A) sends the file to RCS User (B).

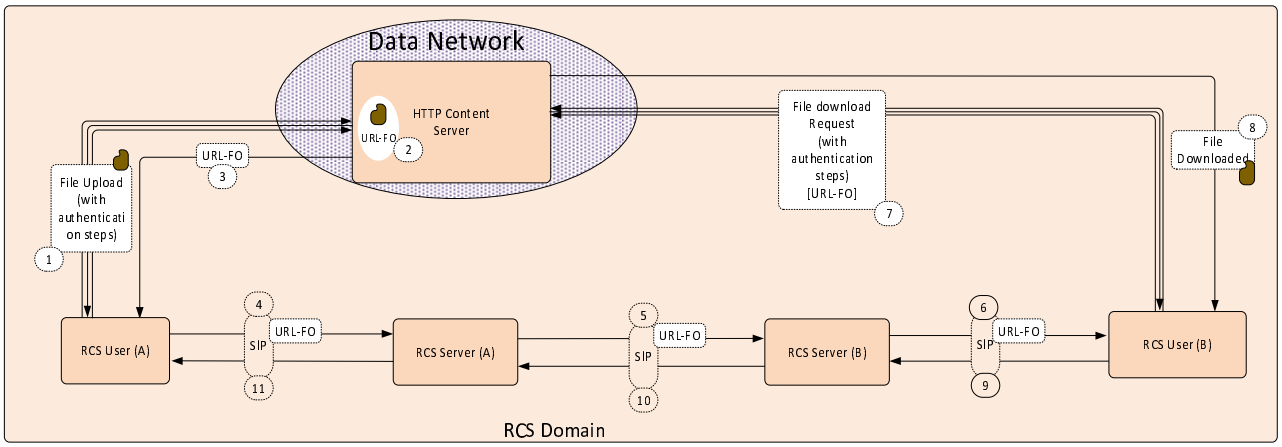


Figure 4.3.1.2-1: File transfer RCS servers in the same domain

Figure 4.3.1.2-1 shows the following file-transfer steps:

1. RCS User (A) uploads the file to the HTTP Content Server. The authentication procedure to perform RCS User (A) authorization to upload the file steps happens here.
2. HTTP Content Server stores the file at the URL shown as URL-FO.
3. HTTP Content Server provides the location where the file is stored (URL-FO) to the RCS User (A).
4. RCS User (A) sends the URL-FO to the RCS Server (A) in a SIP message (note that the signalling passes through the IMS signalling functions).
5. RCS Server (A) forwards the URL-FO to the RCS Server (B) in a SIP message (note that the signalling passes through the IMS signalling functions).
6. RCS Server (B) forwards the URL-FO to the RCS User (B) in a SIP message (note that the signalling passes through the IMS signalling functions).
7. Using URL-FO, the RCS User (B) reaches the HTTP Content Server to download the file. The authentication procedure to perform RCS User (B) authorization to download the file happens here.
8. File is downloaded to RCS User (B).
9. SIP response to 6.
10. SIP response to 5.
11. SIP response to 4.

As illustrated in clause 4.2.8 with various topology scenarios, the RCS Server (A) and RCS Server (B) may be in CSP domain or in a third party RCS service provider domain. In the latter case, the RCS users may also not belong to any CSP domain (see clause 4.2.4).

4.3.1.3 File transfer with the file localization function

Figure 4.3.1.3-1 illustrates the steps involved during the file transfer via content server procedure when the two RCS users are in different RCS domains. Here, a File Localization Function is present at the recipient's end. In the illustration, RCS User (A) sends the file to RCS User (B).

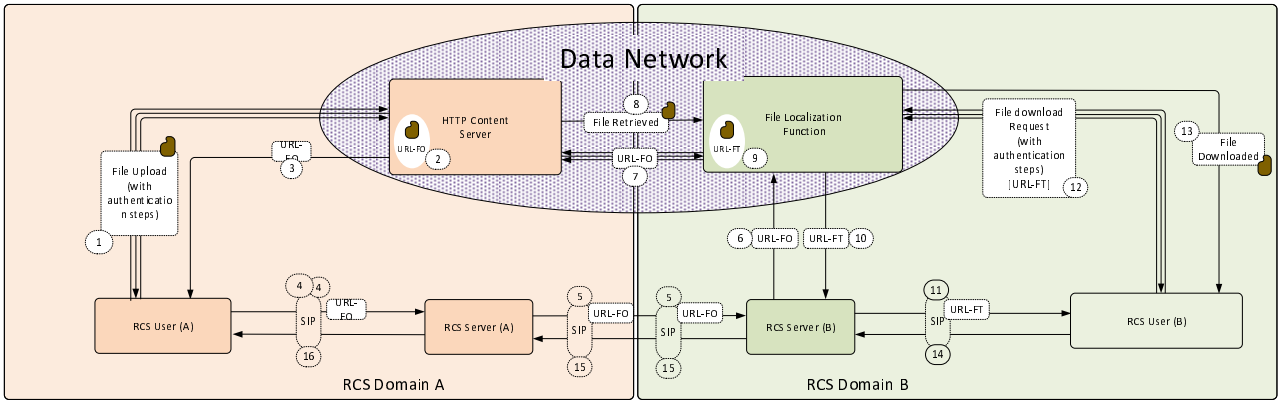


Figure 4.3.1.3-1: File transfer with a file localization function

Figure 4.3.1.3-1 shows the following file-transfer steps:

1. RCS User (A) uploads the file to the HTTP Content Server. The authentication procedure to perform RCS User (A) authorization to upload the file happens here.
2. HTTP Content Server stores the file at the URL shown as URL-FO.
3. HTTP Content Server provides the location where the file is stored (URL-FO) to the RCS User (A).
4. RCS User (A) sends the URL-FO to the RCS Server (A) in a SIP message (note that the signalling passes through the IMS signalling functions).
5. RCS Server (A) forwards the URL-FO to the RCS Server (B) in a SIP message (note that the signalling passes through the IMS signalling functions).
6. RCS Server (B) provides the URL-FO to the File Localization Function to retrieve the file HTTP Content Server.
7. Using URL-FO, the File Localization Function reaches the HTTP Content Server to retrieve the file. The authentication procedure to perform File Localization Function authorization to retrieve the file happens here.
8. File is retrieved by the File Localization Function.
9. File Localization Function stores the retrieved file at the URL shown as URL-FT.
10. File Localization Function provides the location where the file is stored (URL-FT) to the RCS Server (B).
11. RCS Server (B) forwards the URL-FT to the RCS User (B) in a SIP message (note that the signalling passes through the IMS signalling functions).
12. Using URL-FT, the RCS User (B) reaches the File Localization Function to download the file. The authentication procedure to perform RCS User (B) authorization to download the file happens here.
13. File is downloaded to RCS User (B).
14. SIP response to 11.
15. SIP response to 5.

16. SIP response to 4.

As illustrated in clause 4.2.8 with various topology scenarios, the RCS Server (A) and RCS Server (B) may be in CSP domains or in third party RCS service provider domains. In the latter case, the RCS users may also not belong to any CSP domain. A hybrid case (one RCS Server in CSP domain and the other in a third party RCS service provider domain) is also possible.

4.3.1.4 File transfer without the file localization function

Figure 4.3.1.4-1 illustrates the steps involved during the file transfer via content server procedure when the two RCS users are in different RCS domains. Here, a File Localization Function is not present at the recipient's end. In the illustration, RCS User (A) sends the file to RCS User (B).

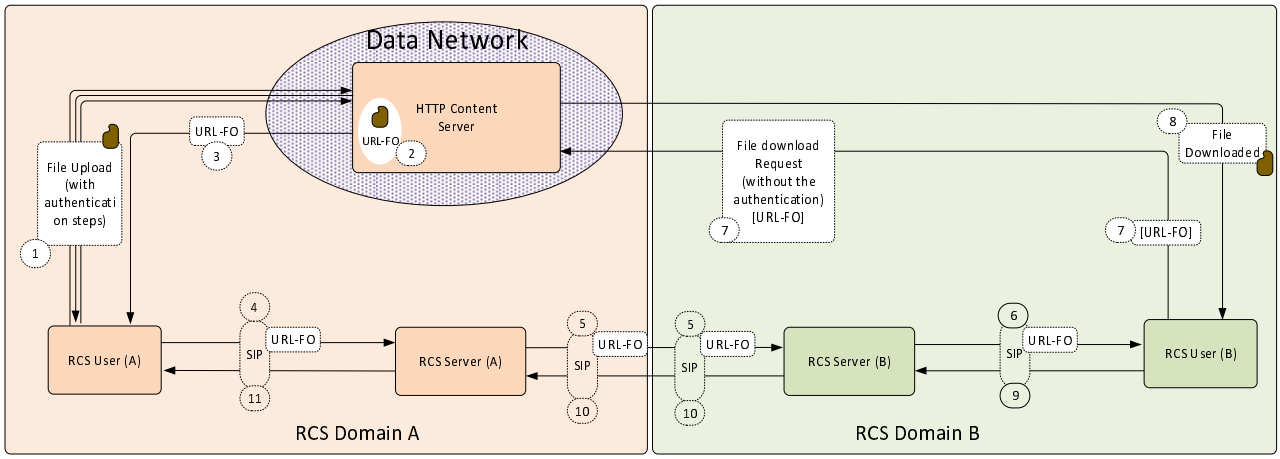


Figure 4.3.1.4-1: File transfer with a file localization function

1. RCS User (A) uploads the file to the HTTP Content Server. The authentication procedure to perform RCS User (A) authorization to upload the file steps happens here.
2. HTTP Content Server stores the file at the URL shown as URL-FO.
3. HTTP Content Server provides the location where the file is stored (URL-FO) to the RCS User (A).
4. RCS User (A) sends the URL-FO to the RCS Server (A) in a SIP message (note that the signalling passes through the IMS signalling functions).
5. RCS Server (A) forwards the URL-FO to the RCS Server (B) in a SIP message (note that the signalling passes through the IMS signalling functions).
6. RCS Server (B) forwards the URL-FO to the RCS User (B) in a SIP message (note that the signalling passes through the IMS signalling functions).
7. Using URL-FO, the RCS User (B) reaches the HTTP Content Server to download the file. No authentication procedure here, since RCS User (B) is in a different domain.
8. File is downloaded to RCS User (B).
9. SIP response to 6.
10. SIP response to 5.
11. SIP response to 4.

As illustrated in clause 4.2.8 with various topology scenarios, the RCS Server (A) and RCS Server (B) may be in CSP domains or in third party RCS service provider domains. In the latter case, the RCS users may also not belong to any CSP domain. A hybrid case (one RCS Server in CSP domain and the other in a third party RCS service provider domain) is also possible.

4.3.2 LI for file transfer

4.3.2.1 General

For file transfer procedures, the LI functions are provided by the RCS Server, HTTP Content Server (at the originating end) and File Localization Function (at the terminating end).

Various topologies concerning the LI functions at the RCS Servers are illustrated in clause 4.2. The same topologies would apply for the file transfer related procedures as well. The RCS Servers would provide IRI-POI, IRI-TF and CC-TF functions. The IRI-POI present in the RCS Servers report the SIP messages used during the file transfer steps. As illustrated in clause 4.2.8, the target non-local ID can apply to the LI functions provided by the RCS Servers when two domains are used.

The TS 33.127 [3] and TS 33.128 [4] defines the LI functions provided at the HTTP Content Server and File Localization Function for file transfer related procedures only at the architectural level. As illustrated in clause 4.1.3, the HTPP Content Server and the File Localization Function (when present) have the LI functions which include IRI-POI and CC-POI. The activation of LI at POIs present in those servers can be done in two ways:

- Via provisioning.
- Via triggering.

Provisioning-method of activation can be used if and only if the user identity provided by the RCS user e.g. SIP/TEL URI, GPSI, Email Address) during the authentication of the user as a part of file upload or download procedures happens to be a target identity provisioned at the POIs in the HTTP Content Server or File Localization Function. To support this, the LIPF in ADMF will have to provision the IRI-POI and CC-POI present in the HTTP Content Server and the File Localization Function.

Triggering-method of activation is used when the RCS User sending or receiving the SIP: MESSAGE happens to a target identity at the respective RCS Servers. The triggering-method also helps for the HTTP Content Server and the File Localization Function to provide the IRI-POI and CC-POI for the file upload and download related procedures when the provisioning-method is not (or cannot be) used. To support the triggering-method, the IRI-TF/CC-TF present in the RCS Servers trigger the IRI-POI/CC-POI present in the HTTP Content Server and File Localization Function, the latter, when present.

The target non-local ID does not apply to the LI functions provided at the HTTP Content Server or File Localization Function.

When both methods (provisioned and triggered) are used for the interception at the HTTP Content Server or File Localization Function, the delivery of LI product to the LEMF is done only once. Within the LI architecture for file-transfer related procedures, detecting and suppressing the redundant delivery is presumed to be a part of MDF as illustrated in figure 4.3.2.1-1 below:

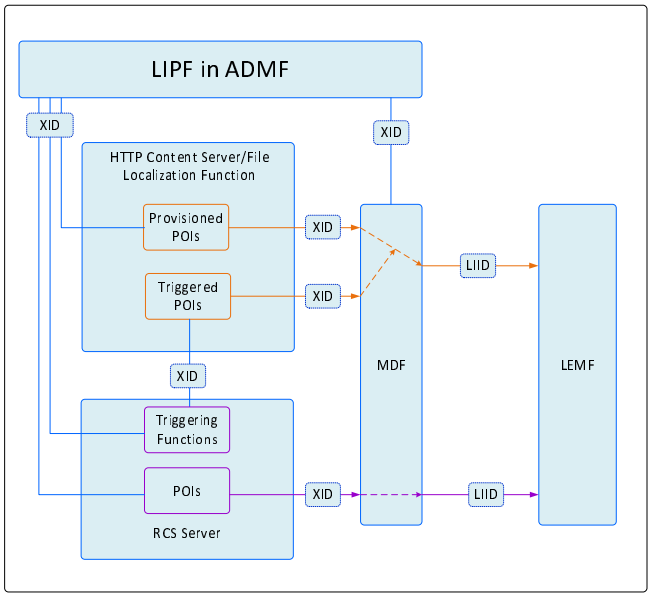


Figure 4.3.2.1-1: Delivery of same LI Product to LEMF is suppressed at MDF

The POIs at the RCS Server report the SIP messaging related xIRI records. These xIRI records are not generated by the POIs present in the HTTP Content Server/File Localization Function. Therefore, the MDF would forward these xIRI records to the LEMF as IRI messages.

If the target identity provisioned at the POIs in the HTTP Content Server/File Localization Function matches with the user identity provided by the RCS User as a part of authentication procedures, then the provisioned POIs generate xIRIs and xCC as and when the file is uploaded or downloaded by that RCS User.

When a trigger is received from the TFs present in the RCS Server, the POIs in the HTTP Content Server/File Localization Function generate xIRI records and xCC if a file is present at the URL given by the TFs as target identity. Based on the fact the received xIRIs and xCC have the same XID, the MDF would forward the xIRIs and xCC to the LEMF only once. However, such a suppression is done if and only if MDF detects that delivered LI products would have been redundant.

Independent of such suppression, all the IRI messages and CC do have the correlation number.

4.3.2.2 LI functions for file transfer in the same domain

4.3.2.2.1 Sender is a target

Figure 4.3.2.2.1-1 extends the illustration shown in figure 4.3.1.2-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (A) is a target in the RCS Server (A) and the HTTP Content Server. Both provisioning-method and triggering-method are considered for the LI at the HTTP Content Server.

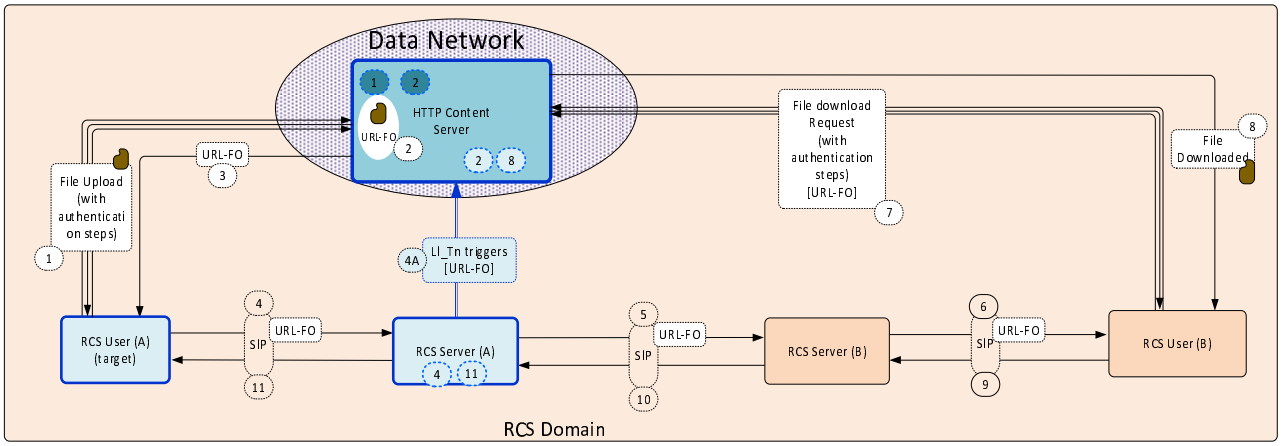


Figure 4.3.2.2.1-1: File transfer in the same domain – LI shown (sender is target)

For the interception of RCS related SIP messages, the RCS Server (A) provides the LI functions when RCS User (A) is a target for RCS.

For file upload and download related steps, the HTTP Content Server provides the LI functions based on the target identity provisioned and based on the trigger received from the TFs present in the RCS Server (A).

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.2-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (A) provides the LI function for the SIP messages received from and sent to the RCS User (A), a target for RCS. The diagram indicates that step 4 and step 11 lead to LI.
- If the identity of RCS User (A) during the authentication procedure (step 1) is a provisioned target identity, then the POIs present in the HTTP Content Server provide the LI functions for the file uploaded by the target. The diagram indicates that step 1 and step 2 lead to LI. Step 2 leads to LI if CC delivery is required.
- When the URL-FO is received from the target (RCS User (A)), the IRI-TF present in the RCS Server (A) sends the LI_T2 trigger to the IRI-POI present in the HTTP Content server.
- Likewise, when the URL-FO is received from the target (RCS User (A)), the CC-TF present in the RCS Server (A) sends the LI_T3 trigger to the CC-POI present in the HTTP Content Server, if CC delivery is required.
- The two triggers are shown in the diagram LI_Tn triggers with URL_FO as the target identifier. Within the diagram, the trigger is shown as step 4A, executed after step 4.
- IRI-POI present in the HTTP Content Server, upon receiving the LI_T2 trigger generates the xIRI record indicating that a file present at the URL-FO is downloaded from that URL. The diagram indicates that step 8 leads to LI.
- The file upload has actually happened before the LI_T2 trigger was received and therefore, the IRI-POI generates the xIRI record based on the fact that an uploaded file is stored at the URL_FO. The xIRI may indicate that a file is present at the URL_FO. The diagram indicates that step 2 leads to LI.
- CC-POI present in the HTTP Content Server, upon receiving the LI_T3 trigger generates the xCC record with the content of the file stored at the URL_FO, the target identifiers received in the LI_T3 trigger. The diagram indicates that step 2 leads to LI.

4.3.2.2.2 Recipient is a target

Figure 4.3.2.2.2-1 extends the illustration shown in figure 4.3.1.2-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (B) is a target in the RCS Server (B) and the HTTP Content Server. Only the provisioning-method can be considered RCS Server (B) is presumed to be not able to send any triggers to the POIs present in the HTTP Content Server.

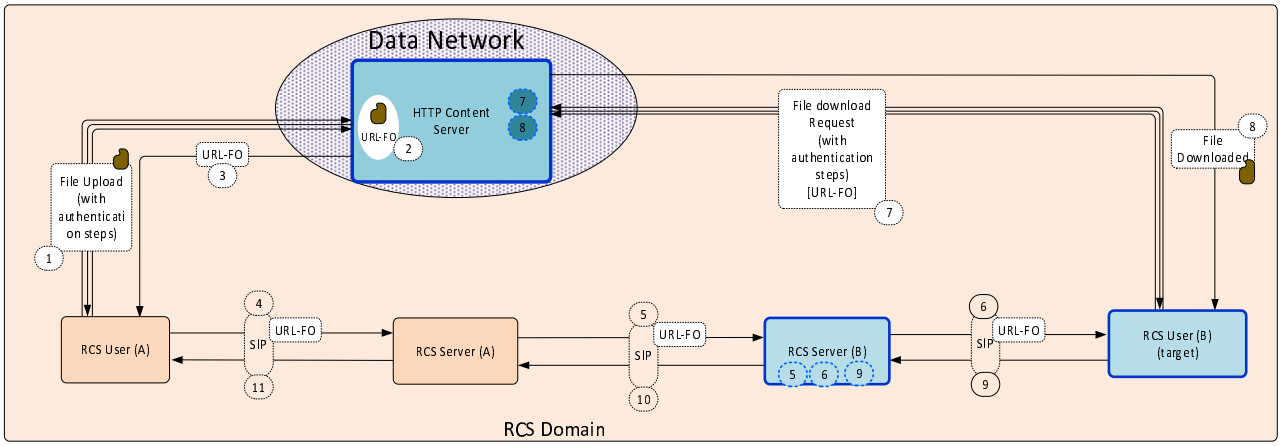


Figure 4.3.2.2.2-1: File transfer in the same domain – LI shown (recipient is target)

For the interception of RCS related SIP messages, the RCS Server (B) provides the LI functions when RCS User (B) is a target for RCS.

With the provisioning method, for file download related steps, the HTTP Content Server provides the LI functions based on the target identity provisioned.

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.2-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (B) provides the LI function for the SIP messages received to and then from and sent to the RCS User (B), a target for RCS. The diagram indicates that steps 5, 6 and 9 lead to LI.
- If the identity of RCS User (B) during the authentication procedure (step 7) is a provisioned target identity, then the POIs present in the HTTP Content Server provide the LI functions for the file downloaded by the target. The diagram indicates that step 7 and step 8 lead to LI. If the CC is delivery is required, then the CC-POI would provide the contents of the file.

4.3.2.3 LI functions for file transfer with file localization function

4.3.2.3.1 Sender is a target

Figure 4.3.2.3.1-1 extends the illustration shown in figure 4.3.1.3-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (A) is a target in the RCS Server (A) and the HTTP Content Server. Both provisioning-method and triggering-method are considered for the LI at the HTTP Content Server.

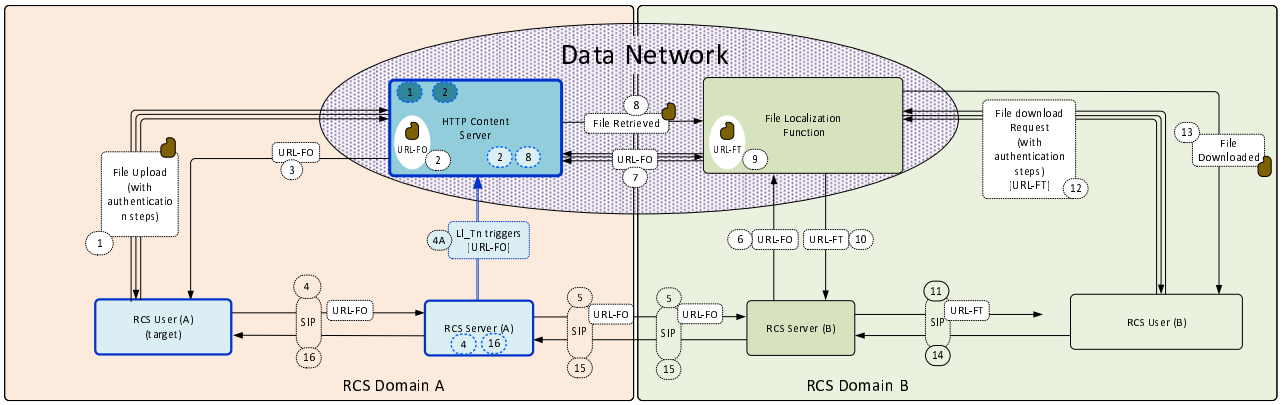


Figure 4.3.2.3.1-1: File transfer with a file localization function – LI shown (sender is a target)

For the interception of RCS related SIP messages, the RCS Server (A) provides the LI functions when RCS User (A) is a target for RCS.

For file upload and retrieval related steps, the HTTP Content Server provides the LI functions based on the target identity provisioned and based on the trigger received from the TFs present in the RCS Server (A).

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.3-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (A) provides the LI function for the SIP messages received from and sent to the RCS User (A), a target for RCS. The diagram indicates that step 4 and step 16 lead to LI.
- If the identity of RCS User (A) during the authentication procedure (step 1) is a provisioned target identity, then the POIs present in the HTTP Content Server provide the LI functions for the file uploaded by the target. The diagram indicates that step 1 and step 2 lead to LI. Step 2 leads to LI if CC delivery is required.
- When the URL-FO is received from the target (RCS User (A)), the IRI-TF present in the RCS Server (A) sends the LI_T2 trigger to the IRI-POI present in the HTTP Content server.
- Likewise, when the URL-FO is received from the target (RCS User (A)), the CC-TF present in the RCS Server (A) sends the LI_T3 trigger to the CC-POI present in the HTTP Content Server, if CC delivery is required.
- The two triggers are shown in the diagram LI_Tn triggers with URL_FO as the target identifier. Within the diagram, the trigger is shown as step 4A, executed after step 4.
- IRI-POI present in the HTTP Content Server, upon receiving the LI_T2 trigger generates the xIRI record indicating that a file present at the URL-FO is downloaded from that URL. The diagram indicates that step 8 leads to LI.
- The file upload has actually happened before the LI_T2 trigger was received and therefore, the IRI-POI generates the xIRI record based on the fact that an uploaded file is stored at the URL_FO. The xIRI may indicate that a file is present at the URL_FO. The diagram indicates that step 2 leads to LI.
- CC-POI present in the HTTP Content Server, upon receiving the LI_T3 trigger generates the xCC record with the content of the file stored at the URL_FO, the target identifiers received in the LI_T3 trigger. The diagram indicates that step 2 leads to LI.

One may observe that the LI steps at the HTTP Content Server are same as shown in the illustration of clause 4.3.2.2.1.

4.3.2.3.2 Recipient is a target

Figure 4.3.2.3.2-1 extends the illustration shown in figure 4.3.1.3-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (B) is a target in the RCS Server (B) and the File Localization Function. Both provisioning-method and triggering-method are considered for the LI at the File Localization Function.

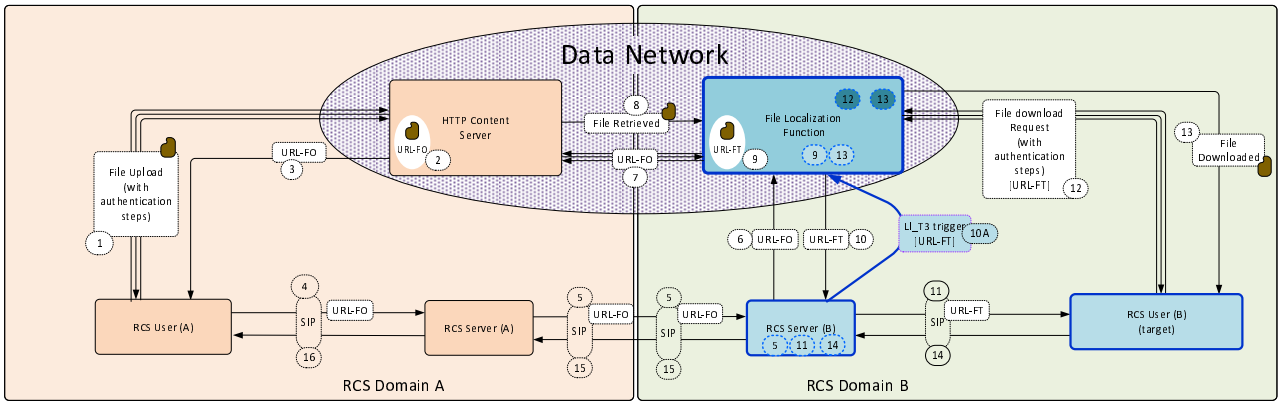


Figure 4.3.2.3.2-1: File transfer with a file localization function – LI shown (recipient is target)

For the interception of RCS related SIP messages, the RCS Server (B) provides the LI functions when RCS User (B) is a target for RCS.

For file retrieval and file download related steps, the File Localization Function provides the LI functions based on the target identity provisioned and based on the trigger received from the TFs present in the RCS Server (B).

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.3-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (B) provides the LI function for the SIP messages received to and then from and sent to the RCS User (B), a target for RCS. The diagram indicates that steps 5, 11 and 14 lead to LI.
- If the identity of RCS User (B) during the authentication procedure (step 12) is a provisioned target identity, then the POIs present in the File Localization Function provide the LI functions for the file downloaded by the target. The diagram indicates that step 12 and step 13 lead to LI. If the CC is delivery is required, then the CC-POI would provide the contents of the file.
- When the URL-FT is received from File Localization Function, if the destination (i.e. the recipient, the RCS User (B)) is a target, then the IRI-TF present in the RCS Server (B) sends the LI_T2 trigger to the IRI-POI present in the File Localization Function.
- Likewise, when the URL-FT is received from File Localization Function, if the destination (i.e. the recipient, the RCS User (B)) is a target, then the CC-TF present in the RCS Server (B) sends the LI_T3 trigger to the CC-POI present in the File Localization Function, if CC delivery is required.
- The two triggers are shown in the diagram LI_Tn triggers with URL_FO as the target identifier. Within the diagram, the trigger is shown as step 10A, executed after step 10.
- IRI-POI present in the File Localization Function, upon receiving the LI_T2 trigger generates the xIRI record indicating that a file present at the URL-FT is downloaded from that URL. The diagram indicates that step 13 leads to LI.
- The file retrieval has actually happened before the LI_T2 trigger was received and therefore, the IRI-POI generates the xIRI record based on the fact that retrieved file is stored at the URL_FT. The xIRI may indicate that a file is present at the URL_FT. The diagram indicates that step 9 leads to LI.
- CC-POI present in the File Localization Function, upon receiving the LI_T3 trigger generates the xCC record with the content of the file stored at the URL_FT, the target identifiers received in the LI_T3 trigger. The diagram indicates that step 9 leads to LI.

4.3.2.3.3 Recipient is a target non-local ID

Figure 4.3.2.3.3-1 extends the illustration shown in figure 4.3.1.3-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (B) is a target non-local ID in the RCS Server (A). Only triggering-method is considered for the LI at the HTTP Content Server.

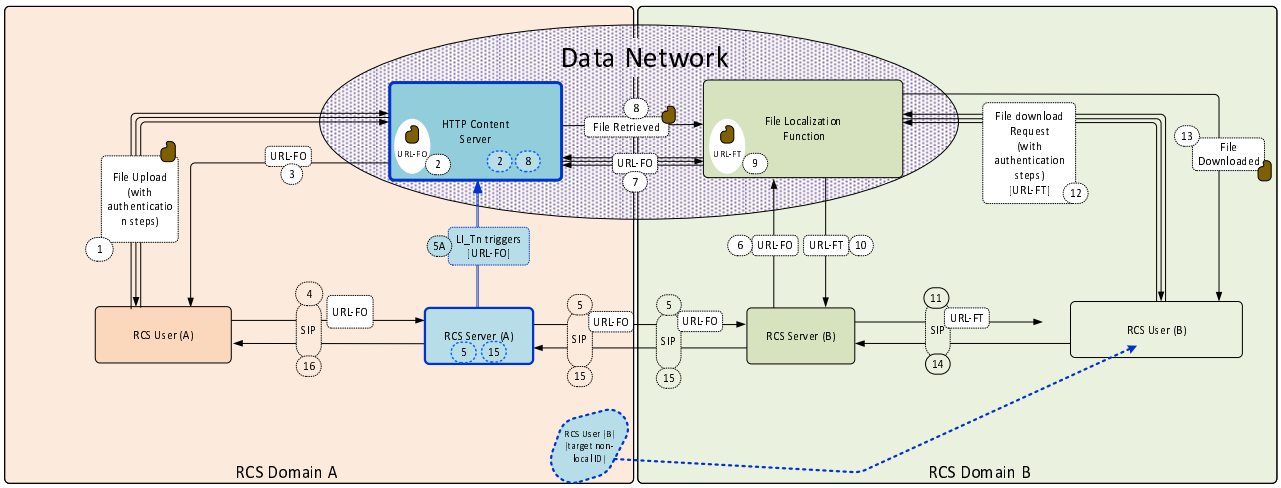


Figure 4.3.2.3.3-1: File transfer with a file localization function – LI shown (recipient is a target non-local ID)

For the interception of RCS related SIP messages, the RCS Server (A) provides the LI functions when RCS User (B) is a target non-local ID for RCS.

For file upload and retrieval related steps, the HTTP Content Server provides the LI functions based on the trigger received from the TFs present in the RCS Server (A).

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.3-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (A) provides the LI function for the SIP messages sent to or received from the direction of RCS User (B), a target non-local ID in RCS. The diagram indicates that step 5 and step 15 lead to LI.
- When the URL-FO is sent to the RCS Server (B) in the direction of RCS User (B) target non-local ID), the IRI-TF present in the RCS Server (A) sends the LI_T2 trigger to the IRI-POI present in the HTTP Content server.
- Likewise, when the URL-FO is sent to the RCS Server (B) in the direction of RCS User (B) target non-local ID), the CC-TF present in the RCS Server (A) sends the LI_T3 trigger to the CC-POI present in the HTTP Content Server, if CC delivery is required.
- The two triggers are shown in the diagram LI_Tn triggers with URL_FO as the target identifier. Within the diagram, the trigger is shown as step 5A, executed after step 5.
- IRI-POI present in the HTTP Content Server, upon receiving the LI_T2 trigger generates the xIRI record indicating that a file present at the URL-FO is downloaded from that URL. The diagram indicates that step 8 leads to LI.
- The file upload has actually happened before the LI_T2 trigger was received and therefore, the IRI-POI generates the xIRI record based on the fact that an uploaded file is stored at the URL_FO. The xIRI may indicate that a file is present at the URL_FO. The diagram indicates that step 2 leads to LI.
- CC-POI present in the HTTP Content Server, upon receiving the LI_T3 trigger generates the xCC record with the content of the file stored at the URL_FO, the target identifiers received in the LI_T3 trigger. The diagram indicates that step 2 leads to LI.

One may observe that the LI steps at the HTTP Content Server are same as shown in the illustration of clause 4.3.2.3.1. The POIs in HTTP Content Server provide LI based on URL-FO as the target identity. The fact that the TFs in RCS Server (A) sent the triggers for the case of a target non-local ID is not of interest to the POIs in the HTTP Content Server.

4.3.2.3.4 Sender is a target non-local ID

Figure 4.3.2.3.4-1B extends the illustration shown in figure 4.3.1.3-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (A) is a target non-local ID in the RCS Server (B). Only triggering-method are considered for the LI at the File Localization Function.

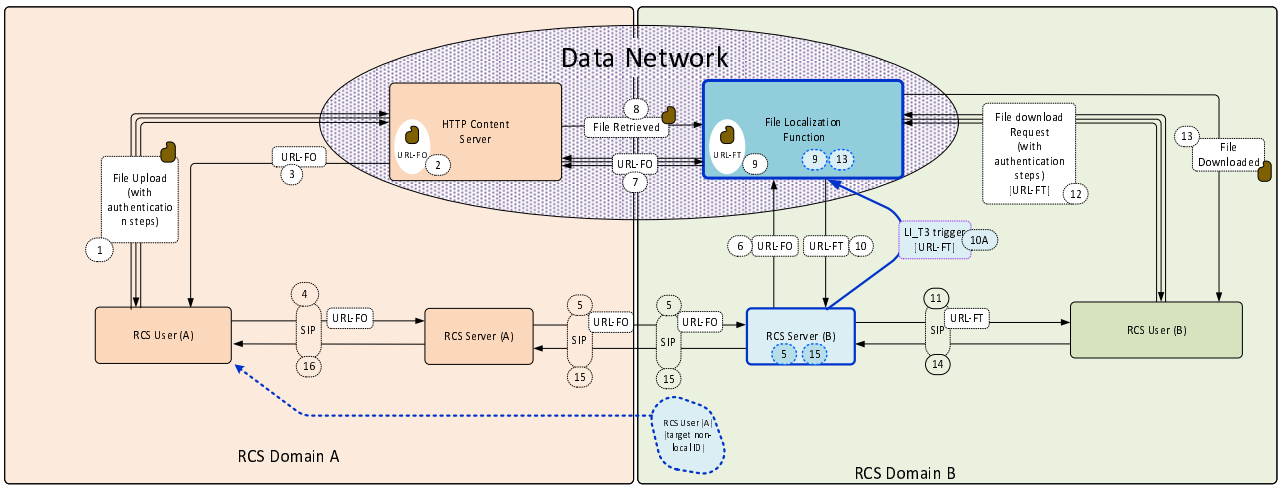


Figure 4.3.2.3.4-1: File transfer with a file localization function – LI shown (sender is a target non-local ID)

For the interception of RCS related SIP messages, the RCS Server (B) provides the LI functions when RCS User (B) is a target for RCS.

For file retrieval and file download related steps, the File Localization Function provides the LI functions based on the target identity provisioned and based on the trigger received from the TFs present in the RCS Server (B).

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.3-1. The LI specific steps a target non-local ID for RCS.

For file retrieval and file download related steps, the File Localization Function provides the LI functions based on the trigger received from the TFs present in the RCS Server (B).

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.3-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (B) provides the LI function for the SIP messages received from and sent to the RCS User (A), a target non-local ID for RCS. The diagram indicates that steps 5 and 15 lead to LI.
- When the URL-FT is received from File Localization Function, if the originating end (i.e. the sender, the RCS User (A)) is a target non-local ID, then the IRI-TF present in the RCS Server (B) sends the LI_T2 trigger to the IRI-POI present in the File Localization Function.
- When the URL-FT is received from File Localization Function, if the originating end (i.e. the sender, the RCS User (A)) is a target non-local ID, then the CC-TF present in the RCS Server (B) sends the LI_T3 trigger to the CC-POI present in the File Localization Function, if CC delivery is required.
- The two triggers are shown in the diagram LI_Tn triggers with URL_FT as the target identifier. Within the diagram, the trigger is shown as step 10A, executed after step 10.
- IRI-POI present in the File Localization Function, upon receiving the LI_T2 trigger generates the xIRI record indicating that a file present at the URL-FT is downloaded from that URL. The diagram indicates that step 13 leads to LI.
- The file retrieval has actually happened before the LI_T2 trigger was received and therefore, the IRI-POI generates the xIRI record based on the fact that retrieved file is stored at the URL_FT. The xIRI may indicate that a file is present at the URL_FT. The diagram indicates that step 9 leads to LI.
- CC-POI present in the File Localization Function, upon receiving the LI_T3 trigger generates the xCC record with the content of the file stored at the URL_FT, the target identifiers received in the LI_T3 trigger. The diagram indicates that step 9 leads to LI.

4.3.2.4 LI functions for file transfer without the file localization function

4.3.2.4.1 Sender is a target

Figure 4.3.2.4.1-1 extends the illustration shown in figure 4.3.1.4-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (A) is a target in the RCS Server (A) and the HTTP Content Server. Both provisioning-method and triggering-method are considered for the LI at the HTTP Content Server.

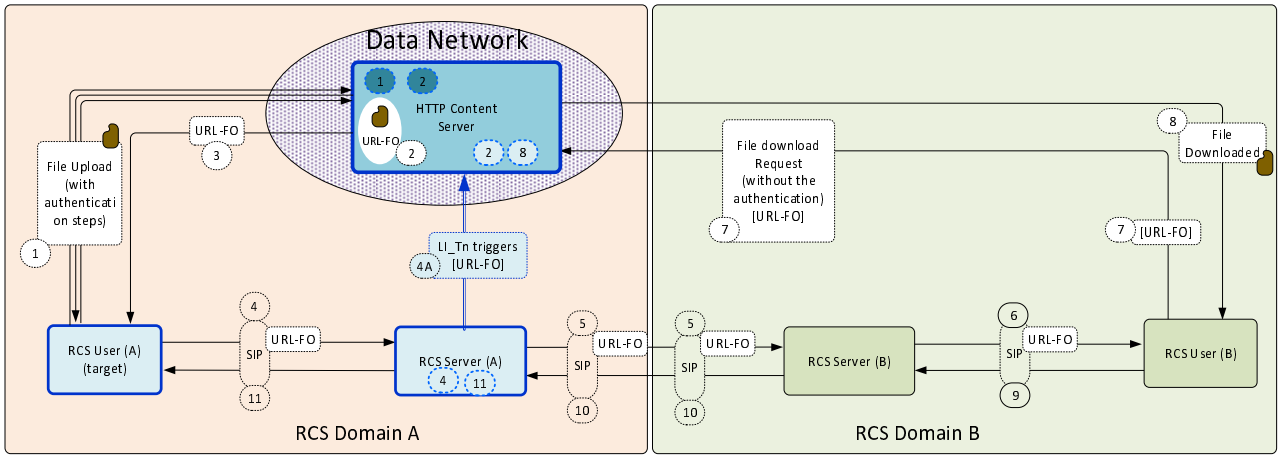


Figure 4.3.2.4.1-1: File transfer without a file localization function – LI shown (sender is a target)

For the interception of RCS related SIP messages, the RCS Server (A) provides the LI functions when RCS User (A) is a target for RCS.

For file upload and retrieval related steps, the HTTP Content Server provides the LI functions based on the target identity provisioned and based on the trigger received from the TFs present in the RCS Server (A).

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.4-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (A) provides the LI function for the SIP messages received from and sent to the RCS User (A), a target for RCS. The diagram indicates that step 4 and step 11 lead to LI.
- If the identity of RCS User (A) during the authentication procedure (step 1) is a provisioned target identity, then the POIs present in the HTTP Content Server provide the LI functions for the file uploaded by the target. The diagram indicates that step 1 and step 2 lead to LI. Step 2 leads to LI if CC delivery is required.
- When the URL-FO is received from the target (RCS User (A)), the IRI-TF present in the RCS Server (A) sends the LI_T2 trigger to the IRI-POI present in the HTTP Content server.
- Likewise, when the URL-FO is received from the target (RCS User (A)), the CC-TF present in the RCS Server (A) sends the LI_T3 trigger to the CC-POI present in the HTTP Content Server, if CC delivery is required.
- The two triggers are shown in the diagram LI_Tn triggers with URL_FO as the target identifier. Within the diagram, the trigger is shown as step 4A, executed after step 4.
- IRI-POI present in the HTTP Content Server, upon receiving the LI_T2 trigger generates the xIRI record indicating that a file present at the URL-FO is downloaded from that URL. The diagram indicates that step 8 leads to LI.
- The file upload has actually happened before the LI_T2 trigger was received and therefore, the IRI-POI generates the xIRI record based on the fact that an uploaded file is stored at the URL_FO. The xIRI may indicate that a file is present at the URL_FO. The diagram indicates that step 2 leads to LI.
- CC-POI present in the HTTP Content Server, upon receiving the LI_T3 trigger generates the xCC record with the content of the file stored at the URL_FO, the target identifiers received in the LI_T3 trigger. The diagram indicates that step 2 leads to LI.

One may observe that the LI steps at the HTTP Content Server are same as shown in the illustration of clause 4.3.2.2.1.

4.3.2.4.2 Recipient is a target

Figure 4.3.2.4.2-1 extends the illustration shown in figure 4.3.1.4-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (B) is a target in the RCS Server (B).

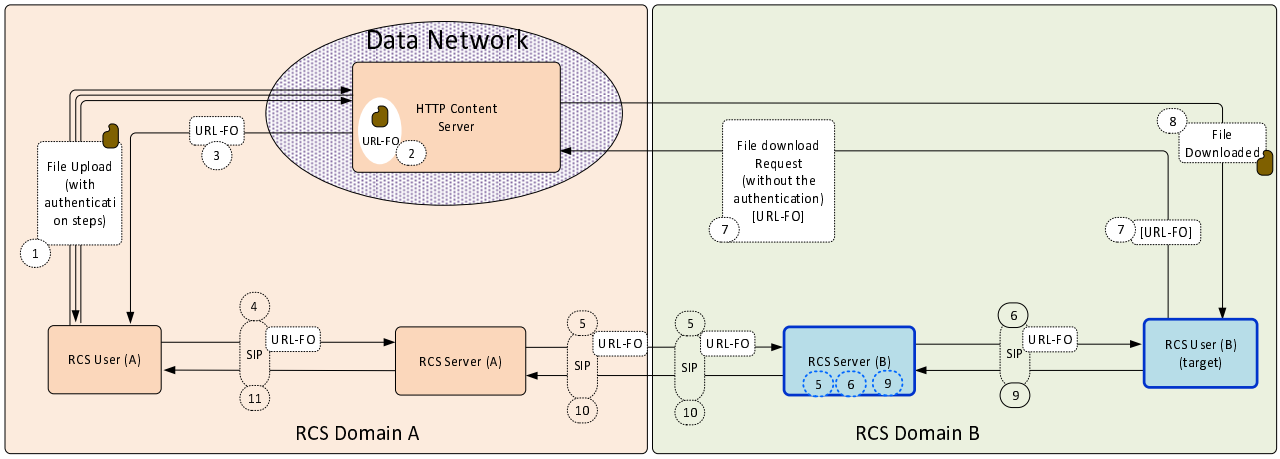


Figure 4.3.2.4.2-1: File transfer without a file localization function – LI shown (recipient is target)

For the interception of RCS related SIP messages, the RCS Server (B) provides the LI functions when RCS User (B) is a target for RCS.

There is no LI for file retrieval and file download related steps since File Localization Function is not present in this case.

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.3-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (B) provides the LI function for the SIP messages received to and then from and sent to the RCS User (B), a target for RCS. The diagram indicates that steps 5, 6 and 9 lead to LI.

4.3.2.4.3 Recipient is a target non-local ID

Figure 4.3.2.4.3-1 extends the illustration shown in figure 4.3.1.4-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (B) is a target non-local ID in the RCS Server (A). Only triggering-method is considered for the LI at the HTTP Content Server.

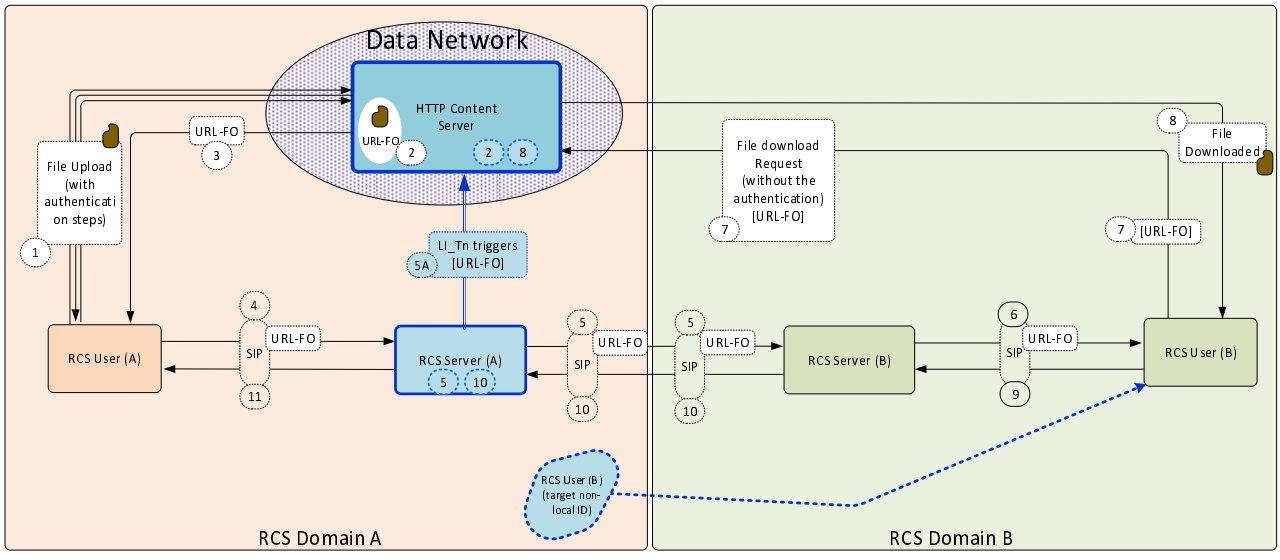


Figure 4.3.2.4.3-1: File transfer without a file localization function – LI shown (recipient is a target non-local ID)

For the interception of RCS related SIP messages, the RCS Server (A) provides the LI functions when RCS User (B) is a target non-local ID for RCS.

For file upload and retrieval related steps, the HTTP Content Server provides the LI functions based on the trigger received from the TFs present in the RCS Server (A).

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.4-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (A) provides the LI function for the SIP messages sent to or received from the direction of RCS User (B), a target non-local ID in RCS. The diagram indicates that step 5 and step 10 lead to LI.
- When the URL-FO is sent to the RCS Server (B) in the direction of RCS User (B) target non-local ID), the IRI-TF present in the RCS Server (A) sends the LI_T2 trigger to the IRI-POI present in the HTTP Content server.
- Likewise, when the URL-FO is sent to the RCS Server (B) in the direction of RCS User (B) target non-local ID), the CC-TF present in the RCS Server (A) sends the LI_T3 trigger to the CC-POI present in the HTTP Content Server, if CC delivery is required.
- The two triggers are shown in the diagram LI_Tn triggers with URL_FO as the target identifier. Within the diagram, the trigger is shown as step 5A, executed after step 5.
- IRI-POI present in the HTTP Content Server, upon receiving the LI_T2 trigger generates the xIRI record indicating that a file present at the URL-FO is downloaded from that URL. The diagram indicates that step 8 leads to LI.
- The file upload has actually happened before the LI_T2 trigger was received and therefore, the IRI-POI generates the xIRI record based on the fact that an uploaded file is stored at the URL_FO. The xIRI may indicate that a file is present at the URL_FO. The diagram indicates that step 2 leads to LI.
- CC-POI present in the HTTP Content Server, upon receiving the LI_T3 trigger generates the xCC record with the content of the file stored at the URL_FO, the target identifiers received in the LI_T3 trigger. The diagram indicates that step 2 leads to LI.

One may observe that the LI steps at the HTTP Content Server are same as shown in the illustration of clause 4.3.2.3.1. The POIs in HTTP Content Server provide LI based on URL-FO as the target identity. The fact that the TFs in RCS Server (A) sent the triggers for the case of a target non-local ID is not of interest to the POIs in the HTTP Content Server.

4.3.2.4.4 Sender is a target non-local ID

Figure 4.3.2.4.4-1 extends the illustration shown in figure 4.3.1.4-1 by including the LI steps performed along with the file transfer related steps. Here, RCS User (A) is a target non-local ID in the RCS Server (B).

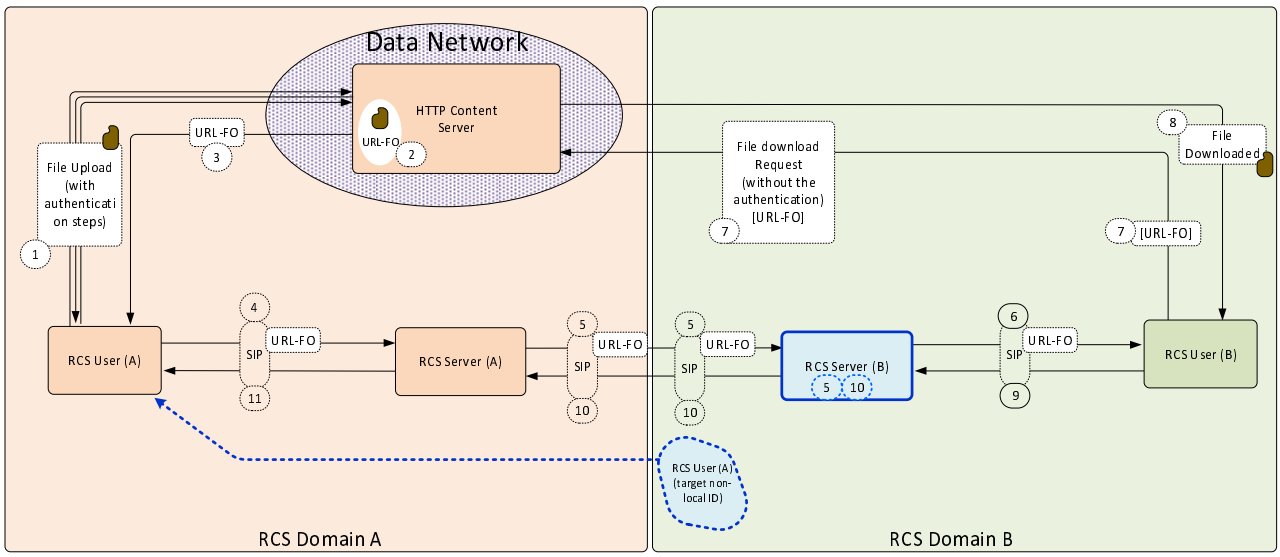


Figure 4.3.2.4.4-1: File transfer without a file localization function – LI shown (sender is a target non-local ID)

For the interception of RCS related SIP messages, the RCS Server (B) provides the LI functions when RCS User (A) is a target non-local ID for RCS.

The steps involved in the file transfer procedure same as the steps illustrated in figure 4.3.1.4-1. The LI specific steps are as listed below:

- The IRI-POI present in the RCS Server (B) provides the LI function for the SIP messages received from and sent to the RCS User (A), a target non-local ID for RCS. The diagram indicates that steps 5 and 10 lead to LI.

5 Example call flows of LI for IMS-based RCS

5.1 General

The subsequent clauses provide flow-diagrams that illustrate the RCS LI. The flows are drawn separately from perspective of originating domain and terminating domain. When the RCS users are in the same domain, the flow will be the same except that IBCFs and TrGWs are not present.

The events that trigger the generation of xIRIs and xCC observe the same principles as in the case of LI for IMS-based services (TS 33.127 [3]. TS 33.128 [4], TR 33.929-1 [1]).

5.2 RCS session establishment related flows

5.2.1 Originating domain

5.2.1.1 User-A (target) initiates an RCS session with MSRP media

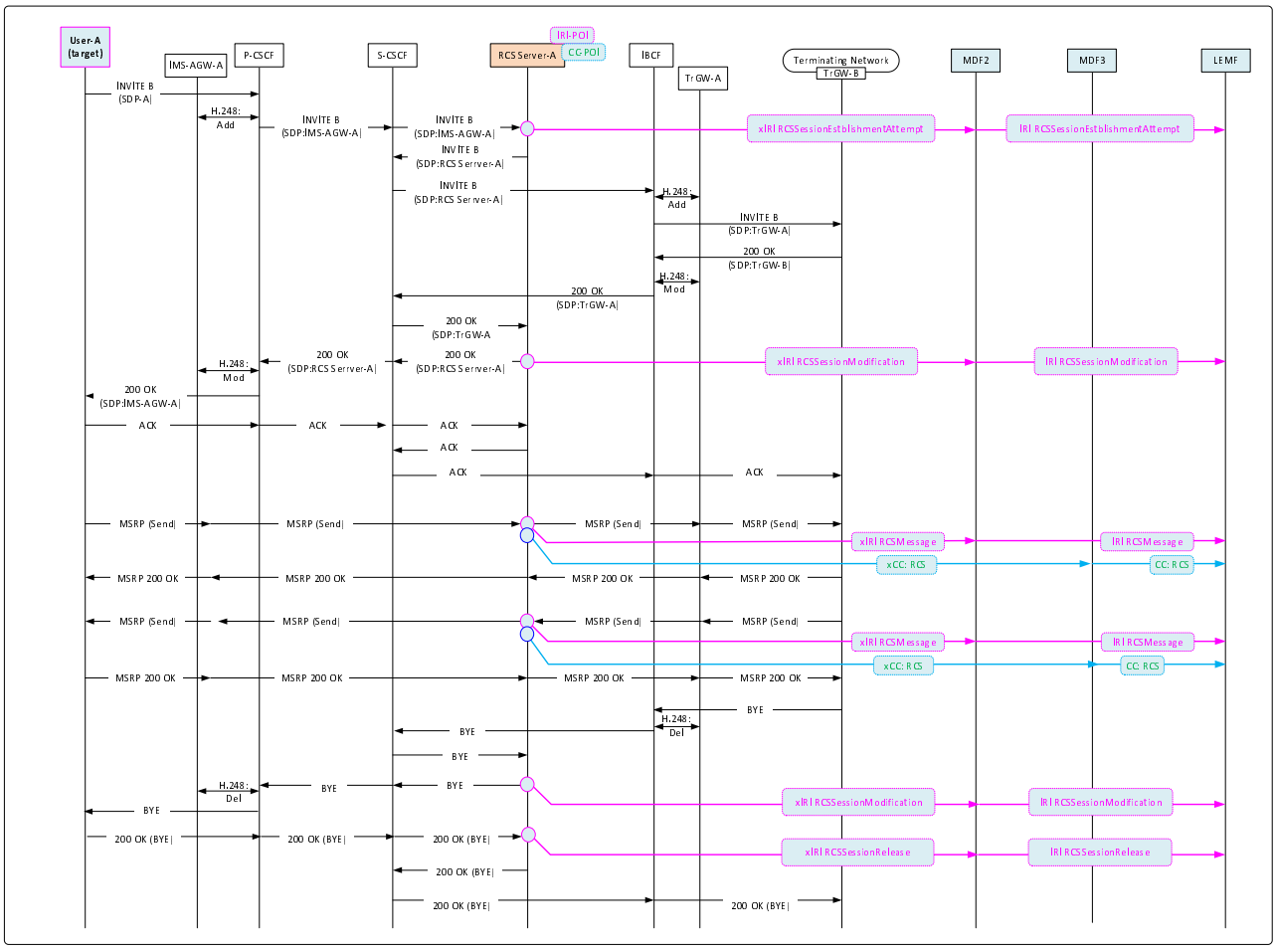


Figure 5.2.1.1-1: User-A initiates an RCS session with MSRP media

In the flow shown in figure 5.2.1.1-1, User-A is a target for RCS in the RCS Server-A. In the illustration, User-A (target) initiates an RCS 1:1 chat session.

SDP answer is received in the SIP 200 OK response for the initial SIP INVITE. After the RCS session is established, the User-A sends an MSRP and receives an MSRP. At the end, the session is released from the terminating end.

5.2.1.2 User-A initiates an RCS session with MSRP media to B to target non-local ID

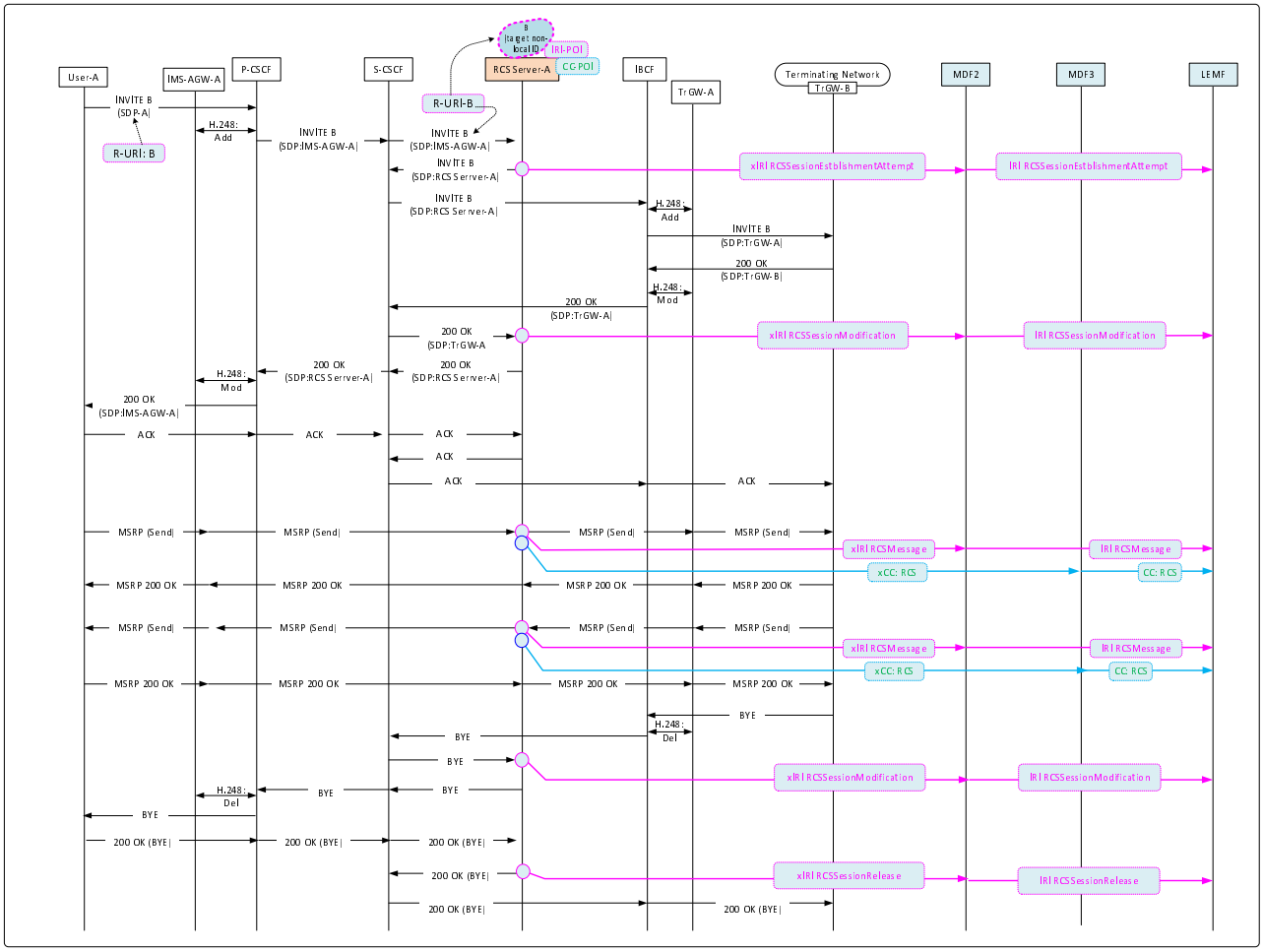


Figure 5.2.1.2-1: User-A initiates an RCS session with MSRP media to target non-local ID

In the flow shown in figure 5.2.1.2-1, the destination (B) is a target non-local ID for RCS in the originating domain (i.e. in RCS Server-A). In the illustration, User-A initiates an RCS 1:1 chat session.

The SDP answer is received in the SIP 200 OK response for the initial SIP INVITE. After the RCS session is established, the User-A sends an MSRP and receives an MSRP. At the end, the session is released from the terminating end.

5.2.2 Terminating domain

5.2.2.1 User-B (target) receives an RCS session with MSRP media

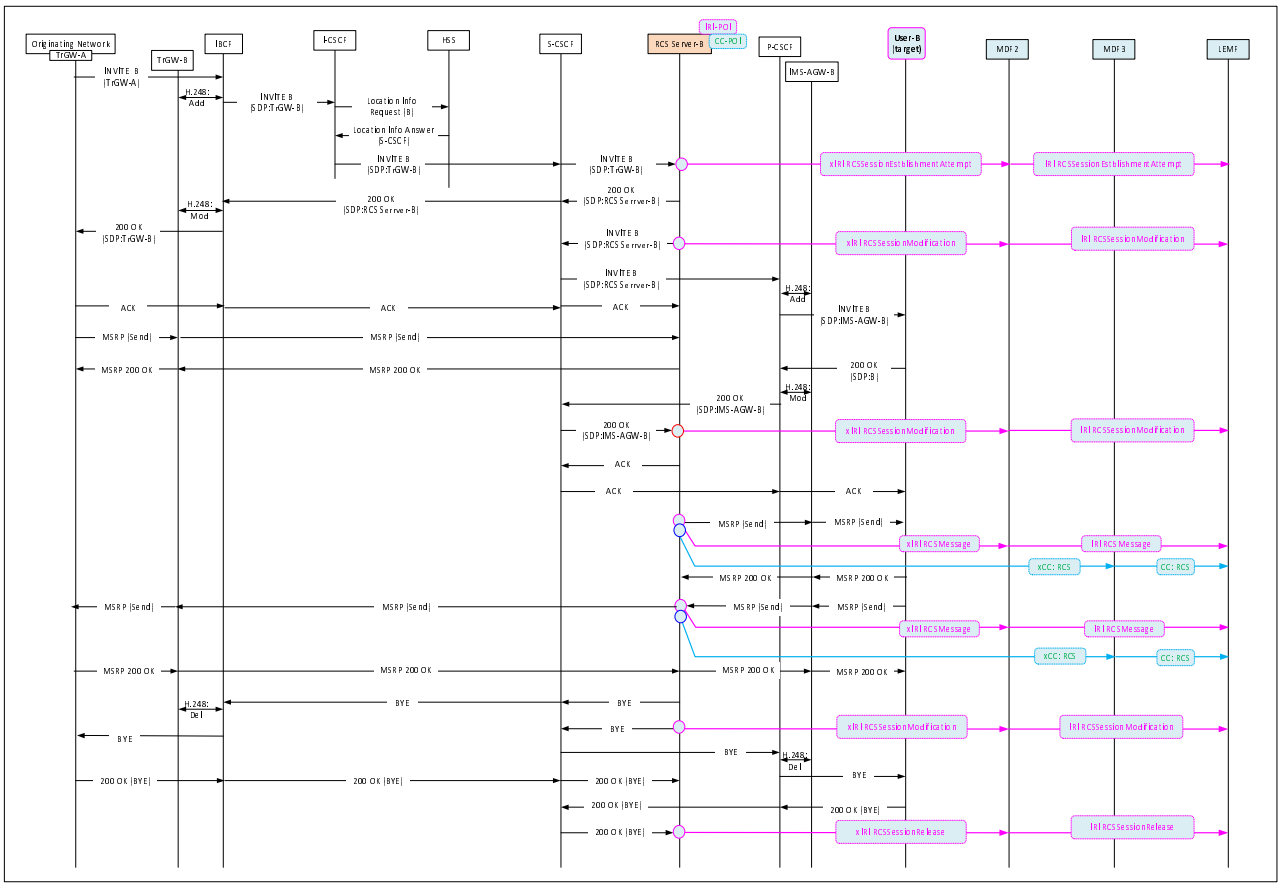


Figure 5.2.2.1-1: RCS session to User-B (target) with MSRP media

In the flow shown in figure 5.2.2.1-1, User-B is a target for RCS in the RCS Server-A. In the illustration, User-B (target) receives an incoming 1:1 chat session request.

In the flow-diagram, the RCS-Server would respond with SIP 200 OK to the SIP: INVITE received from the originating network before forwarding the SIP: INVITE to the User-B (target). The SIP 200 OK carries the SDP answer.

After the RCS session is established, the User-B receives an MSRP from the originating end and sends an MSRP to the originating end. At the end, the session is released from the RCS Server-B.

5.2.2.2 User-B receives an RCS session with MSRP media from target non-local ID

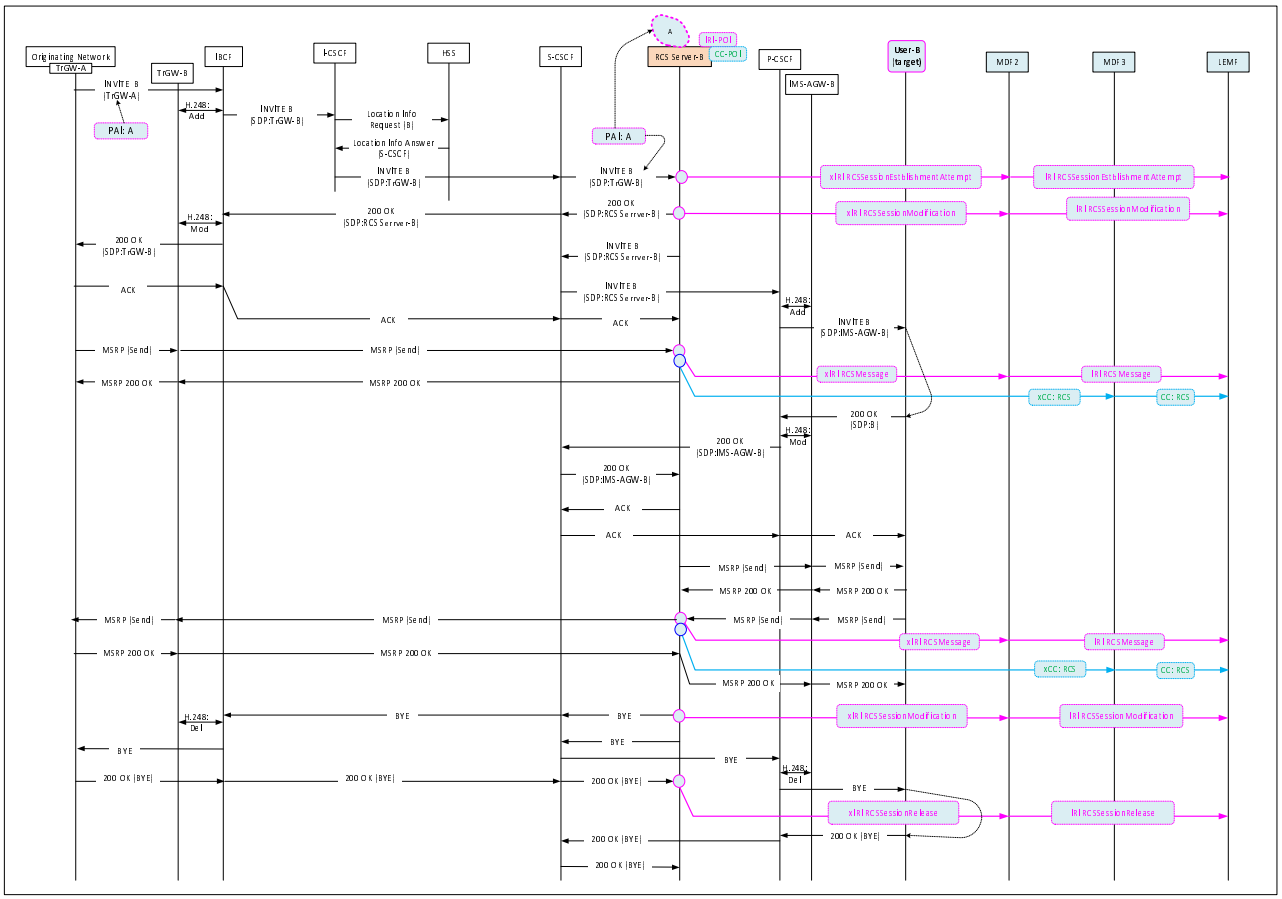


Figure 5.2.2.2-1: RCS session to User-B with MSRP media from target non-local ID

In the flow shown in figure 5.2.2.2-1, the originating user A is a target non-local ID for RCS in the terminating domain (i.e. RCS Server-B). In the illustration, User-B receives an incoming 1:1 chat session request from A (target non-local ID).

In the flow-diagram, the RCS-Server would respond with SIP 200 OK to the SIP: INVITE received from the originating network before forwarding the SIP: INVITE to the User-B. The SIP 200 OK carries the SDP answer. After the RCS session is established, the User-B receives an MSRP from the originating end and sends an MSRP to the originating end. At the end, the session is released from the RCS Server-B.

Annex A:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-07	SA3LI#94	S3i240509				Initial version	0.0.1
2025-07	SA3LI#98	S3i250443				pCRs from SA3LI#98 implemented	0.0.2
2026-07	SA3LI#102	S3i260361				Agreed for Change Control	0.0.3
2026-08	SA#113	SP-260732				Submitted to SA#113 Plenary for Approval	1.0.0
2026-09	SA#113	-				Approved by SA#113 Plenary for Change Control	19.0.0

History

Version	Date	Status
V19.0.0	October 2026	Publication